ATLAS Note

ANA-EXOT-2021-37-INT1

14th November 2025

Draft version 0.1

Unveiling fundamental symmetry violations with dileptons: a test of Lorentz and CPT invariance

The ATLAS Collaboration

This is the first analysis at the ATLAS experiment probing violations of Lorentz and CPT invariance. Dilepton production on the Z-boson pole provides an unparalleled lens for capturing minute and elusive signatures of quark-level Lorentz and CPT violation. Central among these signatures are temporal oscillations of the integrated cross section at harmonics of the Earth's sidereal rotation rate. We perform a comprehensive analysis of generic sidereal signatures in the ATLAS detector frame. Comparison with the ATLAS luminosity profile permits the extraction of several first and leading constraints on effective coefficients for Lorentz and CPT violation.

Contents

1 Motivation

1.1 Target

This analysis probes the fundamental principles of Lorentz and CPT invariance. Recent theoretical breakthroughs have laid a foundation for experimental searches of quark-sector Lorentz- and CPT-violating signatures in collider processes, including the Drell-Yan process. The LHC provides unique and ideal conditions for investigating much of this uncharted landscape.

Our target is to provide first results on Standard Model (SM) quark-sector Lorentz- and CPT-violating effects from ATLAS measurements. Preliminary investigations suggest several first constraints on effects modifying the free propagation and electromagnetic interaction vertex of the first two quark generations will result from this target.

This analysis is planned as the first stage of a new search program studying sidereal oscillations of event observables, including the counting of Z bosons in dilepton final states. Sidereal oscillations are often described as "smoking gun" signatures of Lorentz and CPT violation. They arise from violations of rotation invariance (recall that rotations form a subgroup of the Lorentz group). Most Lorentz- and CPT-violating effects produce sidereal oscillations. Probing these signatures requires dedicated time-dependent analyses involving the binning of measurements based on when they were recorded. As such, most prior works providing constraints on Lorentz- and CPT-violating signatures are indirect, phenomenological, and limited to the smaller subset of time-independent "rotation-invariant" or "boost-violating" signatures. A significant thrust of our program involves developing new methodologies capable of capturing these comparatively numerous and challenging to directly access sidereal signatures.

These latter methodologies are further motivated by the fact that many previous studies have limited attention to the renormalizable modifications and degrees of freedom that can be observed over long durations (e.g. trapped electrons) or over large scales (e.g. cosmic microwave photons). These modifications are mostly orthogonal to the leading signatures in collider processes involving partons and electroweak bosons. The appearance of large collision energies coupled with nearly continuous event data is fertile ground for probing nonrenormalizable operators of unstable or confined degrees of freedom and their associated higher-order sidereal signatures.

While our focus here involves modifications of the first two quark generations, it is crucial to emphasize that the heavy-quark, gluon, and $W^{\pm}$- and Z-boson sectors are also vastly unexplored. Upon completion of our target, we will extend our program to probe new signatures associated to these sectors.

1.2 Context

Lorentz invariance is a foundational assumption of modern physics. It states that physical laws are independent of the relative orientation and velocity of an experiment in spacetime. For example, in the SM Lorentz invariance applies globally (i.e. the same way at every spacetime point) and Lorentz transformations connect observables from experiments rotated or boosted relative to one another. A deep consequence of this Lorentz invariance for quantum field theories in flat spacetime is the invariance of any process under the combined operation of charge conjugation C, parity P, and time reversal T (CPT). This related discrete symmetry invariance follows from the CPT theorem. When restricted to a local symmetry, Lorentz invariance forms a core component of the Einstein equivalence principle underpinning

General Relativity. These strong theoretical motivations are accompanied by numerous experimental results indicating Lorentz invariance holds over a wide range of energies and systems [**datatables**]. Taken together, Lorentz invariance is among the most well-motivated guiding principles in physics.

Despite the numerous investigations undertaken to date, the Lorentz and CPT properties of some sectors of the SM, including the quark sector, remain virtually unexplored. One reason for the comparatively few studies involving quantum chromodynamics (QCD) degrees of freedom relative to quantum electrodynamics (QED)—or, more generally, stable-particle degrees of freedom—is that, as in the conventional SM scenario, inferring quark-level properties from observables involving initial- and final-state hadrons is highly nontrivial. These technical complications persist when Lorentz and CPT invariance are relaxed. However, the theory STA members of this analysis, Enrico Lunghi and Nathaniel Sherrill, have resolved some of the basic issues on this front. They have constructed a Lorentz- and CPT-violating extension of the leading-order (tree level) parton model and have applied it to Drell-Yan dilepton production [**Lunghi˙2021**]. This analysis shows that a class of chiral quark-sector operators challenging to isolate in other experiments can be studied in large momentum transfer pp collisions where the participating quarks couple to $Z, W^{\pm}$ bosons. The estimated size of these corrections are shown in Fig. 1. Details on the SME parameters mentioned in the figure caption will be discussed in the following Sec. 2.

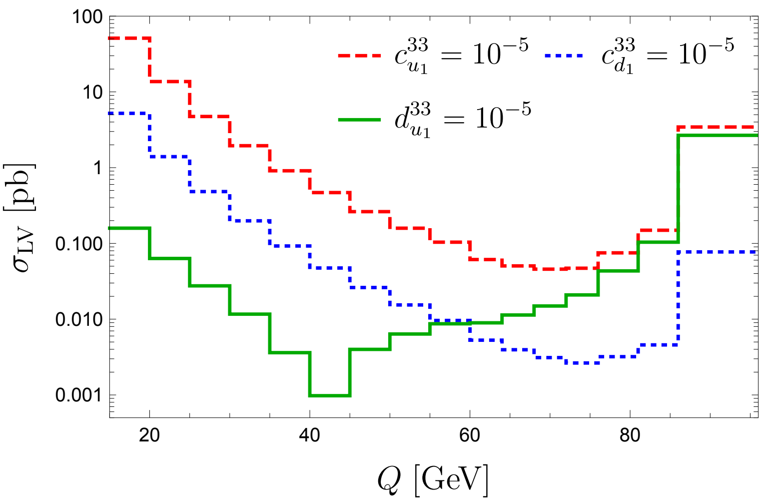

Figure 1: Order-of-magnitude estimates of sensitivity to laboratory Standard-Model Extension (SME) coefficients as a function of Q from predictions in Ref. [**Lunghi˙2021**]. The y-axis displays the Lorentz-violating contribution to the total cross section in picobarns as a function of invariant mass Q from three coefficients, $c_{u_1}^{33}$, $c_{d_1}^{33}$ and $d_{u_1}^{33}$ fixed to 10^{-5}. The subscripts u_1, d_1 denote the first-generation quarks, u, d. The superscripts 33 denote the 33-component of $c_{u_1}^{\mu\nu}$, etc. It is observed that the c-type coefficients have most sensitivity to low Q while the d-type coefficient has most sensitivity on the Z^0 pole at $Q = m_Z$.

2 Theoretical background

2.1 The motivation for and structure of realistic LIV

The absence of a predictive theory of quantum gravity with accompanying low-energy signals remains an open problem. Early work in string theory demonstrated that if additional tensor-valued fields $T^{\mu\nu\cdots}$ obtained a nonzero vacuum expectation value (vev) this could lead to the spontaneous violation of Lorentz and CPT symmetry [**Kostelecky:1988zi**, **Kostelecky:1991ak**]. These vevs would generate additional interactions in an effective fermion theory at low energies with schematic form [**Kostelecky:1991ak**]

$$\mathcal{L}_{\text{LV}} \sim \frac{\lambda}{M^k} \langle T \rangle \cdot \bar{\psi}\Gamma(i\partial)^k\psi + \text{h.c.}, \tag{1}$$

where λ is a coupling constant, M is the scale of new physics (perhaps, but not necessarily, the Planck scale), k is a positive integer and $\langle T \rangle \equiv \langle T^{\mu\nu\cdots} \rangle$ represents tensor-valued vev contracted with the gamma-matrix structure Γ (with suppressed Lorentz indices) in the fermion bilinear product to its right, and h.c. stands for Hermitian conjugation. The vev $\langle T \rangle$ breaks the four-dimensional Lorentz invariance because it has Lorentz indices which "point" in some direction. Experiments with sufficient sensitivity to measure the combination $\lambda \langle T \rangle / M^k$ could indirectly probe certain classes of string theories.

Sustained interest in the possibility of Lorentz violation as a realistic low-energy signal of Planck-scale physics has come from other string-based approaches [**Ellis:2008gg**, **Gliozzi:2011hj**, **Hashimoto:2012ig**], loop quantum gravity [**Gambini:2014kba**, **Rovelli:2010ed**], noncommutative field theories [**Carroll:2001ws**, **Carlson:2001sw**, **Calmet:2004ii**, **Calmet:2004dn**, **Bailey:2018ifc**], high-energy electrodynamics and cosmic rays [**Carroll:1989vb**, **Coleman:1998ti**], and modified theories of gravity [**Kostelecky:2020hbb**, **Kostelecky:2021xhb**, **Kostelecky:2021tdf**, **deRham:2014zqa**, **Horava:2009uw**, **Bluhm:2004ep**, **Bluhm:2007bd**, **Bluhm:2014oua**]. Here we use the adjective "realistic" in the sense of those high-energy mechanisms/models generating LIV that map onto low-energy effective field theories with desirable properties including energy-momentum conservation, locality, the SM gauge structure, etc., except (particle) Lorentz and CPT invariance. We work at the level of realistic effective field theory and do not assume any particular high-energy mechanism inducing LIV.

LIV terms like (1) have some unconventional properties that are best understood by example. Consider the following term:

$$\mathcal{L}_b = -b_\mu \bar{\psi}\gamma_5\gamma^\mu\psi. \tag{2}$$

Here, the coefficient $b_\mu = (b_0, b_j)$ may be thought of as a vector vev. Comparing $\mathcal{L}_b$ to the generic $\mathcal{L}_{\text{LV}}$ in Eq. (1) shows that $b_\mu \leftrightarrow \lambda \langle T \rangle / M^0 = \lambda \langle T \rangle$ and $\Gamma(i\partial)^0 \leftrightarrow \gamma_5\gamma^\mu$. In natural units, power counting indicates the mass dimension of the coefficient is $[b_\mu] =$ GeV. As the Lorentz index μ is contracted, $\mathcal{L}_b \to \mathcal{L}_b$ is invariant under Lorentz transformations of the coordinate system, known as observer Lorentz transformations [**ck97**]. If, in contrast, the *physical system* ψ is rotated, boosted, or some combination of both while leaving the coordinates unaffected, then generically $\mathcal{L}_b \nrightarrow \mathcal{L}_b$. This is known as a particle Lorentz transformation. Under a particle transformation, $b_\mu \to b_\mu$, but $\psi(x)$ transforms in general to a different field $\psi'(x)$ evaluated at the same spacetime point x. Thus, the full term $\mathcal{L}_b$ transforms to a distinct term $\mathcal{L}_b' \neq \mathcal{L}_b$. The reason b_μ is unaffected by particle transformations is because it can be viewed as a background vector-valued field at every point in the vacuum, analogous to the SM Higgs scalar background but including an orientation. Terms like Eq. (2) therefore only describe Lorentz-violating effects associated to the noninvariance under particle Lorentz transformations. To reiterate, $\mathcal{L}_b$ is invariant under observer transformations. This is a desirable feature, because coordinate choices (i.e. changes in the observer's perspective) are arbitrary—they should not affect the physics. The ability to distinguish between particle and observer transformations implies Lorentz invariance is violated.

Equation (2) also describes a CPT-violating effect because $\bar{\psi}\gamma_5\gamma^\mu\psi \to -\bar{\psi}\gamma_5\gamma^\mu\psi$ under the operation of CPT and $b_\mu \to b_\mu$ because the CPT operation is necessarily a type of particle transformation. Thus, $\mathcal{L}_b \to -\mathcal{L}_b$ under a CPT transformation and is therefore CPT odd.

The equations of motion of a system follow from the Euler-Lagrange equations. Performing particle transformations can change the Lagrange density of terms like $\mathcal{L}_b$ in a nontrivial way. Thus, the system ψ can experience different equations of motion depending on its relative orientation to the background field b_μ. For example, one could perform a measurement where the laboratory apparatus (containing the particle content associated to ψ) is initially upright. A second measurement could then be performed except with an the apparatus in a horizontal position. By comparing the two measurements one could, in principle, infer the existence of b_μ. This general idea will be expounded upon when we discuss sidereal oscillations of event observables in the following subsections.

2.2 Standard-Model Extension (SME)

The Lagrange density $\mathcal{L}_b$ in Eq. (2) is actually one among many that appear in the model-independent effective field theory [**Weinberg:2009bg**] for Lorentz and CPT violation, known as the Standard-Model Extension (SME) [**ck96**, **ck98**]. By construction, the SME provides a model-independent parameterization of all possible Lorentz-violating effects constructed from SM and gravitational fields. Each term in the SME can be thought of as the product of a coefficient for Lorentz violation [1] of mass dimension $4 - d$ with a Lorentz-violating field operator of dimension d. For example, in $\mathcal{L}_b$ the coefficient for Lorentz violation is b_μ and the Lorentz-violating field operator is $-\bar{\psi}\gamma_5\gamma^\mu\psi$. Because this field operator is mass dimension $d = 3$, the coefficient for Lorentz violation is of mass dimension $4 - 3 = 1$, as claimed. CPT violation is also parametrized by the SME, since for local and unitary quantum field theories, when CPT invariance is violated, so too is Lorentz invariance [**ck97**, **owg**]. All SME terms violate Lorentz invariance, but the general pattern is that those coefficients with odd numbers of Lorentz indices also violate CPT invariance[2].

Our program will be based on the conservative approach of the SME. The SME Lagrange density can be written as

$$\mathcal{L}_{\rm SME} = \mathcal{L}_{\rm SM} + \delta\mathcal{L}_{\rm LV}, \tag{3}$$

where $\mathcal{L}_{\rm SM}$ is the SM Lagrange density and $\delta\mathcal{L}_{\rm LV}$ represents all possible Lorentz-violating terms consistent with the SM field content and $SU(3) \times SU(2) \times U(1)$ gauge structure. Experimental results suggest it is reasonable to take $\delta\mathcal{L}_{\rm LV}$ as perturbations with respect to $\mathcal{L}_{\rm SM}$. This is equivalent to saying the coefficients for Lorentz violation are treated as perturbative quantities with respect to the mass scales appearing in the process of interest.

SME operators of mass dimension $d \leq 4$ are power-counting renormalizable and part of the *minimal* SME. There are a finite number parameters in the minimal SME. This is the theory obtain by taking the SM and relaxed the conditions of particle Lorentz and CPT invariance. In contrast, operators with $d > 4$

[1] For brevity we collectively refer to the coefficients for Lorentz (and possibly) CPT violation as “SME coefficients”. The SME coefficients are sometimes called Wilson coefficients, given they appear in an effective field theory. We refrain from using this latter terminology to avoid confusion with it usage in Lorentz-invariant effective field theories.

[2] This is true in the vast majority of cases. There are two known exceptions, but these cases involve SME coefficients which are irrelevant for our study.

are power-counting nonrenormalizable and infinite in number. They are part of the *nonminimal* SME. Just as with Lorentz invariant nonrenormalizable theories like the Standard Model Effective Field Theory (SMEFT), the nonminimal SME takes on a particular importance in high-energy process like collider processes. Our program considers both minimal and nonminimal SME effects.

To summarize, the SME provides a rigorously justified analysis framework for probing generic Lorentz- and CPT violating effects in experiments. This is evidenced by the hundreds of constraints placed using the SME over the past two decades [**datatables**]. Despite the impressive progress, some sectors of the SME, including the quark- and electroweak sectors, remain largely unexplored. In the proceeding sections, we describe recent progress on this front and the role existing ATLAS Drell-Yan measurements will play in expanding the search for Lorentz violation.

2.3 Summary of SME studies involving quarks, gluons, and hadrons

Briefly reviewing existing studies involving Lorentz-violating effects in quarks and gluons (partons) and hadrons puts our search into context.

Beginning at the lowest mass dimension $d = 3$, "a-type" effects $-a_\mu \bar{\psi}\gamma^\mu\psi$ have been constrained over a range of ten orders $\lesssim 10^{-12}$-10^{-22} GeV from K, D, B_d, and B_s meson oscillations, and from mapping quark-level coefficients to meson-level effective coefficients using chiral perturbation theory (χPT) [**Kostelecky:1997mh**, **Kostelecky:1999bm**, **Kostelecky:2000ku**, **Isgur:2001yz**, **Nguyen:2001tg**, **Babusci:2013gda**, **Kostelecky:2001ff**, **Aubert:2007bp**, **Kostelecky:2010bk**, **Abazov:2015ana**, **Aaij:2016mos**, **Roberts:2017tmo**, **Schubert:2016xul**, **Altschul:2019beo**]. Similar techniques have led to $d = 5$ constraints on K, D, B_d, and B_s meson effects [**Edwards:2018lsn**, **Edwards:2019lfb**]. The combination of χPT with astrophysical measurements have produced various levels of sensitivity to $d = 4$ "c-type" effects $c_{\mu\nu}\bar{\psi}i\partial^\mu\gamma^\nu\psi$ on light quark (u, d) coefficients and "k-type" gluon effects $-\frac{1}{4}k^{\mu\nu\alpha\beta}G^a_{\mu\nu}G^a_{\alpha\beta}$, respectively [**Kamand:2016xhv**, **Kamand:2017bzl**, **Altschul:2019beo**, **Noordmans:2016pkr**, **Noordmans:2017kji**]. Similar constraints on light quarks have been placed considering the absence of photon decay and vacuum Cherenkov radiation [**Gagnon:2004xh**, **Altschul:2006uw**, **Kostelecky:2013rta**, **Schreck:2017isa**]. Constraints on heavy quarks (the second and third generation) are few in comparison and are typically substantially weaker than those on light quarks. A few bounds have resulted from LEP data [**Karpikov:2016qvq**], $t\bar{t}$ production at Fermilab and the LHC [**Berger:2015yha**, **Abazov:2012iu**, **Carle:2019ouy**], radiative corrections to the photon propagator [**Satunin:2017wmk**], and potentially stable top-flavored hadrons [**Altschul:2020wkw**]

A number of leading constraints in the electron and photon sectors have been placed with collider data. However, these constraints are limited to the rotationally invariant combinations of the SME coefficients [**Altschul:2009xh**, **Altschul:2010na**, **kl19**]. Most of the SME coefficients are not rotationally invariant. Colliders have some advantages over other experiments in constraining the larger set of rotation-violating subsets of SME coefficients. These advantages include the large number of events collected semi-regularly over long durations (many months to years). These features make collider experiments suitable for *sidereal analyses*, which track the oscillation of an observable as a function of the Earth's sidereal period ~ 23 hrs 56 mins. This technique was utilized in Ref. [**Abazov:2012iu**] to place the constraints on several components of the top-quark c- and d-type coefficients (see the following subsection).

Advances in theory and phenomenology (by theory STA members of our program) have enabled arbitrary-dimension d quark-sector effects to be incorporated in perturbative hadronic process, in particular deep inelastic scattering and the Drell-Yan process [**Kostelecky:2016pyx**, **Lunghi:2018uwj**, **Kostelecky:2019fse**, **lssv21**]. We note that gauge-sector Lorentz violation (e.g. involving Lorentz-violating effects in the Z boson itself) have also been considered in Refs. [**Fu:2016fmf**, **Colladay:2017zen**, **Michel:2019tti**], though these considerations lie outside our current goals.

Focusing on the developments in Refs. [**Kostelecky:2016pyx**, **Lunghi:2018uwj**, **Kostelecky:2019fse**, **lssv21**], we show that ATLAS data of Drell-Yan events on the Z-boson pole result in first constraints on the chiral "d-type" coefficients, which are particularly challenging to probe, and derive competitive constraints on the c-type coefficients (introduced above).

2.4 Minimal quark-sector effects

Our primary interest is to investigate the leading quark-sector effects[3]. They are given by three terms in the quark sector of the SME:

$$\mathcal{L} = \tfrac{1}{2} i c_{Q_i}^{\mu\nu} \overline{Q}_i \gamma_\mu \overleftrightarrow{D}_\nu Q_i + \tfrac{1}{2} i c_{U_i}^{\mu\nu} \overline{U}_i \gamma_\mu \overleftrightarrow{D}_\nu U_i + \tfrac{1}{2} i c_{D_i}^{\mu\nu} \overline{D}_i \gamma_\mu \overleftrightarrow{D}_\nu D_i \ , \tag{4}$$

where D_ν is the conventional (SM) covariant derivative. The fields have the usual definitions

$$Q_i = \begin{pmatrix} u_i \\ d_i \end{pmatrix}_L , \quad U_i = (u_i)_R \, , \quad D_i = (d_i)_R \, . \tag{5}$$

The index $i = 1, 2, 3$ denotes flavors $u_i = (u, c, t)$ and $d_i = (d, s, b)$ and the SME coefficients are given by $c_{Q_i}^{\mu\nu}$, $c_{U_i}^{\mu\nu}$, and $c_{D_i}^{\mu\nu}$. They may be taken as symmetric and traceless matrices, leaving nine independent components per type Q, U, D. It is customary and useful to define the combinations

$$\begin{aligned} c_{u_i}^{\mu\nu} &= (c_{Q_i}^{\mu\nu} + c_{U_i}^{\mu\nu})/2, \quad & c_{d_i}^{\mu\nu} &= (c_{Q_i}^{\mu\nu} + c_{D_i}^{\mu\nu})/2 \, , \\ d_{u_i}^{\mu\nu} &= (c_{Q_i}^{\mu\nu} - c_{U_i}^{\mu\nu})/2, \quad & d_{d_i}^{\mu\nu} &= (c_{Q_i}^{\mu\nu} - c_{D_i}^{\mu\nu})/2 \, , \end{aligned} \tag{6}$$

with the constraint $c_{u_i}^{\mu\nu} - c_{d_i}^{\mu\nu} = d_{d_i}^{\mu\nu} - d_{u_i}^{\mu\nu}$. As suggested by the subscripts, we refer to these latter coefficients as the c- and d-type coefficients. When expressing the Lagrange density (58) in terms of mass eigenstates, the d-type coefficients are contracted with a Lorentz-violating operator $\bar{\psi}(i\partial_\mu)\gamma_5\gamma_\nu\psi$ and therefore are associated with a chiral current.

2.5 LIV effects in the Drell-Yan process

Direct probes of the c- and d-type coefficients are typically frustrated by QCD confinement. However, perturbative QCD factorization is known to be maintained in the presence of these coefficients. Analytical results for the scattering cross sections are thus possible and are available for analysis. The d-type coefficients introduce additional complications relative to the c-type coefficients. In contrast to the c-type coefficients, the d-type coefficients modify the polarization properties of quarks. It turns out this makes them significantly harder to probe. For example, the presence of virtual gluons inside a hadron imply a

[3] By "leading" we mean those SME operators of most relevance to the considered process of dilepton production. For more information see Sec. A.1.

given quark does not have a well-defined spin. For a spin-zero meson, the net spin from valence quarks must average to zero. This means that even if the d-type effects are present they average to zero and are unobservable in spinless mesons like pions. It has also been emphasized [**ls18**] that unpolarized hadronic processes mediated by a photon or gluon are also insensitive to d-type coefficients, due to the opposite-sign contributions in the scattering amplitudes.

The only known ways to probe the d-type coefficients are with unpolarized beams exchanging a W or Z boson, or though polarized beamlines. Since no currently operating colliders fall into the latter category, this leaves the LHC as the sole experiment for probing the d-type coefficients.

With regard to the c-type coefficients, the bounds we expect to obtain [**ls18**, **klsv19**] will not be competitive with existing first-generation limits. However, we will also probe distinct linear coefficient combinations and the second-generation (s, c). The latter constraints will exceed the few existing ones by by a two or so orders of magnitude [**Karpikov:2016qvq**]. As suggested, the d-type coefficients are the most motivated and uniquely suitable for study using ATLAS data of the Drell-Yan process on the Z-boson pole.

2.6 Dilepton distribution

This section summarizes the cross section calculations relevant for our analysis and described in Refs. [**Kostelecky:2019fse**, **lssv21**].

The leading-order Drell-Yan dilepton distribution $d\sigma/dQ^2$ including photon, photon-Z interference, and pure Z exchange including the effects from Eq. (58) is [4][5]

$$\frac{d\sigma}{dQ^2} = \frac{4\pi\alpha^2}{3N_c}\sum_f\left[\frac{e_f^2}{2Q^4} + \frac{1-m_Z^2/Q^2}{(Q^2-m_Z^2)^2+m_Z^2\Gamma_Z^2}\frac{1-4\sin^2\theta_W}{4\sin^2\theta_W\cos^2\theta_W}e_f g_{fL}\right.$$
$$\left.+\frac{1}{(Q^2-m_Z^2)^2+m_Z^2\Gamma_Z^2}\frac{1+(1-4\sin^2\theta_W)^2}{32\sin^4\theta_W\cos^4\theta_W}g_{fL}^2\right]\int_\tau^1 dx\frac{\tau}{x}\widehat{\sigma}'_f\left(x,\tau/x,c_{fL}^{\mu\nu}\right) + (L\to R), \qquad (7)$$

where the number of colors $N_c = 3$, m_Z and Γ_Z are Z boson mass and width, $e_f = \{2/3, -1/3, -1/3, 2/3\}$ is the fractional quark charge for flavor $f = \{u, d, s, c\}$, and the left and right chiral couplings are $g_{fL} = I_W^{(3)} - e_f\sin^2\theta_W$, $g_{fR} = -e_f\sin^2\theta_W$. The Drell-Yan scaling variable is $\tau \equiv Q^2/s$ and

$$\widehat{\sigma}'_f\left(x,\tau/x,c_{fL}^{\mu\nu}\right) \equiv \left(1+\frac{2}{s}c_{fL}^{\mu\nu}(1+x^2/\tau)(p_{1\mu}p_{1\nu}+p_{1\mu}p_{2\nu}+(p_1\leftrightarrow p_2))\right)f_{fS}(x,\tau/x)$$
$$+\frac{2}{s}c_{fL}^{\mu\nu}\left(xp_{1\mu}p_{1\nu}+\frac{\tau}{x}p_{1\mu}p_{2\nu}+(p_1\leftrightarrow p_2)\right)f'_{fS}(x,\tau/x), \qquad (8)$$

where the flavor-symmetric PDF products are defined as

$$f_{fS}(x,\tau/x) \equiv f_f(x)f_{\bar{f}}(\tau/x) + f_f(\tau/x)f_{\bar{f}}(x), \qquad (9)$$
$$f'_{fS}(x,\tau/x) \equiv f_f(x)f'_{\bar{f}}(\tau/x) + f'_f(\tau/x)f_{\bar{f}}(x). \qquad (10)$$

[4] Recalling the assumption that the LIV effects are small relative to the SM contributions, we work at first-order in the SME coefficients, disregarding the highly suppressed quadratic (and all higher-order) corrections.

[5] Note this is Eq. (3.14) in Ref. [**lssv21**], except here the prefactor for the interference term now correctly reads $+(1-4\sin^2\theta_W)/4$, whereas the Ref. [**lssv21**] has the typos $-(1+4\sin^2\theta_W)/12$.

The first, second, and third terms in the brackets in Eq. (7) are associated with the electromagnetic, interference, and Z-exchange contributions, respectively. Note that $d\sigma/dQ^2$ is a Lorentz observer scalar. It can thus be evaluated in any observer frame. In whatever frame is chosen, the SME coefficients are those expressed in that frame. Application to ATLAS dilepton data means the ATLAS detector frame is the relevant observer frame. This frame will be discussed in detail in the following subsections.

2.7 Sidereal signals, the Sun-Centered frame, and the ATLAS detector frame

Fundamental physical laws imply equations of motion for particles and fields that are valid in inertial frames. However, experiments are typically described with Earth-based coordinate systems. These are nonintertial frame, primarily due to the Earth's rotation. Noninertial effects like the centrifugal force $F_\mathrm{c} = m\omega^2 r$ are usually several orders of magnitude smaller than the gravitational force $F_\mathrm{g} \simeq mg$. As F_g is considerably weaker than the electroweak and strong forces at the scale of collider experiments, noninertial effects are safe to neglect.

Noninertial effects cannot be disregarded in the presence of LIV. Because the Earth is rotating, the orientation of the laboratory with respect to the SME coefficients, which may be viewed as background tensor fields, changes with time. The dominant contribution to this time variation comes from the Earth's axial rotation. This will produce oscillations at harmonics of the Earth's sidereal frequency $\omega_\oplus = 2\pi/(23\text{ hr } 56\text{ min } 4.091\text{ sec})$ in laboratory observables. Furthermore, as no two experiments (in principle) have identical coordinates and measurement apparatus orientations, every experiment is sensitive to a unique linear combination of background effects.

For a straightforward comparison of results from distinct experiments, it is useful to introduce a fixed and nonrotating inertial frame. To a good approximation, the SME coefficients can be taken as fixed in this frame [**ak04**]. The commonly adopted frame for SME analyses is the Sun-centered Frame (SCF) [**kd16**, **km02**, **k2000**, **klv17**, **ls18**, **km09**], which we describe in detail.

Expressing laboratory observables in terms of coefficients defined in the SCF introduces explicit time dependence. The explicitly time-dependent laboratory coefficients are re-expressed in terms of the constant SCF coefficients at the cost of introducing sinusoidal functions of the sidereal phase.

The SCF spatial axes $X^J \equiv (X, Y, Z)$ are centered on the Sun such that the Z axis is parallel to the Earth's rotation axis, the X axis points toward the 2000 vernal equinox, and the Y axis completes a right-handed coordinate system (see Fig. 2). Analogously, the time coordinate of the SCF is denoted T. The spacetime coordinates of the SCF are then $X^\mu = (T, X^J)$. As the definitions would suggest, the time $T = 0$ is chosen to coincide with the date and time of the 2000 vernal equinox. We identify the 2000 vernal equinox as occuring on March 20th, 2000 at 7:35 am Universal Coordinated time (UTC).

In the following few subsections we introduce a number of frames and coordinate transformations that taken together will connect the SCF to the ATLAS detector frame. The net transformation will allow us to reexpress the time-dependent ATLAS frame SME coefficients in terms of the constant SCF coefficients. Next, the canonical frames used to describe the laboratory apparatus frame and its connection to the SCF are introduced.

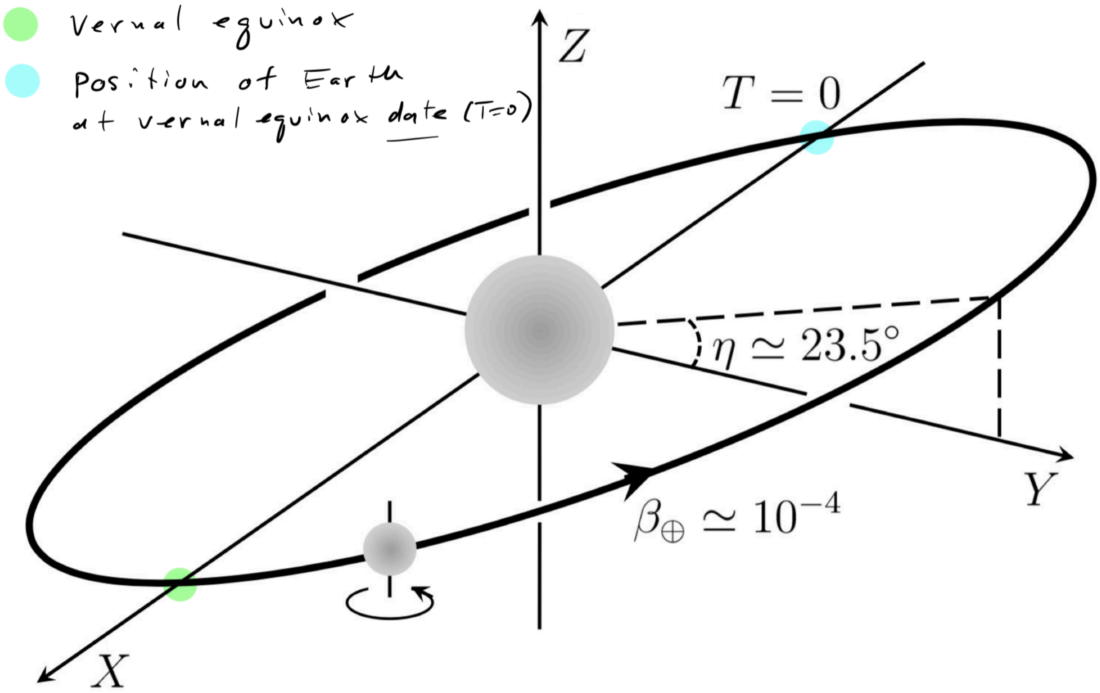

Figure 2: Edited illustration of the SCF (originally image from Ref. [**kd16**]). The Sun is the larger shaded disk in the center and and the Earth the smaller disk revolving about the Sun. When looking down the Sun Z axis, the Earth is seen revolving counterclockwise in an approximately circular orbit and rotating counterclockwise about its rotation axis. The SCF axes are chosen such that Z axis is parallel with the Earth's rotation axis. The orbit of the Earth in this frame is inclined relative to the X-Y plane by $\eta \simeq 23.5°$. The X-Y plane is parallel to the Earth's equatorial plane. The vernal equinox is the point where the X axis intersects the Earth's orbit (green dot). On the date of the 2000 vernal equinox the Earth is positioned at the light-blue dot near $T = 0$. At this latter dot, the X axis intersects the Earth and points from the Earth to the Sun. At this moment, there is one position on the Earth's equator where an observer would see the Sun directly overhead (toward their local zenith) at noon local time. The longitude of this location is given by λ_0 from Eq. (27).

2.8 Standard laboratory frame

The *standard laboratory frame* is a fixed frame on the surface of the Earth, and hence rotates with respect to the SCF. The coordinates of this frame are defined as $x^j \equiv (x, y, z)$, where the x axis points toward local south, the y axis toward local east, and the z axis toward the local zenith. It is convenient to identify the standard laboratory frame time coordinate t with the local sidereal time $T_\oplus$. The local sidereal time is defined to have an origin $T_\oplus = 0$ when the laboratory y axis is parallel to the Sun Y axis, $y||Y$.

Under the approximation that the Earth moves in a circular orbit around the Sun, the transformation from the SCF to the standard laboratory frame is given by a rotation R^{jJ} such that

$$x^j = R^{jJ} X^J, \tag{11}$$

where the sum over the repeated index J is implied. To determine R^{jJ}, we imagine that the coordinates x^j and X^J are initially parallel and with coincident origins. From the viewpoint of X^J, the coordinates x^j

appear to be rotating. So, first a rotation about the Z axis by an angle $\omega_\oplus T_\oplus$ is performed:

$$R_Z = \begin{pmatrix} \cos\omega_\oplus T_\oplus & \sin\omega_\oplus T_\oplus & 0 \\ -\sin\omega_\oplus T_\oplus & \cos\omega_\oplus T_\oplus & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{12}$$

In looking down the Z axis, the x and y axes will be rotating counterclockwise in the X-Y plane. This is consistent with the fact that the Earth rotates counterclockwise about its own axis when viewed from the northern hemisphere. Next, a second rotation is needed to make $\widehat{z} \cdot \widehat{Z} = \cos\chi$, where χ is the colatitude of the standard laboratory[6]. This is achieved by a counterclockwise rotation about the Y axis by an angle χ:

$$R_Y = \begin{pmatrix} \cos\chi & 0 & -\sin\chi \\ 0 & 1 & 0 \\ \sin\chi & 0 & \cos\chi \end{pmatrix}. \tag{13}$$

Thus, the net rotation is now $R^{jJ} = R_Y \cdot R_Z$ and given by

$$R^{jJ} = \begin{pmatrix} \cos\chi\cos\omega_\oplus T_\oplus & \cos\chi\sin\omega_\oplus T_\oplus & -\sin\chi \\ -\sin\omega_\oplus T_\oplus & \cos\omega_\oplus T_\oplus & 0 \\ \sin\chi\cos\omega_\oplus T_\oplus & \sin\chi\sin\omega_\oplus T_\oplus & \cos\chi \end{pmatrix}. \tag{14}$$

Note this is Eq. (44) in Ref. [**kd16**]. For more information on the standard laboratory frame, see Ref. [**km02**]. An illustration depicting the rotation (14) is given in Ref. [**k2000**].

Connecting other laboratory frames, such as the ATLAS detector frame, to the SCF can be achieved by applying additional rotations following $R^{jJ} = R_Y \cdot R_Z$ given in Eq. (14).

2.9 ATLAS detector frame

We now describe additional rotations that must be performed to connect the ATLAS frame to the SCF.

Recall that the standard laboratory frame (positive) y axis points to local east. At the ATLAS detector interaction point the collider ring has its beamlines oriented at some angle ψ north of east. The following matrix rotates the y axis to this direction:

$$\begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{15}$$

Note that y is now pointing along the ATLAS "beamline 1" direction. As is typically chosen, the beam-direction axis is in the $\widehat{z}$ direction. A transformation that swaps the y and z axes is

$$\begin{pmatrix} \pm 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & \mp 1 & 0 \end{pmatrix}. \tag{16}$$

The upper (lower) signs orient $\widehat{z}$ antiparallel (parallel) to the prior $\widehat{y}$ direction. For either sign the final $\widehat{y}$ direction points toward the local zenith. Applying Eqs. (15),(16) after Eq. (14) results in the rotation

[6] the hat notation indicates unit vectors

used in studies involving the HERA/EIC colliders in [**klv17**, **ls18**]. For ATLAS, one final transformation needed because the ATLAS detector is slightly inclined relative to the $\hat{z}$ direction. Performing a rotation of $-\delta$ around the $\hat{x}$ direction achieves this[7]:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(-\delta) & \sin(-\delta) \\ 0 & -\sin(-\delta) & \cos(-\delta) \end{pmatrix}. \tag{17}$$

In total, the net rotation converting the SCF coordinates to the ATLAS coordinates is found to be

$$\begin{aligned} R(T_\oplus) = &\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\delta & -\sin\delta \\ 0 & \sin\delta & \cos\delta \end{pmatrix} \begin{pmatrix} +1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &\times \begin{pmatrix} \cos\chi\cos\omega_\oplus T_\oplus & \cos\chi\sin\omega_\oplus T_\oplus & -\sin\chi \\ -\sin\omega_\oplus T_\oplus & \cos\omega_\oplus T_\oplus & 0 \\ \sin\chi\cos\omega_\oplus T_\oplus & \sin\chi\sin\omega_\oplus T_\oplus & \cos\chi \end{pmatrix}. \end{aligned} \tag{18}$$

The chosen sign in the $y \leftrightarrow z$ swap matrix (Eq. (16)) identifies the beam $\hat{z}$ direction with the ATLAS "beam 2 direction"—see the ATLAS New Small Wheel TDR, Section 3.6.1, pg. 24. The numerical values of the relevant angles are

$$\chi = 43.764290859^\circ, \tag{19}$$
$$\psi = 168.7^\circ, \tag{20}$$
$$\delta = |-\delta| = |-0.704^\circ| = +0.704^\circ. \tag{21}$$

Note the minus sign in δ is taken into account in Eq. (18). For an expanded discussion of the SCF, standard laboratory frame, and laboratory apparatus frame, see Ref. [**km09**].

2.9.1 Explicit example

As a simple example, consider again our archetypal LIV effect: $\mathcal{L}_b = -b_\mu \bar{\psi}\gamma_5\gamma^\mu\psi$ from Eq. (2). Let us imagine we are working in the ATLAS detector frame. Then $b_\mu = (b_0, b_j)$ has components decomposed in the coordinate basis describing this frame. Noting Eq. (18), we wish to express b_μ in terms of the fixed SCF frame coefficients, which we call $b^\mu_{\text{SCF}} = (b^T, b^X, b^Y, b^Z)$. Since the ATLAS detector and SCF differ by a coordinate transformation (Lorentz observer transformation), the coeffcient b_μ transforms like a usual four-vector:

$$b^\mu = [R(T_\oplus)]^\mu{}_\nu b^\nu_{\text{SCF}}, \tag{22}$$

where $[R(T_\oplus)]^\mu{}_\nu$ is the four-dimensional version of the spatial rotation matrix $R(T_\oplus)$ from Eq. (18). The additional elements are $R^0{}_0 = 1, R^0{}_j = R^j{}_0 = 0$. As a simplifying limit, and without loss of generality, consider $\delta \longrightarrow 0$. It is then found that

$$\begin{pmatrix} b^0 \\ b^1 \\ b^2 \\ b^3 \end{pmatrix} = \begin{pmatrix} b^T \\ -b^Z\sin\chi\cos\psi + \cos(\omega_\oplus T_\oplus)(b^X\cos\chi\cos\psi + b^Y\sin\psi) + \sin(\omega_\oplus T_\oplus)(b^Y\cos\chi\cos\psi - b^X\sin\psi) \\ b^Z\cos\chi + b_X\cos(\omega_\oplus T_\oplus)\sin\chi + b_Y\sin(\omega_\oplus T_\oplus)\sin\chi \\ -b^Z\sin\chi\sin\psi + \cos(\omega_\oplus T_\oplus)(b^X\cos\chi\sin\psi - b^Y\cos\psi) + \sin(\omega_\oplus T_\oplus)(b^Y\cos\chi\sin\psi + b^X\cos\psi) \end{pmatrix}. \tag{23}$$

[7] Our conventions for rotations are that a positive angle is associated with a counterclockwise rotation. Here we have a clockwise rotation around the x axis, hence the minus sign.

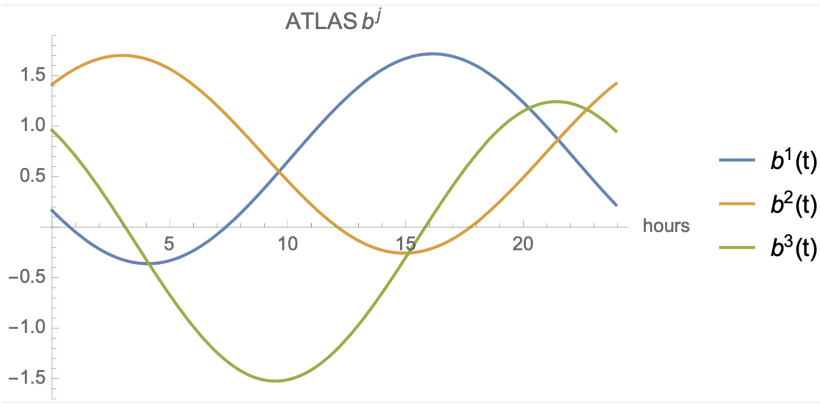

Figure 3: Components b^j from Eq. (23) plotted over a sidereal day for $b^J_{SCF} \equiv 1$. The (small) effects of setting $\delta = |-0.704°|$ are also included.

We now observe that the time dependence has been made explicit through the functions $\cos(\omega_\oplus T_\oplus), \sin(\omega_\oplus T_\oplus)$. See Fig. 3 for the resulting time dependence of the ATLAS components. Note that all components have a distinct shape and that even in the absence of time-dependent effects there is a unique dependence on the laboratory location through the angles χ, ψ.

For more complicated LIV backgrounds, like c- and d-type coefficients introduced in Sec. 2.4, additional applications of the net rotation matrix must be performed. For example,

$$(c_{u_i})^{\mu\nu} = [R(T_\oplus)]^\mu{}_\alpha \, [R(T_\oplus)]^\nu{}_\beta \, (c_{u_i})^{\alpha\beta}_{\text{SCF}}, \tag{24}$$

where the parentheses () have been introduced for clarity.

2.9.2 Another explicit example

We now revisit the cross section $d\sigma/dQ^2$ from Eq. (7). It involves combinations of the coefficients described in Eq. (6). Evaluation of $d\sigma/dQ^2$ in the ATLAS detector frame reveals that the only time-dependent coefficients that appear are c_f^{33}, d_f^{33}. We must re-express these coefficients in terms of the SCF coefficients using, e.g., Eq. (24). We find

$$\begin{aligned}
c_f^{33} =& \frac{1}{2}(c_f^{XX} + c_f^{YY}) \left(\cos^2\chi \sin^2\psi + \cos^2\psi\right) + c_f^{ZZ} \sin^2\chi \sin^2\psi \\
& - 2c_f^{XZ} \sin\chi \sin\psi \, [\cos\chi \sin\psi \cos(\omega_\oplus T_\oplus) + \cos\psi \sin(\omega_\oplus T_\oplus)] \\
& - 2c_f^{YZ} \sin\chi \sin\psi \, [\cos\chi \sin\psi \sin(\omega_\oplus T_\oplus) - \cos\psi \cos(\omega_\oplus T_\oplus)] \\
& + c_f^{XY} \left[(\cos^2\chi \sin^2\psi - \cos^2\psi) \sin(2\omega_\oplus T_\oplus) - \cos\chi \sin(2\psi) \cos(2\Omega_\oplus T_\oplus)\right] \\
& + \frac{1}{2}(c_f^{XX} - c_f^{YY}) \left[(\cos^2\chi \sin^2\psi - \cos^2\psi) \cos(2\omega_\oplus T_\oplus)\right. \\
& \left. + \cos\chi \sin(2\psi) \sin(2\omega_\oplus T_\oplus)\right],
\end{aligned} \tag{25}$$

and similarly for d_f^{33}. We see that the coefficients with SCF superscript indices $XX + YY$ and ZZ are only associated to time-independent effects. Those with indices $XZ, YZ, XY, XX - YY$ introduce sidereal time-dependent effects and are of primary interest. Inserting Eq. (25) into the laboratory-frame expression of Eq. (7) and noting the constraint

$$c_{u_i}^{\mu\nu} - c_{d_i}^{\mu\nu} = d_{d_i}^{\mu\nu} - d_{u_i}^{\mu\nu}, \tag{26}$$

results in the observable of interest for our analysis.

2.10 Initial conditions and shifts

The previous sections focused on the spatial transformation from the SCF to the ATLAS detector frame and the implications for observables. In this section, we describe an important temporal transformation that must be accounted for.

We begin by asking:

> *Where on the equator at the vernal equinox 2000 ($T = 0$) is the Sun directly overhead (at noon local time)?*

At this location, the standard laboratory frame z axis is oriented such that $\widehat{z} \parallel \widehat{X}$, $\widehat{y} \parallel \widehat{Y}$, and $T = T_{\oplus} = 0$, assuming the origin of $T_{\oplus}$ is chosen to occur on the equinox date. This location is found by noting there are $(4 + \frac{25}{60} \simeq 4.417)$ local hours that must elapse for the Sun to be approximately overhead at noon in the UTC timezone at longitude 0° (recall that the equinox occurs at 7:35 am UTC). Therefore, the Sun is overhead on the equator at a longitude *advanced* by

$$\begin{aligned} \lambda_0 &= 4.417 \text{ hr} \cdot \frac{2\pi}{24\text{hr}} \cdot \frac{180}{\pi} \\ &= 66.25°, \end{aligned} \tag{27}$$

which is in the Indian Ocean west of the Maldives, matching the claim in Ref. [**kd16**]. Note it is only at λ_0 that a (hypothetical) laboratory with $T = T_{\oplus}$ has its local zenith pointing toward the Sun.

For a laboratory positioned at any other longitude, $\widehat{y} \parallel \widehat{Y}$ at an earlier or later time than the equinox. Given the direction of Earth's rotation, for $\lambda < \lambda_0$, $T - T_{\oplus} > 0$, and for $\lambda > \lambda_0$, $T - T_{\oplus} < 0$. The time it takes for such laboratory to advance depends on a linear longitudinal shift, and the time for $\widehat{y}$ to continually realign with $\widehat{Y}$ is a sidereal day with period 23 hr 56 min 4.091 s $\simeq$ 23.934 hr. This latter dependence on the sidereal day, as opposed to a 24-hour solar day, is because the Earth is slightly changing its angular position relative to the Sun by an amount quantified by the sidereal day. We find

$$T - T_{\oplus} = \frac{\lambda_0 - \lambda}{360°} \cdot (23.93446972\bar{2} \text{ hr}), \tag{28}$$

which is Eq. (43) in Ref. [**kd16**] reported with higher precision. Note that this result is independent of the latitude of the laboratory. Also note that $T - T_{\oplus}$ can more generally depend on a integer number n of sidereal days, such that an additional shift $\frac{2\pi}{\omega_{\oplus}} n$ needs to be included on the right-hand side of Eq. (28) if the origin $T_{\oplus} = 0$ *isn't* chosen on the 2000 vernal equinox date. As a relevant example, for ATLAS

$$\lambda = 6.055279086°, \tag{29}$$

giving

$$T - T_{\oplus}\Big|_{\text{ATLAS}} = 4.002 \text{ hr}. \tag{30}$$

event	calendar date	T_{Unix}	T	$T_\oplus$	$\oplus\angle$	$\odot\angle$
1	01/01/1970 00:00:00	0	-953537717	-953552124	0.371415	0.517083
2	01/01/2000 12:00:00	946728000	-6809717	-6824124	0.35914	0.0170833
3	20/03/2000 07:35:17	953537717	0	-14407	0.857198	0.833252
4	20/03/2000 11:35:24	953552124	14407	0	0	0
5	25/01/2021 18:49:42	1611600582	658062865	658048458	0.590071	0.301597
6	25/03/2021 18:49:42	1616698182	663160465	663146058	0.117588	0.301597

Table 1: Times other than the calendar date (GMT) are given in seconds. Sidereal ($\oplus\angle$) and solar ($\odot\angle$) phases are calculated with respect to $T_\oplus$ for ATLAS.

Note that 2×10^{-3} hr = 7.28725 s. Thus, rounding to the nearest second, for the ATLAS experiment $\widehat{y} \parallel \widehat{Y}$ on March 20th, 2000 at 11:35:24 UTC. It is this date and time that we identify with $T_\oplus|_{\text{ATLAS}} = 0$. All data timestamps are assigned a unique $T_\oplus$ which is measured relative to this initial condition.

Since we have chosen to initialize our laboratory clock with the local sidereal time, $t = T_\oplus$, every laboratory measurement comes with a time stamp $T_\oplus$. However, suppose that events are instead known with respect to a different time origin, for instance "Unix time", where $T_{\text{Unix}} = 0$ is defined as January 1st, 0:00:00 UTC, 1970. To convert from Unix time to local time, we must subtract off the difference

$$(T_\oplus = 0) - (T_{\text{Unix}=0}) = 953552124 \text{ s} \tag{31}$$

from each event to compute how much time has elapsed since $T_\oplus = 0$. In Tab. 1 we compare the various dates and times discussed so far, as well as the sidereal and solar phases with reference to $T_\oplus$ for ATLAS. Alternatively, the bin for a given event $T_\oplus$ may be calculated from

$$\text{bin \#} = \text{mod}\left[\frac{T_\oplus - \text{mod}[T_\oplus, \frac{2\pi}{n\omega_\oplus}]}{\frac{2\pi}{n\omega_\oplus}}, n\right] + 1, \tag{32}$$

where n is the number of bins (intervals) the sidereal day has been split into. For example, for $n = 1$ the length of the interval is the sidereal day, which is 23.94 hr. For $n = 2$, there are two intervals of duration 23.94/2 = 11.97 hr, and so on.

2.11 Minimal SME coefficients probed in this analysis

As shown in Sec. 2.9.2, for a given flavor f there are four coefficient types $XZ, YZ, XY, XX - YY$ that induce sidereal time dependence in the Drell-Yan dilepton distribution $d\sigma/dQ^2$. In the basis of c- and d-coefficients (Eq. (6)) there are three independent coefficient types per generation. This gives $3 \times 4 = 12$ coefficients for (u, d) quarks and 12 more for (s, c) quarks. Note that we are not sensitive to the third generation (b, t). Together this gives 24 SME coefficients. For the first generation, the functional dependence attached to each coefficient is given by

$$d_u^{XZ} \quad : \quad 6.28069\cos(\omega_\oplus T_\oplus) - 41.0569\sin(\omega_\oplus T_\oplus) \tag{33}$$

$$d_u^{YZ} \quad : \quad 41.0569\cos(\omega_\oplus T_\oplus) + 6.28069\sin(\omega_\oplus T_\oplus) \tag{34}$$

$$d_u^{XX-YY} \quad : \quad 77.6067\cos(2\omega_\oplus T_\oplus) + 24.3128\sin(2\omega_\oplus T_\oplus) \tag{35}$$

$$d_u^{XY} \quad : \quad -48.6256\cos(2\omega_\oplus T_\oplus) + 155.213\sin(2\omega_\oplus T_\oplus) \tag{36}$$

$$c_u^{XZ} \quad : \quad 8.084\cos(\omega_\oplus T_\oplus) - 52.8451\sin(\omega_\oplus T_\oplus) \tag{37}$$

$$c_u^{YZ} \quad : \quad 52.8451\cos(\omega_\oplus T_\oplus) + 8.084\sin(\omega_\oplus T_\oplus) \tag{38}$$

$$c_u^{XX-YY} \quad : \quad 99.8891\cos(2\omega_\oplus T_\oplus) + 31.2935\sin(2\omega_\oplus T_\oplus) \tag{39}$$

$$c_u^{XY} \quad : \quad -62.5869\cos(2\omega_\oplus T_\oplus) + 199.778\sin(2\omega_\oplus T_\oplus) \tag{40}$$

$$c_d^{XZ} \quad : \quad 0.181551\cos(\omega_\oplus T_\oplus) - 1.1868\sin(\omega_\oplus T_\oplus) \tag{41}$$

$$c_d^{YZ} \quad : \quad 1.1868\cos(\omega_\oplus T_\oplus) + 0.181551\sin(\omega_\oplus T_\oplus) \tag{42}$$

$$c_d^{XX-YY} \quad : \quad 2.24331\cos(2\omega_\oplus T_\oplus) + 0.702788\sin(2\omega_\oplus T_\oplus) \tag{43}$$

$$c_d^{XY} \quad : \quad -1.40558\cos(2\omega_\oplus T_\oplus) + 4.48662\sin(2\omega_\oplus T_\oplus) \tag{44}$$

and similarly for $(u, d) \leftrightarrow (s, c)$. Figure 4 shows the signature of these coefficients as a function of sidereal phase $\omega_\oplus T_\oplus$. Note that each coefficient type is strictly associated with either first or second harmonics of the sidereal phase $\omega_\oplus T_\oplus$.

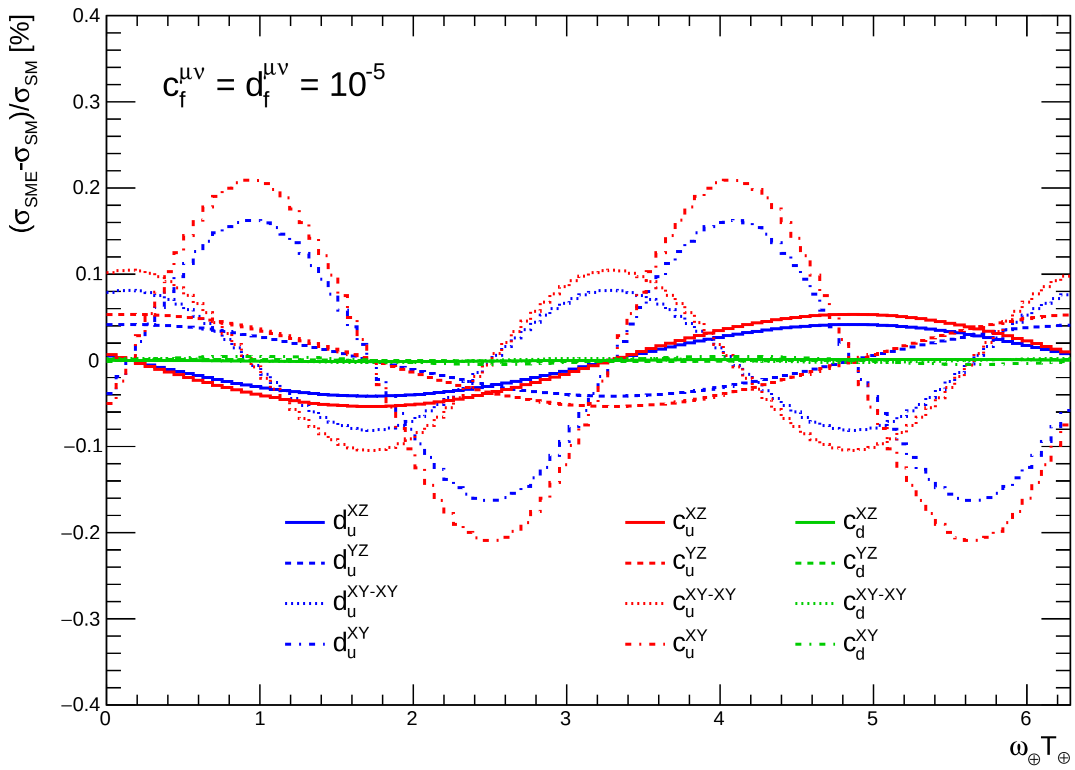

Figure 4: Functional forms of the time dependence associated with each SME coefficient, as expressed in Eqs. 33-44.

	HERA	LHC
$\lvert a^{(5)TXX}_{\mathrm{S}u} - a^{(5)TYY}_{\mathrm{S}u}\rvert$	4.3×10^{-7}	1.5×10^{-8}
$\lvert a^{(5)TXY}_{\mathrm{S}u}\rvert$	6.5×10^{-7}	2.7×10^{-9}
$\lvert a^{(5)TXZ}_{\mathrm{S}u}\rvert$	1.1×10^{-6}	7.2×10^{-9}
$\lvert a^{(5)TYZ}_{\mathrm{S}u}\rvert$	7.4×10^{-7}	7.0×10^{-9}

Table 2: Comparison of HERA and LHC sensitivity to $a^{(5)}$-type coefficients. The limits from HERA come from a recent ZEUS analysis [**ZEUS:2022msi**] and coincide with the positive upper limit report in units of GeV^{-1}. The quoted values for LHC come from simulations assuming uncorrelated systematic uncertainties. Table adapted from Refs. [**Kostelecky:2019fse**],[**ZEUS:2022msi**].

2.12 Nonminimal quark-sector effects

Thus far we have focused on the minimal SME in the quark sector. From an effective field theory viewpoint, nonrenormalizable operators of mass dimension $d > 4$ parametrize the effects of new physics suppressed by the associated characteristic energy scale. The more energetic a given process the more relevant nonrenormalizable operators become. For example, a $d = 5$ operator has a SME coefficient with dimension GeV^{-1}. The dominant (tree-level) insertions of this operator introduces an expansion parameter $\delta = (E/\Lambda)$, where E is the process scale (e.g. the energy of the initial-state partons in the Drell-Yan procees), and Λ the scale of new physics (e.g the Planck scale). The larger E is, the more relevant the expansion parameter. A $d = 6$ operator would include a leading-order correction controlled by δ^2. Though clearly suppressed relative to the $d = 5$ operator, since Λ is fixed the influence of the $d = 6$ operator grows more rapidly. Nonrenormalizable operators are extensively studied in Lorentz-invariant new-physics searches at the LHC because they generically parametrize new physics too heavy to be produced on-shell by the colliding beamlines.

Nonrenormalizable effects from the nonminimal SME have also been investigated in Drell-Yan dilepton production [**Kostelecky:2019fse**]. The $d = 5$ Lagrange density was considered:

$$\mathcal{L}_{a^{(5)}} = -\frac{1}{2}\sum_f a_f^{(5)\mu\alpha\beta}\bar{\psi}_f\gamma_\mu iD_{(\alpha}iD_{\beta)}\psi_f + \text{h.c.}, \tag{45}$$

where the parenthesis surrounding α, β denotes index symmetrization and h.c. is shorthand for Hermitian conjugation. We will refer to the SME coefficients that appear as the $a^{(5)}$-type coefficients. From Table 2 it is observed that the LHC could expect up to two orders of magnitude more sensitivity to these coefficients than what was obtained by the ZEUS Collaboration [**ZEUS:2022msi**]. This increased sensitivity has its origin in the increased collision energy of the LHC relative to HERA.

The predictions of Table 2 are limited to photon exchange. When electroweak bosons are exchanged, the $a^{(5)}$-type coefficients cannot be treated in isolation. The reasons are exactly the same as for why the c-type coefficients can not be treated independently of the d-type coefficients when a $W^\pm$ or Z is exchanged, as discussed in Sec. 2.5. Analogous to the d-type coefficients, the other set of coefficients relevant for the nonminimal effects probed here are called the $b^{(5)}$-type coefficients.

Predictions for Drell-Yan on the Z pole and including the $b^{(5)}$-type coefficients are currently being calculated by the theory STA members [**LS24**]. Because the $a^{(5)}$- and $b^{(5)}$-type coefficients are analogous in structure to the c- and d-type coefficients, we expect a similar enhancement of nonminimal LIV signatures on the Z

pole. Therefore, it is reasonable to expect a greater relative sensitivity to these coefficients than estimated in Table 2.

Since the $a^{(5)}$- and $b^{(5)}$-type coefficients have an odd number of Lorentz indices, they also produce striking CPT-violating effects. For example, in deep inelastic scattering, when summing the antiquark contributions these coefficients appear with opposite signs to the quark contributions. Thus, for the sea quarks in the proton the effect cancels out. The ZEUS analysis was therefore only sensitive to the u and d quark $a^{(5)}$ coefficients. In contrast, the leading Drell-Yan diagram involves $q\bar{q}$ fusion. The appearance of particle and antiparticle in this diagram does not produce the same cancellation effect as in deep inelastic scattering. Therefore, the Drell-Yan process can probe the CPT properties of quarks. Our analysis will be the first probe of the CPT properties of sea quarks.

3 Methodology

3.1 Use of the Z-counting luminosity measurement

The results to be used for this analysis are those presented in the Z-counting luminosity measurement [**ATL-DAPR-PUB-2021-001**][8]. This Section contains a brief introduction to the methodology used for a time-dependent measurement of the Z-boson production at ATLAS as presented in the Z-counting PUB note. Appendix **??**contains a full overview of the methodology, transferred to a more commonly used ATLAS analysis framework.

The methodology was first established as an alternative measurement of the luminosity recorded by the ATLAS experiment, where the Z-boson decay into dileptons during physics runningis used to provide additional checks against residual time, pile-up, or beam-condition-dependent biases that may affect the baseline ATLAS luminosity values, the determination of which is detailed in Ref [**ATLAS-CONF-2019-021**].

The Drell-Yan production of dilepton pairs is a well understood process, with the clean Z-boson signature of two high-transverse momentum, oppositely charged leptons[9] providing robustness against varying pileup conditions. Thus, a luminosity measurement can be obtained by measuring the number of produced Z-bosons, N_Z, using the formula

$$\int \mathcal{L}_Z dt = \frac{N_Z^{corr}}{\sigma_Z} \tag{46}$$

where σ_Z is the Z production cross section. Experimentally, precise measurements of $\sigma_{Z\to e^+e^-}$ ($\sigma_{Z\to\mu^+\mu^-}$) have been obtained with systematic uncertainties (excluding luminosity) of about 1.0% (1.5%) [**ATLAS:2019zci**]. The value of the $Z \to \ell^+\ell^-$ cross-section is predicted by calculations at next-to-next-to-leading order QCD with a total uncertainty of 3-4% at 90% CL, where the uncertainty is driven by the current knowledge of the proton parton distribution functions (PDFs). As the uncertainties in the baseline ATLAS luminosity are below 2% [**ATLAS-CONF-2019-021**], the Z-counting luminosity is not competitive as an absolute measurement, but has many powerful features as a relative luminometer.

The Z-counting procedure is performed as a function of time, in units of ”luminosity blocks” (LBs), the smallest time interval for which luminosities are measured in ATLAS. These typically correspond to 60 seconds intervals, but occasionally account for a shorter amount of time depending on trigger or data acquisition configurations.

For each lepton flavour the raw rate of detected Zs is measured on a LB-by-LB basis. To obtain the Z production rate, the raw rate is corrected to account for the trigger and reconstruction efficiencies. These data-driven efficiencies are subject to additional corrections, derived as a function of the average number of simultaneous collisions, $\langle\mu\rangle$, from Monte Carlo simulations. A small background subtraction is also applied in order to account for dilepton pairs in the event selection phase space originating from diboson and $t\bar{t}$ events.

N_Z^{corr} is obtained after correcting following effects which described in the public note[**ATL-DAPR-PUB-2021-001**]:

[8] The results originally reported were based on the full Run-2 results obtained using the Tag10 ATLAS official luminosity measurement. The search presented in this Note will use the updated results using the luminosity Tag11.

[9] Z-boson decays to τ leptons or quarks are not considered here, as they are much more difficult to reconstruct and prone to very high background levels.

- The fraction of background in the Z peak region, which is 0.5% in both the electron and muon channel.
- $Z \rightarrow \ell^{+}\ell^{-}$ cross-section of 1970 pb.
- Acceptance of the fiducial phase space.
- Trigger efficiency and reconstruction efficiency measured for each LB.
- Efficiency correction factor to take into account the average pileup, $< \mu >$, dependence.
- duration of each LB.

Each correction is carefully assessed in a per-LB (or per-pileup) basis, obtaining an independent measurement of the instantaneous luminosity recorded by ATLAS, with which we can monitor the time and pileup stability of the nominal luminosity measurement or, in the case of this analysis, search for Lorentz-invariance violation in the production of the Z-boson cross-section.

The methodology was originally developed to run on the DataQuality ATLAS analysis framework, using the WZFinder tool. This framework, with which the results this analysis will be based on were obtained, runs on Tier-0 ATLAS data.

3.2 Sidereal phase in this analysis

In the actual analysis, we use a double ratio between the N_Z^{corr} and the integrated luminosity ($\mathcal{L}i$).

$$R_D = \frac{N_{Z_i}^{corr}/\Sigma N_{Z_i}^{corr}}{\int \mathcal{L}_i dt/\Sigma \int \mathcal{L}_i dt}, \tag{47}$$

where the index i indicates bin number as function of the sidereal phase. By normalizing with total summation of each value, systematic uncertainties related to normalization are fully canceled, also most of uncertainties on object efficiencies and all theory related uncertainties are canceled. However, any time-dependent effects may remain and need to be evaluated, e.g. arising from the time-dependence of the baseline ATLAS luminosity measurement. These effects are discussed in Section 6.

Some details for these input variables described below.

- **Integrated luminosity for each LB, $\int \mathcal{L}_i dt$**, provided by the ATLAS luminosity group. The measured luminosity of Lucid is used in the analysis. Need to have selection and merging scheme for "bad" LBs and LBs which have shorter duration. As a cross check, the measured luminosity based on the track counting method is used.
- **the corrected number of Z bosons**, $N_{Z_i}^{corr}$, for each LB of i. The luminosity group provides three values with error for each LB, the electron channel, the muon channel and the combined value. We study each channel. The final result will be extracted with the combined value. The error of each measurements includes statistical uncertainty and the systematic uncertainty of the corrections.

year	number of runs	number of LBs	Integrated luminosity (fb^{-1})
2015	66	22,850	3.10
2016	150	91,111	33.28
2017	186	86,471	44.61
2018	194	92,711	58.70
Total	596	293,143	139.69

Table 3: Summary of input data for each year

- **Sidereal bin assignment**, described in Eq.32 in Section 2.9.2. The start and end time stamps of each LB is available in the unix time. We use them and compute a time stamp of middle of each lumi block, and convert it in the sidereial time at ATLAS detector in sun centered frame. The number of bin can be optimized according to data statistics. We set default number of bins as 100 according to a Toy study described in Section 4.3.2.

Summary table of the input LBs for each year is shown in Table 3. Here, need to describe selection of LBs. Remove low duration, zero for one of output, etc..

Once the binned histogram on the double ratio is obtained, we fit it with expected formula discussed in Section A. Also, we are planning to fit with genelic functional form.

3.3 Samples and steps to unblind the LIV search

This analysis is the first search in ATLAS to probe the sidereal effect, we do not look final result until ATLAS exotic group approve the unblinding. The blind the result is established by NOT perform the final fit with probing period with sidereal period (23 hour 56 minutes 04 seconds) with real data. This is checked by TOY study described in Section 6.1. We found that looking at data with 24 hours to verify day-night effect would unblind the data effectively.

Because any physics process could have LIV effect, we would not rely on any specific kinetics region as a "control region". Instead, we check the data behavior with different periods to compute the phase. We have verified that if we use 1 hour as a period to compute the phase, instead of the sidereal period $T_{\text{sid.}}$, any LIV signal can be wiped out. With this approach, we can verify the final statistical analysis procedure and statistical uncertainty. However, it turned out that 1 hour period is too short to reveal the potential issue, i.e. missing measurements on particular time period (any experimental effect or analysis issue could be wiped out as well). Therefore, we propose following steps to unblind:

1. check any bias with data driven TOY sample (randomly generated number of Z with data luminosity measurement)
2. analyze the real data set with scrambling time stamp
3. probe R_D with probing period of 1 hour
4. probe R_D with probing period with 7 hours and 24 hours step by step.

After confirming no bias within statistics and systematics uncertainties, unblind the R_D with the sidereal phase.

Step-1 and -2 are described in Section 6.1 and 5. Exotic group already approved to proceed Step-2 in a LPX meeting in September 2023. Once all statistics tool and systematic studies are done, we would ask to move Step-3 and later.

At the end, we do not use normal ATLAS MC simulation samples. This is because nominal systematic effects from both theoritical and experimental sources are canceled out in the double ratio. Only time dependent systematics are affected in the search. This is discussed in Section 6.

3.4 Statistics analysis on the fitting as function of the sidereal phase

A statistics treament to extract the LIV search result is summarized here. Basically we do a binned fit as function of the probing phase with chisquare minimization. In this fit, we include extra terms to take into account systematic uncertainty.

The chi-squared (χ^2) fit method is used to determine a signal strengh from a givent dataset (a double ratio). This method is commonly employed in various scientific disciplines to assess how well a model describes the observed data. The formula for the chi-squared statistic is:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{\sigma_i^2}, \tag{48}$$

where

- i is the bin index, and it's range is [0-99] becuase we use 100 bins.
- O_i is the observed value for data point i.
- E_i is the expected value for data point i.
- σ_i^2 is the uncertainty (standard deviation) associated with the observed value.

Datasets and theory models:

- Dataset is a double ratio obserable binned in a probing perioud. Total number of bin is 100 (see Section. 6.1). The dataset is eigher toy, scrambled data or real data when this analysis is unblinded.
- Thery models are analytic fuctions predicted by SME theory described in Section 2. Each model is fitted to a dataset independent to other models. and Each model has a free parameter, μ, which is the signal strength.
- Degree of freedom: 100 bins minus number of free parameters which one in our case. So the DoF is 99.

3.4.1 χ^2 fit

For a given theory prediction f with a parameter μ and dataset $\{y_i\}$, a χ^2 estimator is constructed as:

$$\chi^2(\mu) = \sum_{i}^{100} \left(\frac{y_i - f_i(\mu)}{\sigma_i} \right)^2 . \tag{49}$$

Here, the $\{\sigma_i\}$ are the uncertainties carried with the dataset (e.g statistical uncertainties of the dataset). In this analysis, the double ratio observable, R, contains 100 data points within the 0 to 2π range (sidereal phase). Therefore, the index i runs from 1 to 100.

3.4.2 Profiled χ^2 fit

The χ^2 estimator described in the EQ 3.4.1 does not consider a systematic effect. Thus, this section introduces a $\chi 2^2$ estimator with an additional nuisance parameter.

Assume this analysis has a systematic uncertainty **S**. A sigma variation of this systematic will the shifts (absolute shifts with sign) a data point by s_i in ith bin; the data point y_i increase or decrease by s_i. In this case, the dataset and theory predictions are $\{y_i\}$, $f_i(\mu)$, and input uncertainties are $\{\sigma_i\}$ and $\{s_i\}$.

The formula 3.4.1 does not take into account the s_i. We can add one nuisance parameter, β, to the theoretical model to carry a systematic variation. The modified theory will be:

$$F_i(\beta, \mu) = f_i(\mu) + \beta s_i . \tag{50}$$

Here, $\{s_i\}$ are constant (they have plus and minus sign). The β is a standard normal Gaussian distributed variable centered at ZERO and varies within [-1,1] range, which corresponds to a plus-minus one sigma variation of the systematic **S**. In this case, the original χ^2 estimator become a two-parameter estimator (profiled χ^2 estimator):

$$\chi^2(\beta, \mu) = \sum_{i}^{100} \left(\frac{y_i - F_i(\beta, \mu)}{\sigma_i} \right)^2 + \beta^2 . \tag{51}$$

3.4.3 Test with toy systematics

The toy data represents the signal-free data corresponding to full ATLAS Run2 luminosity. Plots in Figure 5 show the Double Ratio (R) of this signal-free dataset. The black color shows the nominal data points, and the gray color shows the shifted data due to a systematic variation (e.g., plus one sigma). The blue line in the right plot is a signal shape and strength used to obtain a shift in data.

To test the profiled χ^2 estimator, three cases for a set of systematic uncertainty are considered:

1. $\beta = 0$. This is a default case where no systematic variations are included. This is for χ^2 fit to the data points with black color in Figure 5, without profiling.

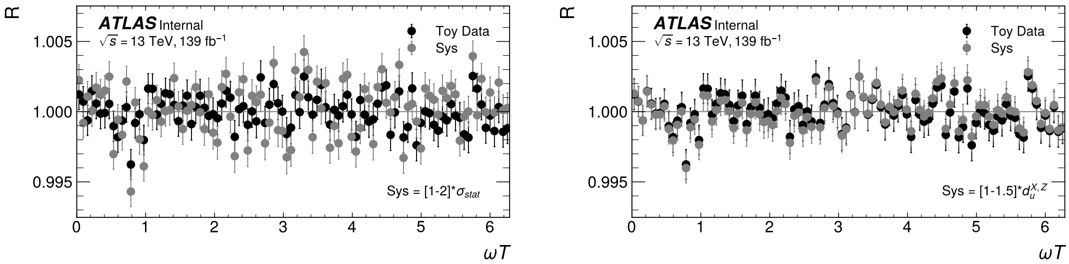

Figure 5: The black color shows nominal data points, and the gray color shows shifted data points due to a systematic variation. The randomized systematic impacts top left (case 2). The systematic effect introduces a shape to data, top right (case 3)

2. $s_i = RandomChoise([-1, 1]) \times RandomNumber([1, 2]) \times \sigma_i$. Both the sign and size of s_i are randomly chosen. Because we assume that the systematic impacts on data points are not correlated.
3. $s_i = RandomNumber([1, 1.5]) \times {d_u^{X,Z}}_i$. Add small signals to mimic shifts in data due to systematics. The size of s_i is randomized between 1-1.5 times the signal in ith point.

Case 1 is one parameter χ^2 fit since $\beta = 0$. To perform the profiled χ^2 fit, $scipy$ python package is used (it is easier to implement in Python rather than in RooFit). The original χ^2 fit in equation 3.4.1 is used to validate $scipy$ and $RooFit$ packages. Results from these two packages agreed very well up to high precision.

In case 2, the size and sign of s_i are randomly chosen so that they don't introduce some shapes (for example, cos- or sin-like oscillation). The s_i are the difference between black and gray points in the left plot in Figure 5.

In case 3, the size and direction of shifts (sign) $\{s_i\}$ are introduced by a signal shape. The $d_u^{X,Z}$ is considered the shape, and the signal strength is set to $\mu = 1E^{-5}$. The $\{s_i\}$ are the difference between black and gray points in the right plot in Figure 5. As we can see in the plot, the shifts in data will introduce a signal-like shape to the dataset.

A profiled χ^2 (Equition 3.4.2) is fitted to Cases 2 and 3. The fit results are shown in Figure 6. The top left plot shows the toy data and fitted model (blue line). Since data is signal free and the systematic impact does not introduce any shapes to data, the fitted signal amplitude (or strength) is small. The bottom left plot shows the corresponding one-sigma contour. The input size for β is [-1, 1] and it is constrained after the fit. The top right plot Figure 6 shows the toy data and fitted model with profiled shape systematics (this systematic introduces signal like shape). From the corresponding contour plot (bottom right), one can see the impact of systematic uncertainty on the parameter estimation.

Fitted signal strength and its uncertainty from three cases are compared in Figure 7. The impact of the systematic is small for randomized uncertainty (Sys(rnd)) because this kind of systematics does not introduce oscillation shapes. The large impact is seen for a systematic that can introduce a signal-like shape (Sys(CORR)) to data.

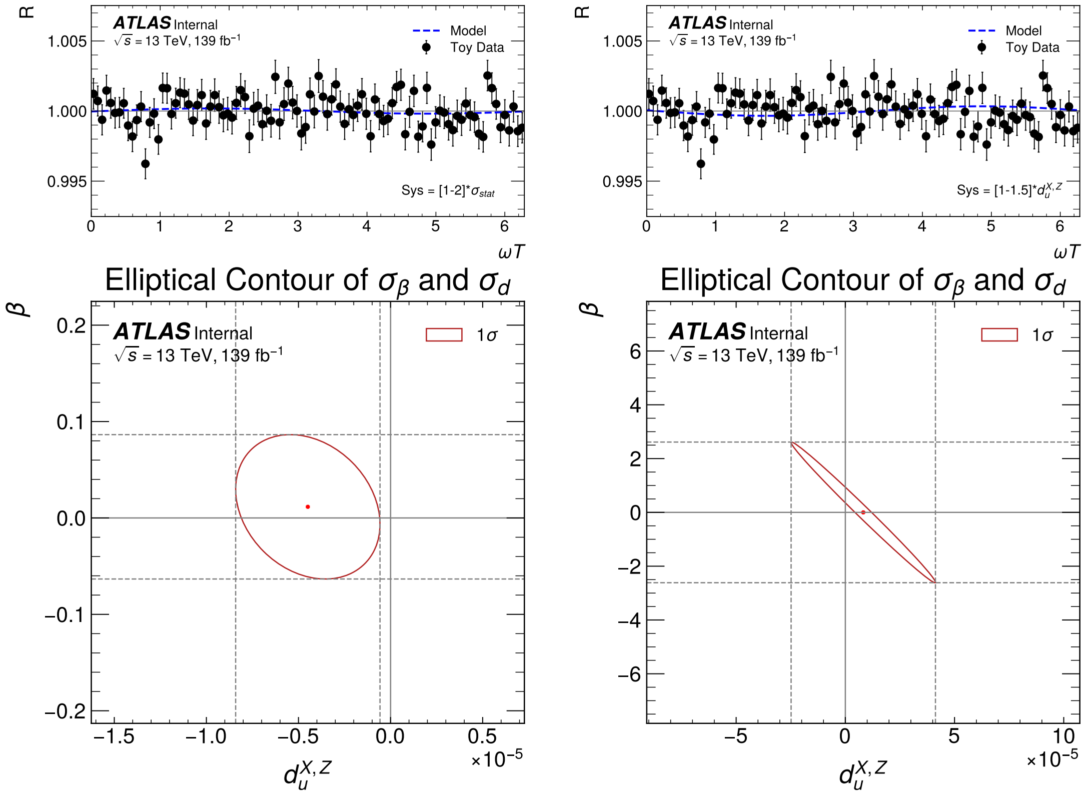

Figure 6: Fit from the randomized systematic impacts top left (case 2). The fit with the systematics introduces a shape to data, top right (case 3). Corresponding one sigma contours from profiled $\chi^2(\beta, \mu)$ in the bottom plots.

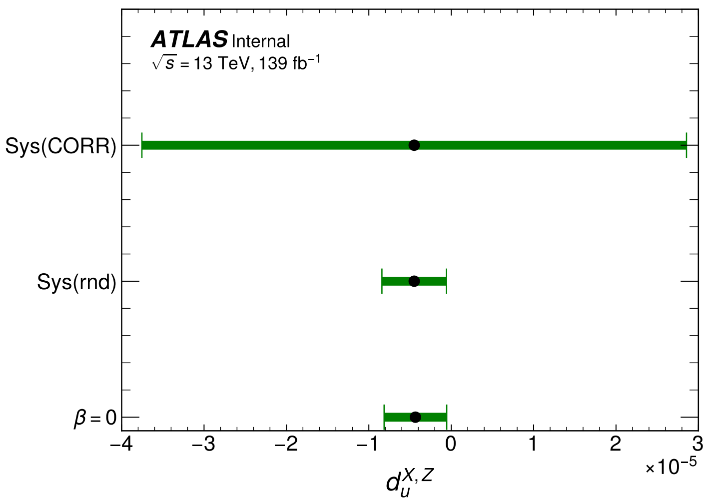

Figure 7: Comparison of the fitted parameter $d_u^{X,Z}$ from χ^2 ($\beta = 0$) and profiled $\chi^2(\beta, \mu)$ fits using toy systematics in Case 2 (Sys(rnd)) and 3 (Sys(CORR)).

4 Luminosity-data driven toy studies

Main purpose of TOY study is to check basic principles of the measurement. This toy study is used for understanding of binning and sidereal phase, and estimating sensitivity also evaluate effect of systematic uncertainties.

In this section, basics of our TOY model is described and show several studies with changing conditions.

4.1 Basic TOY model

One of most important target of this TOY study is to find any structure or non-uniformity in the double ratio as function of the sidereal phase. Because physics runs were taken when LHC was ready and setup, therefore the distribution of the time stamp of LBs is not uniform. A time stamp depends on starting time of a run which depends on LHC beam condition and life time and ATLAS detector status. Also, instatenous luminosity is not uniform. It's not simple to simulate this time structure, therefore we use the actual time stamp of Run-2 data for this toy studies. Fig. 8 (left) shows the time profile of LBs of full Run-2 data, and their duration are plotted in Fig. 8 (right).

As a basic TOY sample, the number of Z is computed each LB according to its duration and compute the Sidereal phase with following condition using only the time stamp of Lumi Block, and not use of other information in the actual data:

1. use actual time stamp of each LB of Run-2 data.
2. a Z fiducial cross sectin of 750 pb. (and inject LIV effect as function of sidereal phase if necessary)
3. a flat instantaneous luminosity of $1.0 \times 10^{34} cm^{-2} s^{-1}$ (as a simplest case).
4. introduce random effect on the computed number of Z by applying poisson statistics in each LB.
5. Siderial period, T, is 23h 56m 04s (~ 365 / 366 days).
6. T_0 is set March 20th 11:35 UTC, which is T_0 of SCF at the center of the ATLAS detector. (See Section 2).

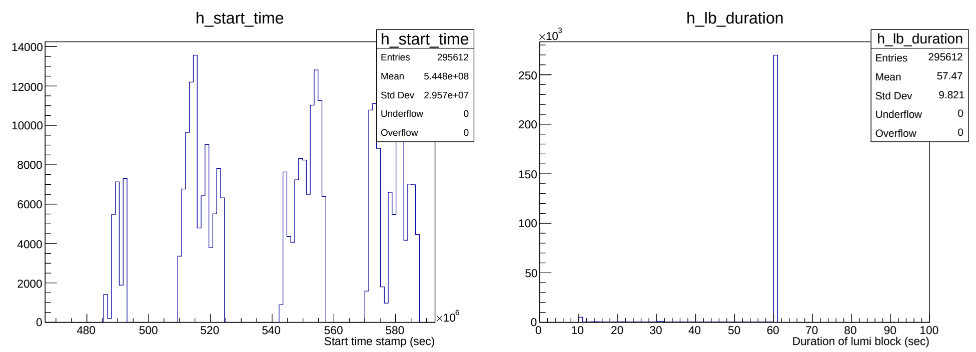

Figure 8: Distribution of the time stamps of each Lumi block (left) and their duration (right).

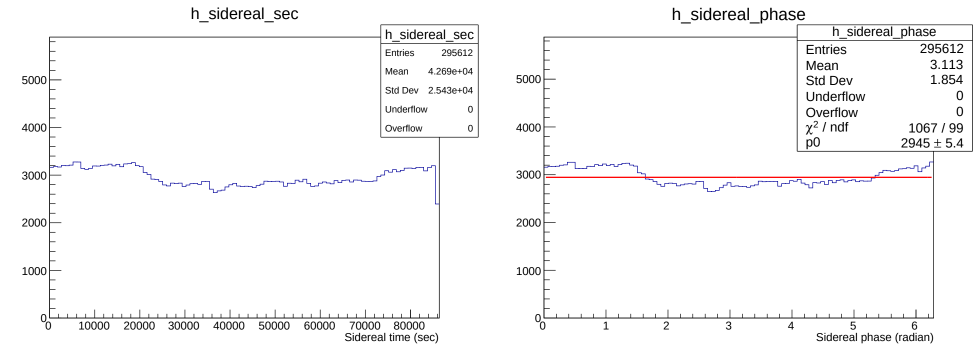

Figure 9: Distribution of Lumi block as function of the sidereal period (left) and the sidereal phase (right).

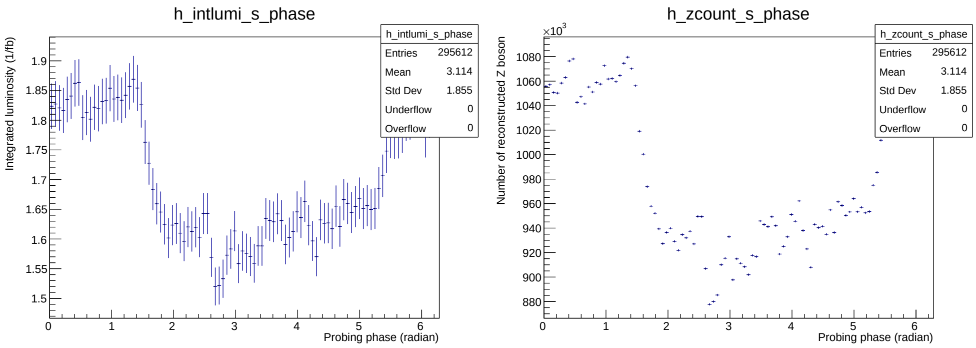

Figure 10: Integrated Lumiminoisty (left) and number of Z boson (right) for each sidereal phase bin (100 bins over 2π.)

Histogram of number of LBs as function of the generated Sidereal period and sidereal phase are shown in Fig. 9. These distributions are not flat because of run time (ATLAS is taking data when LHC makes collision when ATLAS detector is ready.) Distribution of the integrated luminosity as a function of sidereal phase and number of Z boson are shown in Fig. 10. Non-uniform structure of this distribution is due to the non-uniform profile of time stamp of the lumi blocks. For example, a fill in 2018 started at 8:00 and lasted 7 hours and no other fill on the same day. In this case, we have measured the number of Z only 7 hours over 24 hours.

In order to probe tiny effect, we take the double ratio as Eq. 47. The distribution of the double ratio is shown in Fig. 11. On this TOY configuration, $\mathcal{L}_{int} = 178.2 fb^{-1}$, because we assume a flat instanteneous luminosity.

Statistical uncertainties are shown in the error bar in Fig. 11. In Figs. 12, the projection of the double ratio and it's pull are shown. The mean and RMS of the pull distribution are consistent with 1.0, which shows reasonable error propagation has been made.

Figures 13 show the double ratio in case of LIV signal amplitude of 1 pb with sin wave. Just for illustration purpose, prepared with and without statistics fluctuation. The amplitude of no randome case is around 0.00133, which is from the cross section ratio of 1 pb over 750 pb, which is used to generate this Toy experiment.

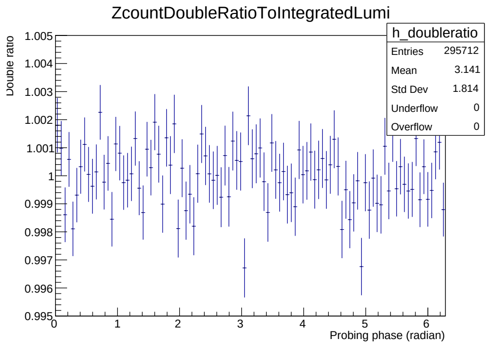

Figure 11: The Double ratio for each sidereal phase bin (100 bins over 2π.)

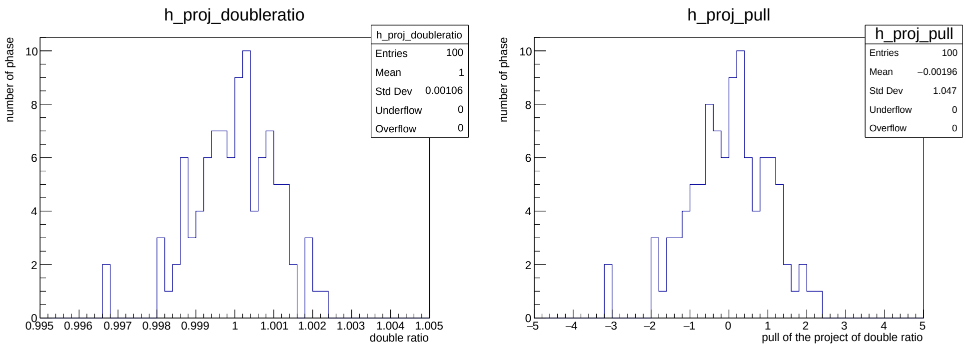

Figure 12: Projection of Fig. 11 (left) and pull distribution.

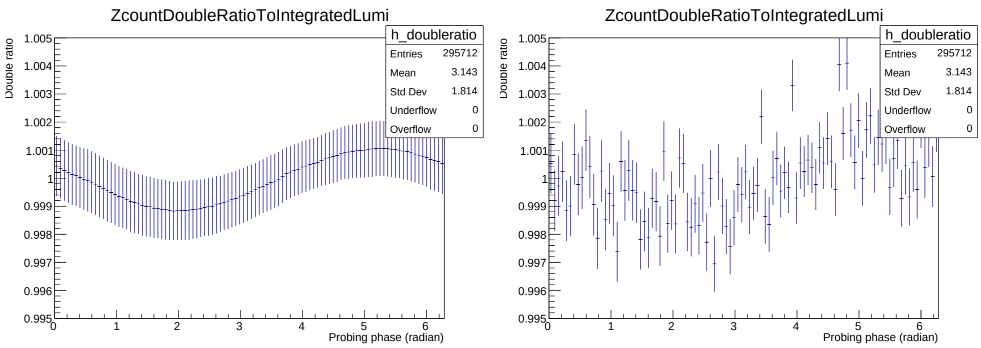

Figure 13: The double ratio for each sidereal phase bin (100 bins over 2π.) with LIV signal amplitude of 1 pb.

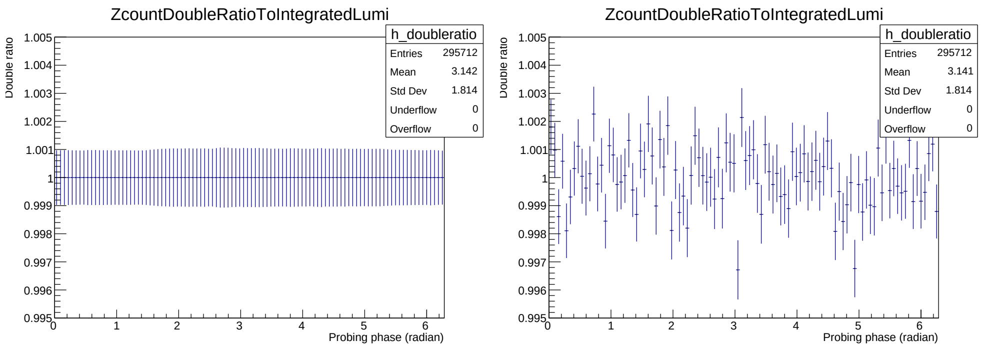

Figure 14: The double ratio for each sidereal phase bin (100 bins over 2π.) with LIV signal amplitude of 0 pb (NULL signal).

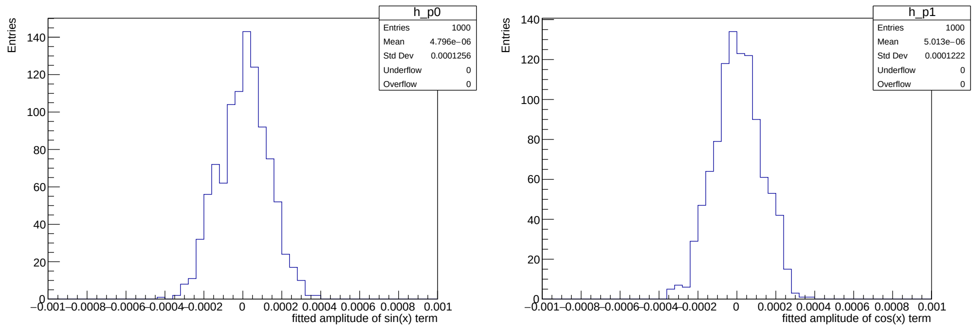

Figure 15: Result of 1000 toy experiement with NULL signal. Fit function is 1 + p0 sin(x)+p1 cos(x). (left) amplitude of sin(x) term, (right) amplitude of cos(x).

4.2 Study with TOY data

By using Toy data, some basic studies are performed.

4.2.1 No signal case

This is test on NULL signal hypothesis. Figure 14 shows the double ratio with NULL signal, for both with and without statistical fluctuation. Based on this configuration, 1000 Toy experiments are generated and fit simple sin and cosin curve. The results of these fit are shown in Fig. 15. Mean value are consistent with zero, and the RMS is 0.00012. From this, we have statistical limit around 10 to -4.

In this test, 1000 toy datasets generated with SM expection and corresponds to Run 2 datase are used. In another words, no signal is injected to these datasets.

Figure 16 show distributions of fit parameters, errors, χ^2/ndf (ndf is number of degree of freedom) and p_0 value of χ^2, from fitting a flat distribution (bkg expection) to the 1000 toy datasets. Since the observable is a Double Ratio, the fitted off-set (fit parameter) is allways one in each fit. The normalzed χ^2 (χ^2/ndf, bottom right plot) distribution is centered around one which means the toy datasets agree with a flat distribution (this is what we expected). If we take p_0 ¿ 0.01 as a threshold to determine if the dataset agree with the SM expectation (flat distribution), then more then 99% of the toy dataset passed this cut (p_0 ¿ 0.01). Since more then 99% of the samples agree with SM expectation we can conclude that random fluctuation can not introduce strange shapes that significantly different then a flat distribution.

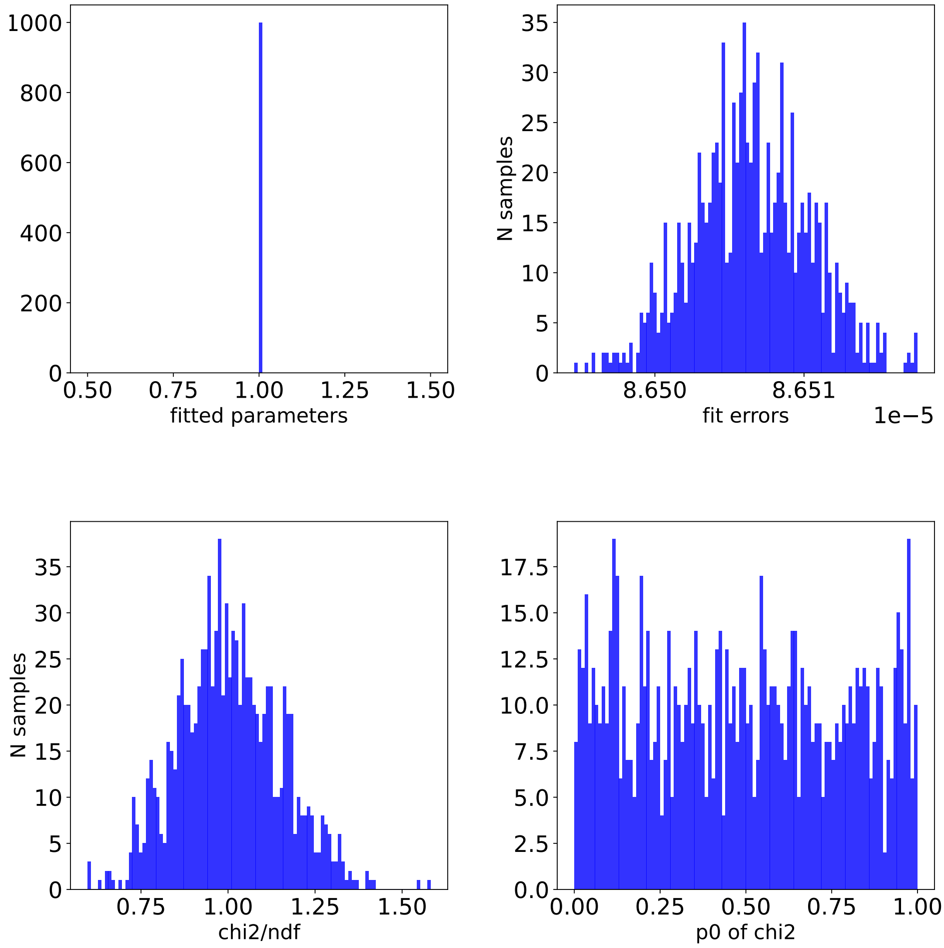

Figure 16: Results of Null Tests on 1000 signal-free toy datasets. Top left plot is estimated parameters, top right is fit errors, bottom left is χ^2/ndf (ndf is number of degree of freedom), bottom right is p_0 of χ^2.

4.3 Spurious signal test

Spurious signals may occur from statistical fluctuations. Therefore, in this section, we test toy datasets to check if there are signal like structures. The toy datasets are signal-free datasets (no signals injected), and generated total of 1000 samples. Each sample (toy dataset) is fitted 12 times, each time with different signal shapes (we have total of 12 signals). Each fit will estimate a coefficient.

Figure 17 shows distributions of 4 example coefficients ($d_u^{XZ} \equiv d_0$, $d_u^{YZ} \equiv d_1$, $d_u^{XX-YY} \equiv d_2$ and $d_u^{XY} \equiv d_3$) estimated from 1000 toy datasets. From each plot we can see that mean of each distribution is very close to zero. The Figure 18 shows mean and standard deviation of the distributions shown in Figure 17, expanded to all 12 coefficients.

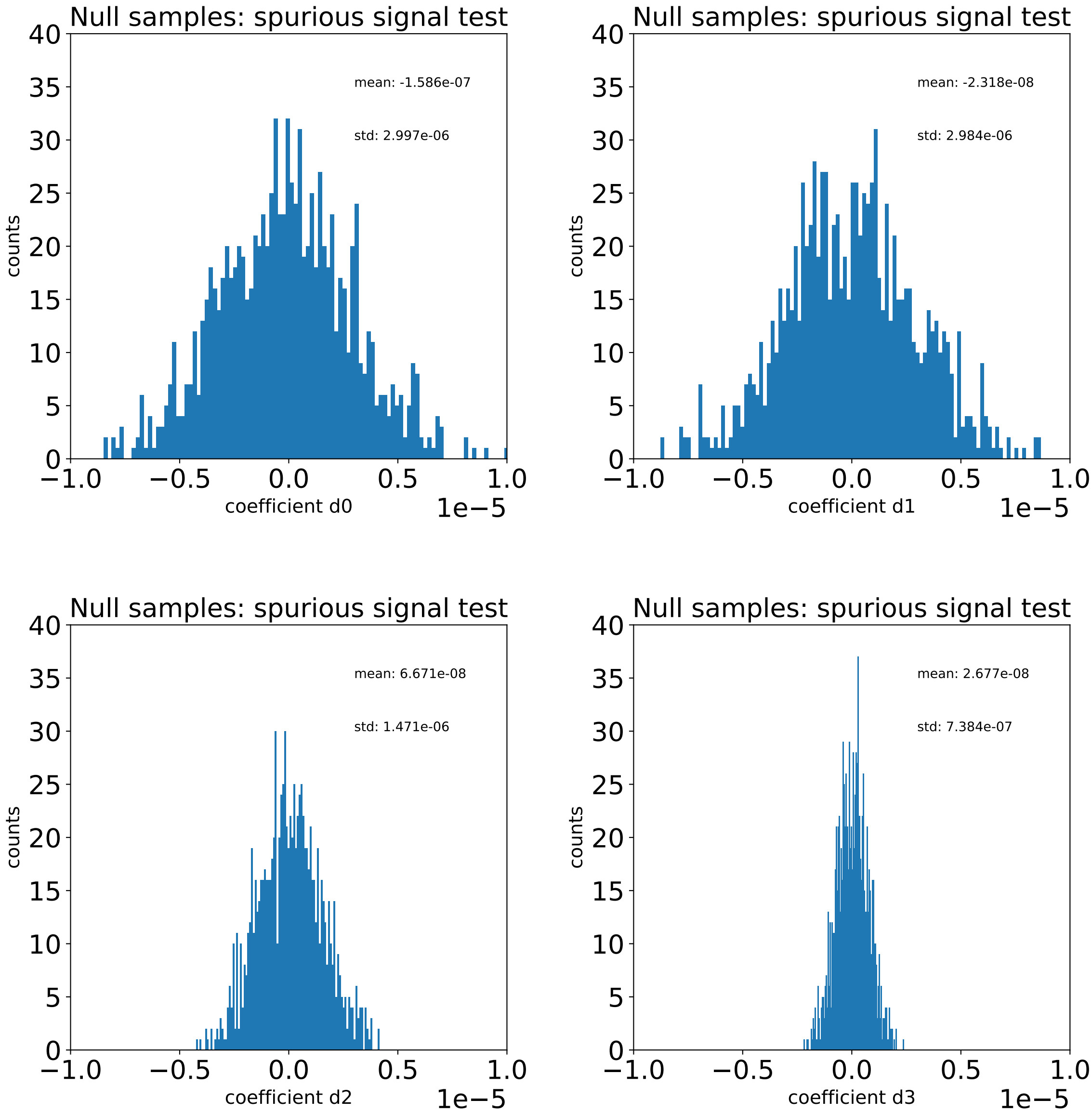

Figure 17: Results of spurious signal tests on 1000 signal-free toy datasets. Top left is a distribution of estimated d_0 parameters, top right is d_1 parameters, bottom left is d_2 parameters and bottom right is d_3 parameters.

4.3.1 Simple LIV effect

This is test with LIV signal amplitude of 1 pb with sin curve. Same as NULL case, 1000 toy experiments are generated as same method described in Section 4.1, and fitted with simple function. Results are shown in Fig. 19 The mean of the fitted amplitude of sin (cos) term is 0.0013 (0.000), which is consistent with generated configuration.

Figure 20 shows same set of result close to 3 sigma level sensitivity (only taken into account statistics uncertainty). The injected signal amplitude is 300 fb.

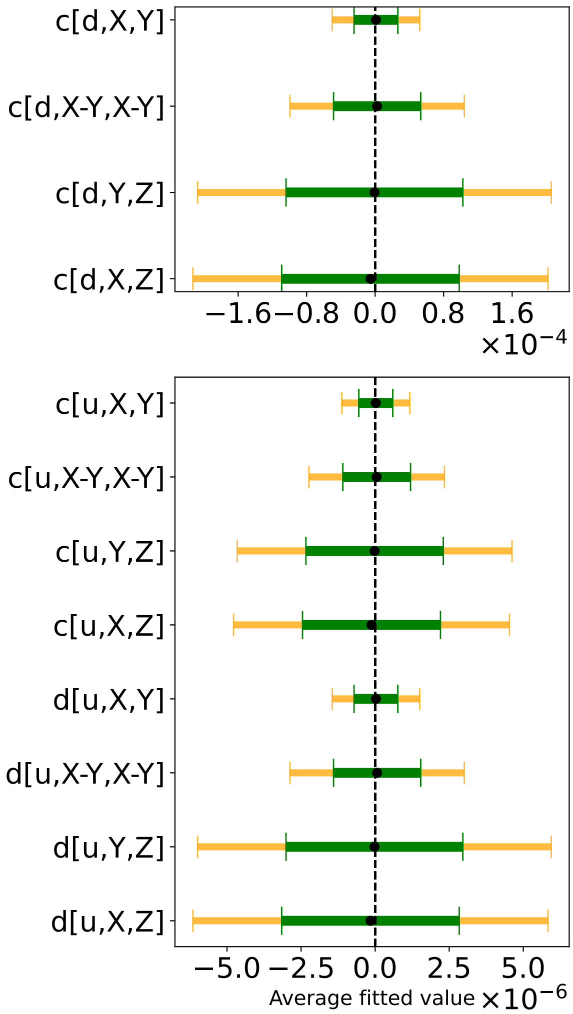

Figure 18: Summary of spurious signal tests on 1000 signal-free toy datasets. Black dot is a numpy mean of corresponding distributions in the Figure 17, the green and orange bands are one and two standard diviations calculated with the numpy function.

4.3.2 Number of bin in sidereal phase

Number of binning in sidereal phase is scanned with simple Toy. Result is shown in Fig. 21, which is performed with LIV signal amplitude of 1 pb. As expecdted, low number of bins shows low significance, this is just simply cancel negative part and positive part of signal due to coarse binning. An interesting observation is there are small bias around number of bins from 10 to 15. It has higher than reaching plateau, after number of bins greater than 25. This indicates it's better to use enough large number of bins. For finer direction, the Run-2 data provides a large number of Z events, therefore we could use very fine binning. Finest limit case would be 1440, which is based on duration of LB (60 seconds) over 24 hours. To avoid too fine bin, we choose number of bins of 100 as the default configuration. Here 1 bin corresponds to about 15 min duration for the sidereal phase.

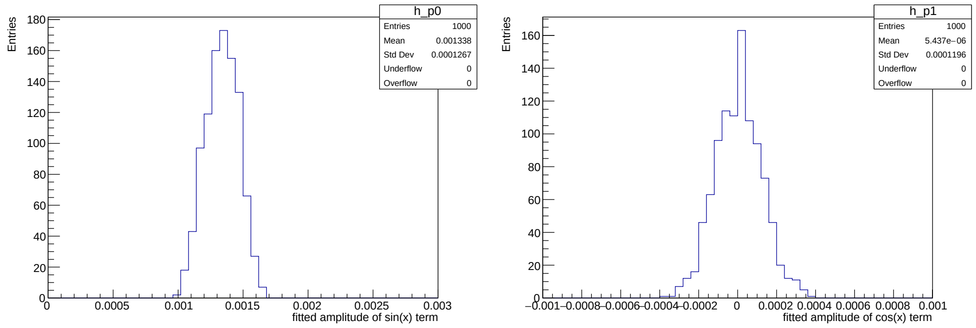

Figure 19: Result of 1000 toy experiement with LIV signal amplitude of 1 pb. Fit function is 1 + p0 sin(x)+p1 cos(x). (left) amplitude of sin(x) term, (right) amplitude of cos(x).

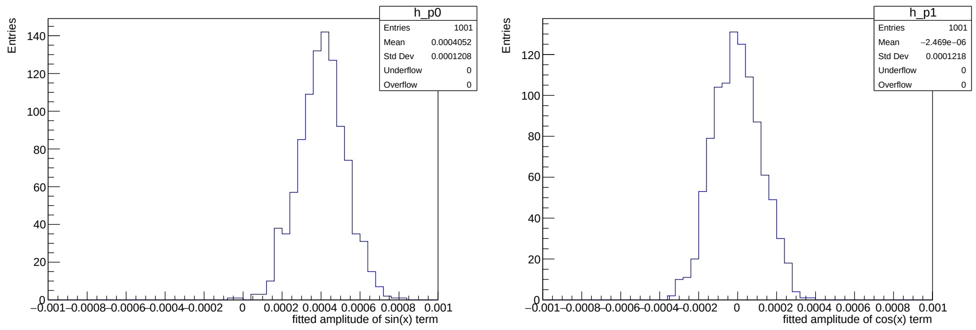

Figure 20: Result of 1000 toy experiement with LIV signal amplitude of 0.3 pb. Fit function is 1 + p0 sin(x)+p1 cos(x). (left) amplitude of sin(x) term, (right) amplitude of cos(x).

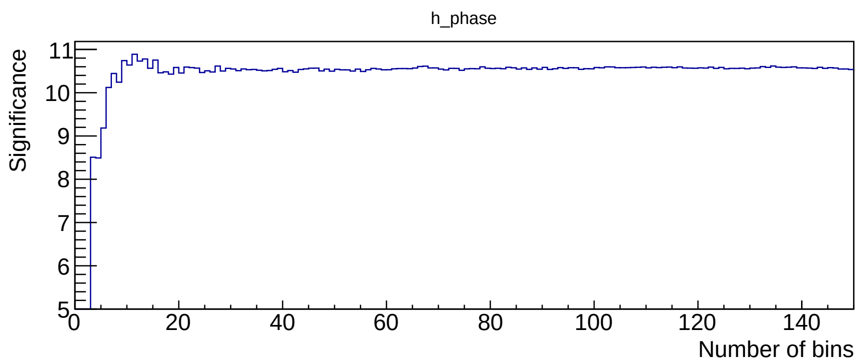

Figure 21: Significance (quadrature sum of diviataion from 1 on the double ratio in sidereal 100 bins) as function of number of bins with no random case. Injected LIV amplitude is 1 pb.

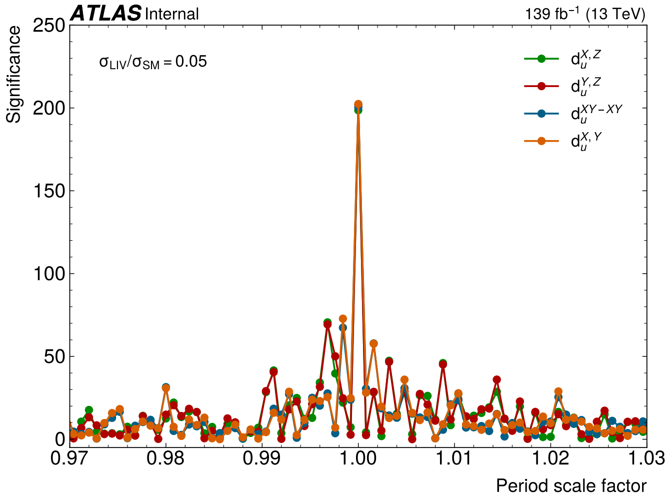

Figure 22: Significance (quadrature sum of deviataion from 1 on the double ratio in sidereal 100 bins) as function of probing period scale factor with no random case. Injected LIV amplitude is 1 pb. The period with scale factor = 1 is the 23h56m04s, which is used to generate signal. TO BE UPDATED with figure from Ablet

4.3.3 Changing probing sidereal period for expected signal

This study was made to check if we could check day-night effect. The difference of duration between solar day and the sidereal day is about 4 minutes, which corresponds to only $\sim 0.3\%$. Looking the day-night effect may be effectively reveal the potential LIV effect. Fig. 22 confirms this hypothesis, we would see LIV signal with probing period with the solar day. In this test, inject signal amplitude is 1 pb sin curve with sidereal phase. And the significance is computed by changing the probing period scale factor, $\alpha_{preriod}$. Here the $\alpha_{preriod}$ is the scale factor on the period to compute phase. To have the period of solar day, $\alpha_{preriod}$ is 1.00274. In the figure, $\alpha_{preriod}$ is scanned from 0.975 to 1.025. Observed periodic peak structure. Wider range of scan is shown in Fig. 23.

Base on this study, we understood we should not look the double ratio with the period of solar day in order to make blind data analysis. We are planning to make step by step unblinding. Once all studies are performed, start to make fit on the double ratio with the probing phase of 1 hour and 7 hours.

4.3.4 Study on the effect of correction

Effect of efficieincy correction is checked also simple TOY study. Above studies are all not taken into account efficiency effect on the Z counting method. As described in Section 3, there are $< \mu >$ dependent efficiency changes, i.e. on lepton reconstruction and trigger. These effects are modeled in Toy generation. Electron efficiency is parameterized as a function of $< mu >$, and generated Toy based on it. Figure 24(left) shows the double ratio with NULL signal configuration without applying the efficiency effect. The double ratio shows a flat dependence. Once efficiency effect taken into accout as the middle figure of Fig. 24, structure showing up. This is because profile of $< mu >$ as function of sidereal phase is not flat due to limited statistics. This is small effect, but it become large effect on the double ratio.

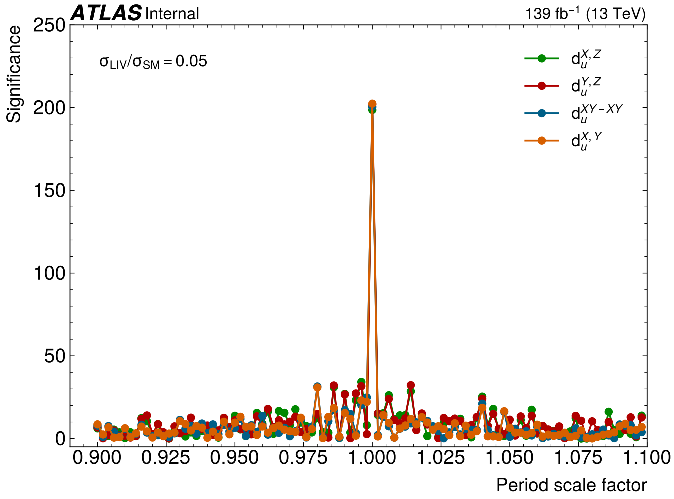

Figure 23: Significance as function of probing period scale factor with no random case. Injected LIV amplitude is 1 pb. The period with scale factor = 1 is the 23h56m04s, which is used to generate siganl. TO BE UPDATED with figure from Ablet

Figure 24(right) shows same one after applying efficiency correction. In case the modeling of efficiency dependence is well understood, we can minimize the modulation due to $< mu >$ dependent modulation on the double ratio. This could be the largest systematic uncertainty of the search.

4.3.5 One run behavior with TOY experiment

During ATLAS data taking, we are monitoring various physics observables. Number of Z is also one of observable. In case a large LIV effect, we would see large effect on number of Z as function of sidereal time. For example, Fig. **??** shows case of $d_u^{XY} \sim 10^{-3}$, the amplitude in the double ratio would be larger than 15%. This level, we would notice during data taking or data quality check.

Figure 25 shows the expected double ratio with $d_u^{YZ} = 10^{-3}$ in the longest run in 2017 data taking. With this level of signal, just one run of data would be enough to notice. We consider that amplitudes larger than a few % are already excluded.

Relation between significance and signal strength (ratio of the amplitude of LIV effect and SM Z yields) for wilson coeffiicnets shows in Fig. 26.

Peak amplitude of the LIV effect, $(\sigma_{SME} - \sigma_{SM})/\sigma_{SM}$, with wilson coefficients are one are listed Table 4. Scan ranges for each wilson coefficient are set at the peak amplitude at 5% which are summarized in the same Table.

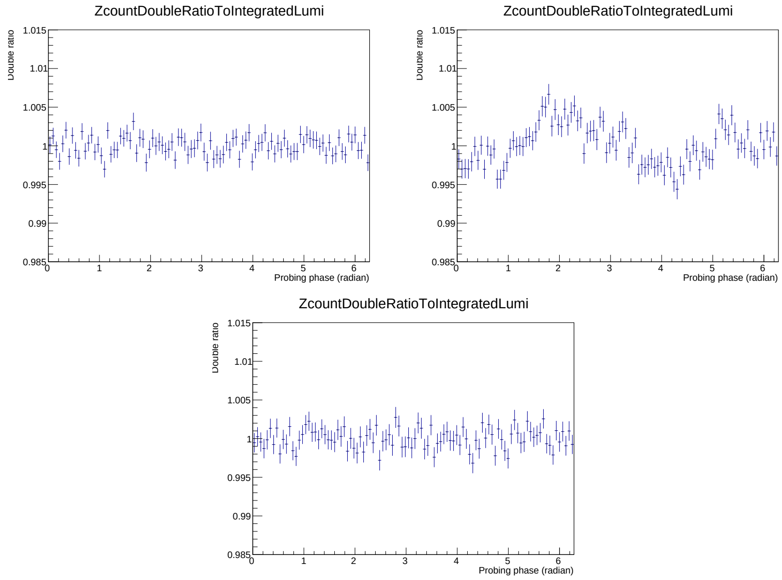

Figure 24: Effect of efficiency correction and its correction to the double ratio. Left: The double ratio in case efficiency=1, Middle: The double ratio with efficiency effect taken into account, Right: the double ratio with efficiency effect after applying its correction. These are generated with NULL signal configuration.

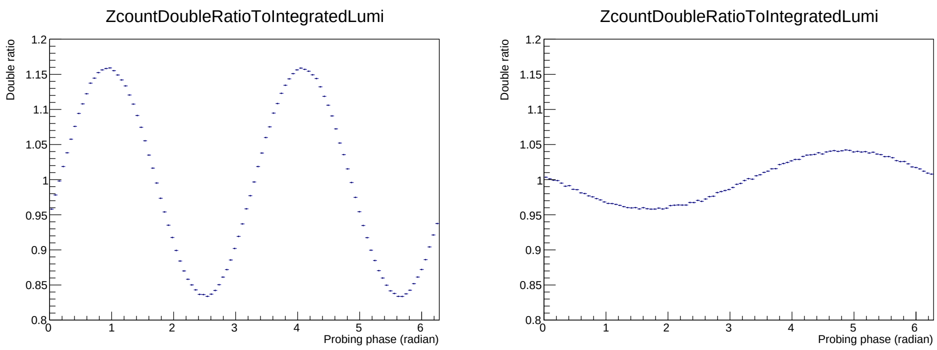

Figure 25: Expected double ration for the case of $d_u^{XY} = 10^{-3}$ (left) and $d_u^{YZ} = 10^{-3}$ (right). Very large oscillation would be observed.

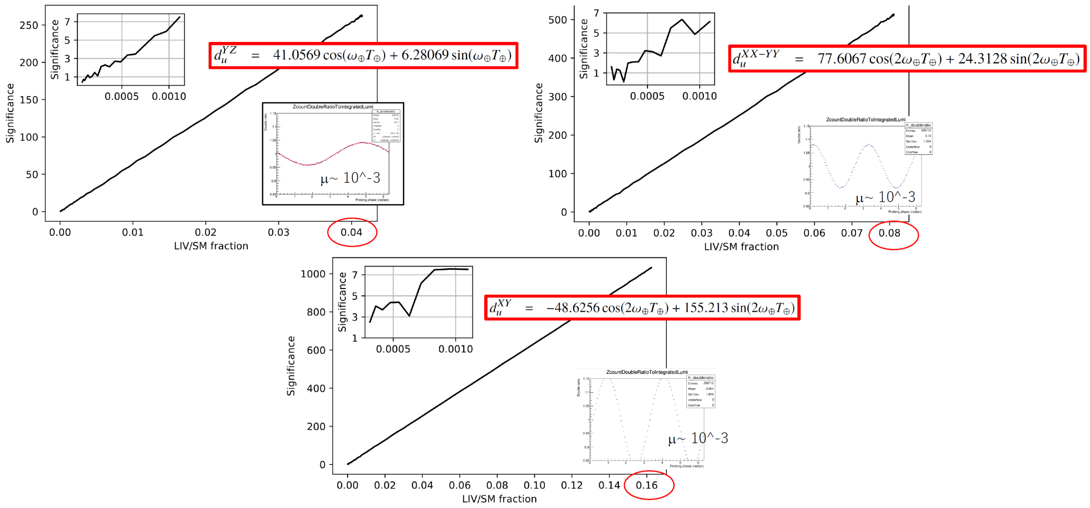

Figure 26: Significance as function of signal strength for wilson coefficients d_u^{YZ} (left), d_u^{XX-YY} (middle) and d_u^{XY} (right).

Coefficient	peak amplitude	upper bound (peak amplide of = 0.05)
d_u^{XZ}	41.53449	1.203819×10^{-3}
d_u^{YZ}	41.53449	1.203819×10^{-3}
d_u^{XX-YY}	81.32592	6.148103×10^{-4}
d_u^{XY}	162.6518	3.074048×10^{-4}
c_u^{XZ}	53.45986	9.352805×10^{-4}
c_u^{YZ}	53.45986	9.352805×10^{-4}
c_u^{XX-YY}	104.6762	4.776644×10^{-4}
c_u^{XY}	209.3524	2.388322×10^{-4}
c_d^{XZ}	1.200602	4.164584×10^{-2}
c_d^{YZ}	1.200602	4.164584×10^{-2}
c_d^{XX-YY}	2.350819	2.126917×10^{-2}
c_d^{XY}	4.701638	1.063459×10^{-2}

Table 4: Peak amplitude of the LIV effect ($(\sigma_{SME}-\sigma_{SM})/\sigma_{SM}$) at each wilson coefficinet is 1, and their uppper bound of the scan range which are set at the amplitude of 5%.

5 Data scrambling studies

5.1 Methodology of data scrambling

In order to check data behavior with blinding on LIV effect, we have studied the data scrambling. This uses the measurement of Z counting for each LB in data. However, we do scrambling the time stamp. Because LIV effect can be probed by looking consequenced data, therefore if we scramble time stamp of each LB, we can not probe the LIV effect.

Actual method is very simple, we scramble time stamp of LBs within each year. No same time stamp is used twice. Actual function to scramble time stamps is the function, "sample", in random library of python3.

By using this dataset, we can check real value from the measurement with the lucid luminosity and the Z counting. In order to check staticial behavior, 1000 (or 10000) set of scrabled data are generated by using different seed in same way. Number of LBs for each year of data is listed in Table. 3. Those are about 90 k LBs[10], therefore we could generate different scriabled dataset by using different seed.

As we have three channels, electron, muon and combined, we check these output.

5.2 Summary of results and comparisons to toys

5.2.1 Results

Fit results are shown for each year, sum of Run 2, and Run 2 - combined datasets. Definition of these datasets are shown in the list bellow.

- year 2015+2016
- year 2017
- year 2018
- Run 2: sum of all year's data.
- Run 2 - Combined: stitch up three datasets in the range of $0 \to 6\pi$. The 2015+2016 dataset is in the $0 \to 2\pi$ range, the 2017 dataset is in the $2pi \to 4\pi$ range, and 2018 dataset is in the $4pi \to 6\pi$ range.

In Figure 63, a comparison between a scrambled data (one of the 1000 scrambled datasets) and one of the fitted models, denoted as $c_u^{X,Y}$ are presented. The top-left plot shows for the years 2015 and 2016, the top-right plot shows the same for the year 2017, the bottom-left plot shows the year 2018, and the bottom-right plot shows for the Run 2.

The comparison of the fitted model ($c_u^{X,Y}$) and Run 2 - combined data is shown in figure 64. The fit range is $0 \to 6\pi$ since this dataset stitch up three individual years.

In the say way as showen above, we fitted each scrabled dataset (total of 1000 dataset in each year). In the end we obtained 1000 fit results for each year. Therefore, in fugures 65, 66, 67, 32, 69 average of 1000 fit

[10] we combine data15 and data16 because data15 has only 22 k LBs

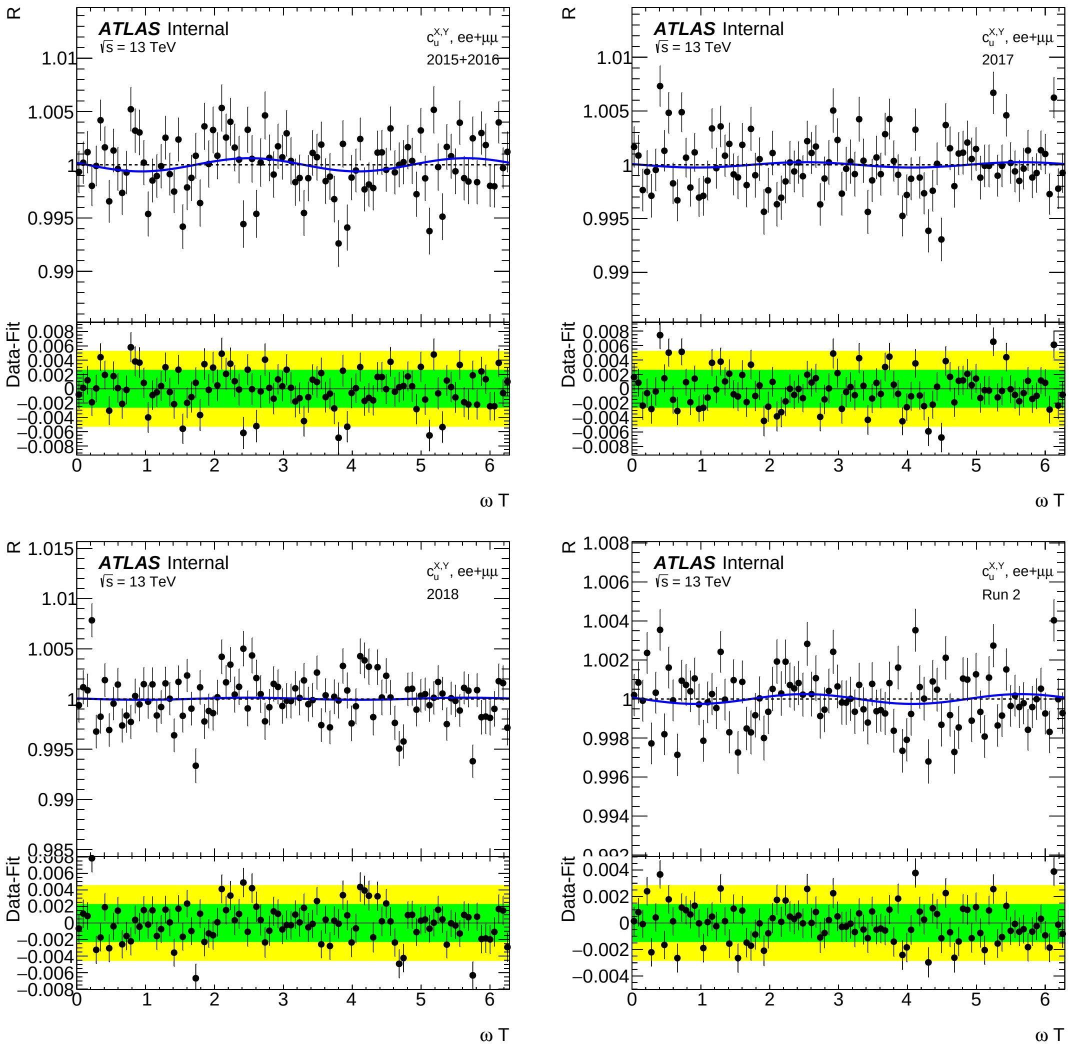

Figure 27: One of the scrambled datasets VS $c_u^{X,Y}$ model. The black dots are the scrambled data, and the blue lines are the fitted model. The 2015+2016 dataset(top left), the 2017 dataset (top right), the 2018 dataset(bottom left), and the Run 2 dataset (bottom right). The Run 2 dataset is the sum of all year's data. The bottom panel of the plots is the residual (data-model). The green and yellow bands correspond to one and two sigma uncertainty bands.

results (black dot) and their one and two standard deviations (green and yellow bands) are presented as plots.

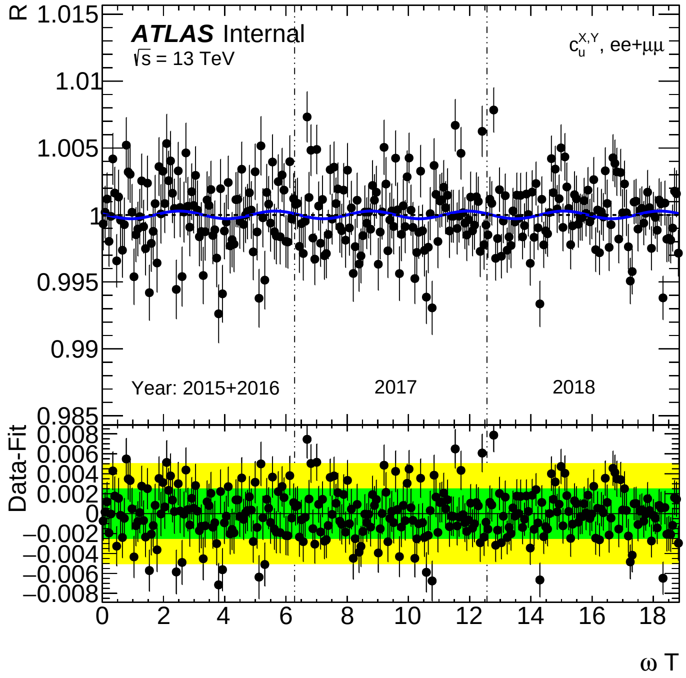

Figure 28: The Run 2 combined dataset (ne of the scrambled datasets) VS $c_u^{X,Y}$ model. The black dots are the scrambled data, and the blue lines are the fitted model. The bottom panel of the plots is the residual (data-model). The green and yellow bands correspond to one and two sigma uncertainty bands.

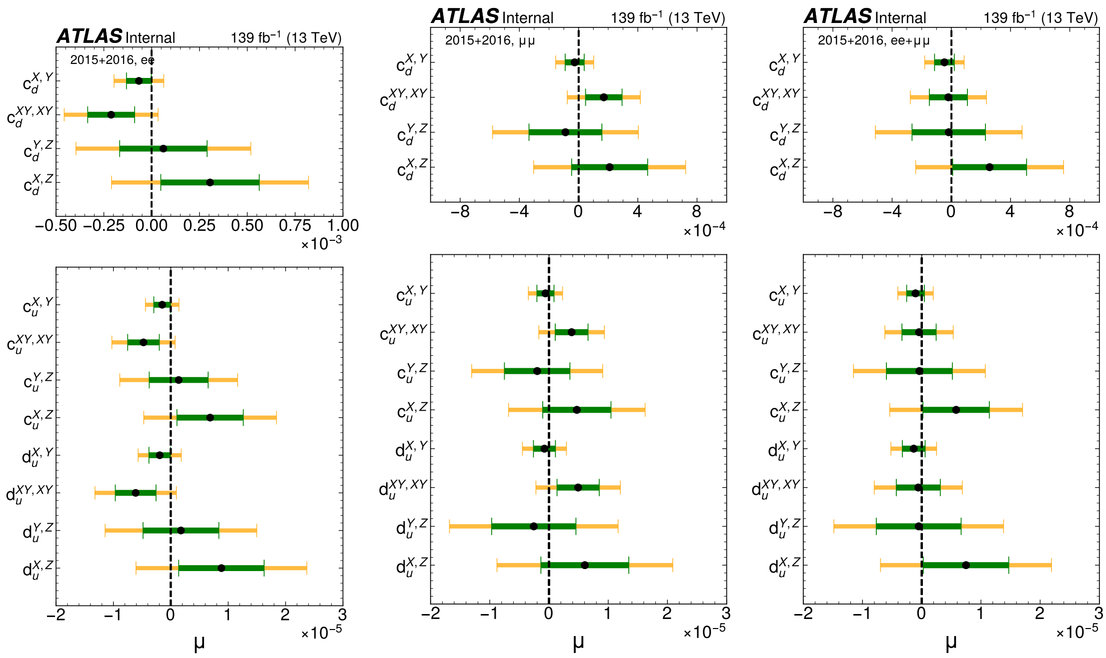

Figure 29: Scrambled datasets collected in 2015 and 2016. Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The ee channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

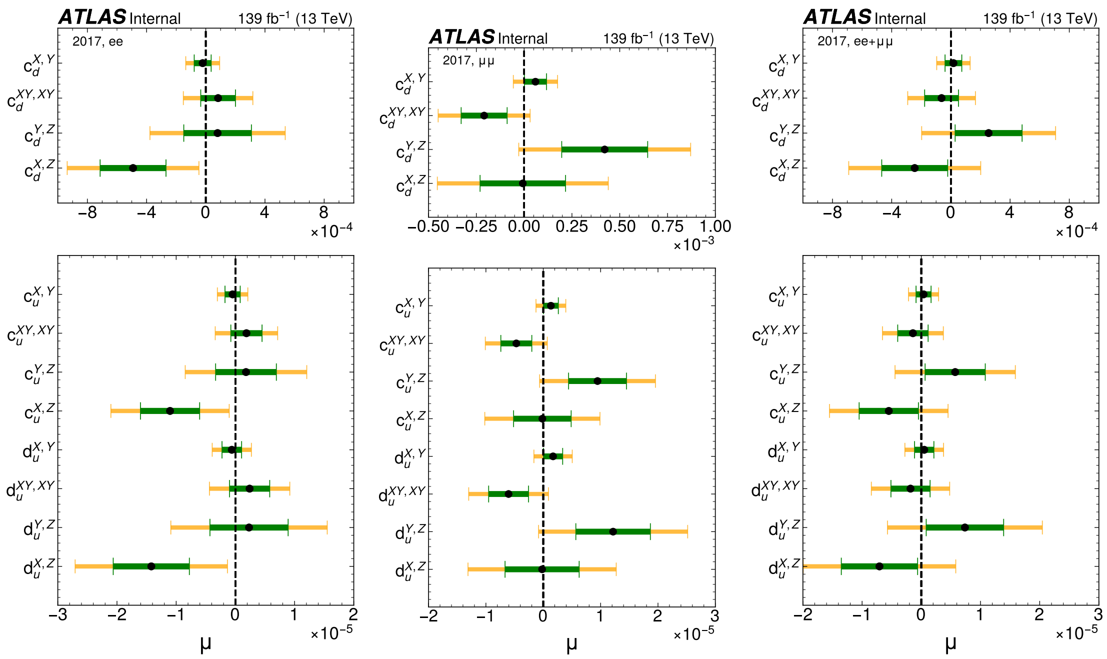

Figure 30: The scrambled datasets collected in 2017. Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The ee channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

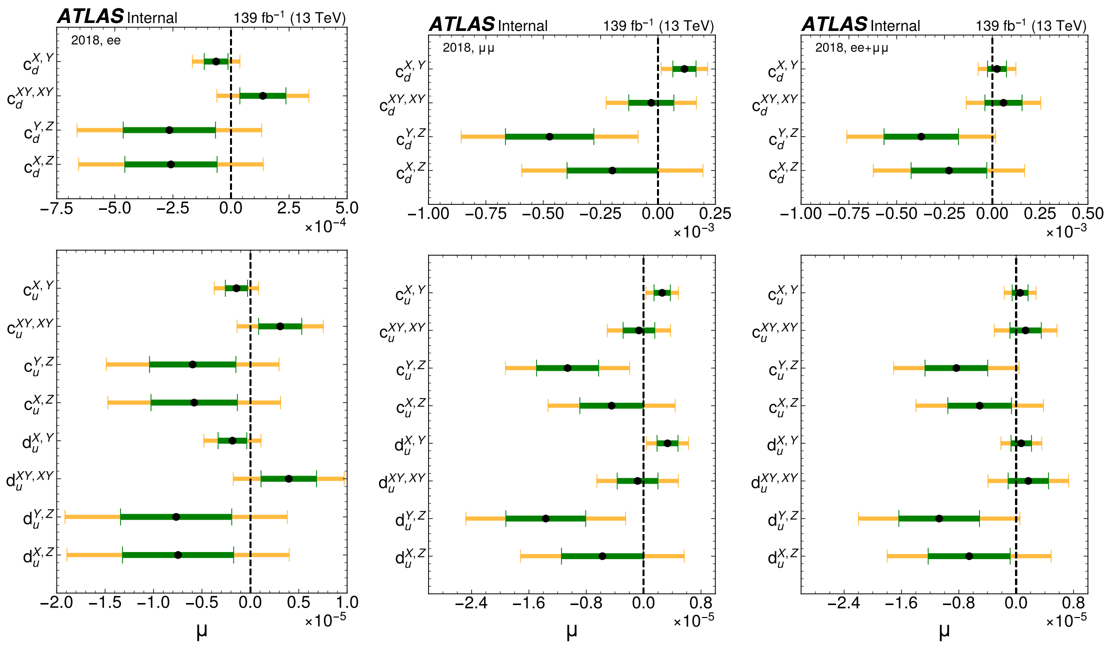

Figure 31: The scrambled datasets collected in 2018. Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The ee channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

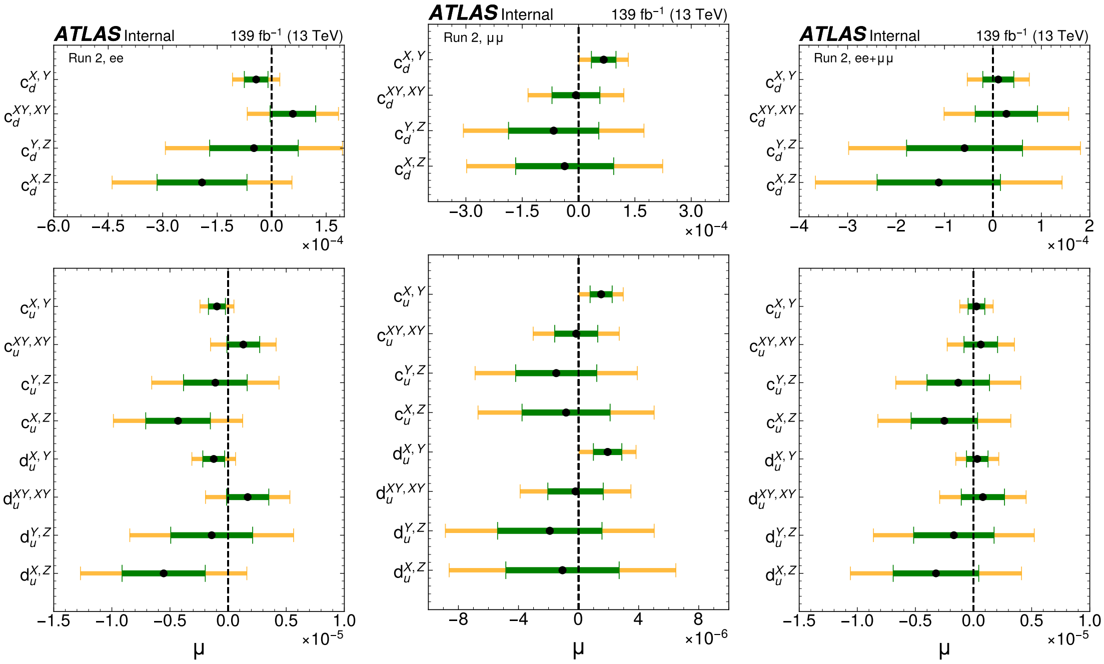

Figure 32: The scrambled datasets in full Run 2 (sum of all years). Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The ee channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

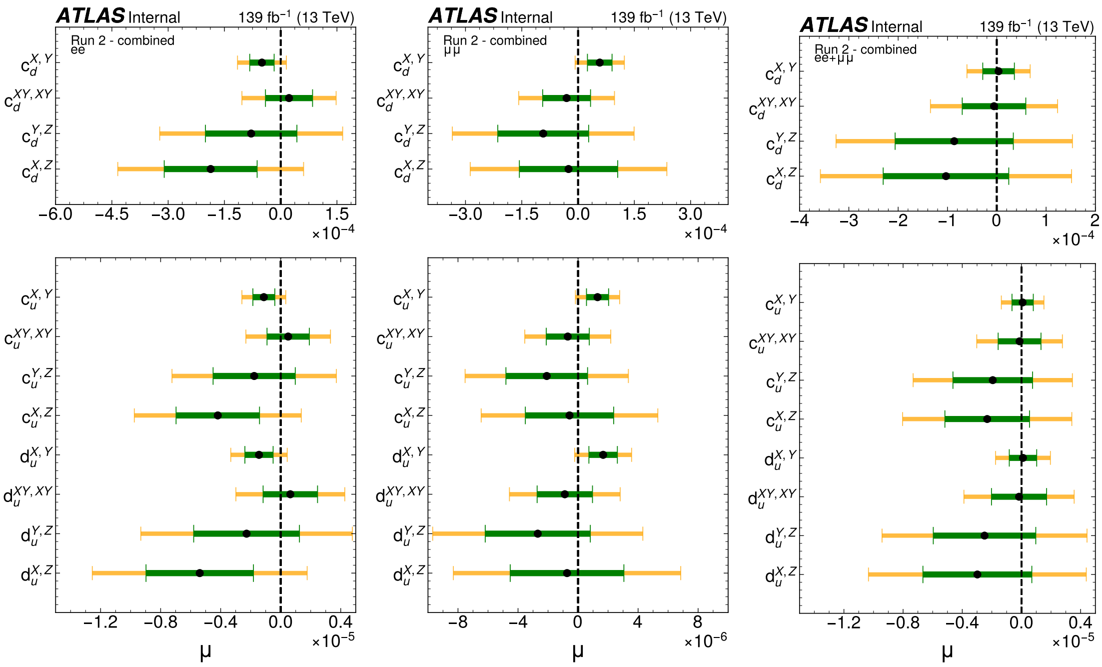

Figure 33: Run2-combined scrambled datasets. Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The ee channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

	2015+2016	2017	2018	full Run2	Run2-Comb. Fit
$d_u^{X,Z}$	8.82e-06 ± 7.44e-06	-1.42e-05 ± 6.43e-06	-7.47e-06 ± 5.74e-06	-5.54e-06 ± 3.58e-06	-5.39e-06 ± 3.58e-06
$d_u^{Y,Z}$	1.77e-06 ± 6.61e-06	2.31e-06 ± 6.61e-06	-7.67e-06 ± 5.74e-06	-1.41e-06 ± 3.52e-06	-2.27e-06 ± 3.52e-06
$d_u^{XY,XY}$	-6.12e-06 ± 3.55e-06	2.41e-06 ± 3.39e-06	3.97e-06 ± 2.87e-06	1.69e-06 ± 1.81e-06	6.42e-07 ± 1.81e-06
$d_u^{X,Y}$	-1.93e-06 ± 1.88e-06	-6.03e-07 ± 1.65e-06	-1.85e-06 ± 1.48e-06	-1.24e-06 ± 9.41e-07	-1.45e-06 ± 9.41e-07
$c_u^{X,Z}$	6.85e-06 ± 5.78e-06	-1.10e-05 ± 5.00e-06	-5.80e-06 ± 4.46e-06	-4.31e-06 ± 2.78e-06	-4.19e-06 ± 2.78e-06
$c_u^{Y,Z}$	1.38e-06 ± 5.13e-06	1.79e-06 ± 5.13e-06	-5.96e-06 ± 4.46e-06	-1.10e-06 ± 2.74e-06	-1.77e-06 ± 2.74e-06
$c_u^{XY,XY}$	-4.76e-06 ± 2.76e-06	1.87e-06 ± 2.64e-06	3.08e-06 ± 2.23e-06	1.31e-06 ± 1.41e-06	4.99e-07 ± 1.41e-06
$c_u^{X,Y}$	-1.50e-06 ± 1.46e-06	-4.69e-07 ± 1.29e-06	-1.44e-06 ± 1.15e-06	-9.60e-07 ± 7.31e-07	-1.12e-06 ± 7.31e-07
$c_d^{X,Z}$	3.05e-04 ± 2.57e-04	-4.91e-04 ± 2.23e-04	-2.58e-04 ± 1.99e-04	-1.92e-04 ± 1.24e-04	-1.87e-04 ± 1.24e-04
$c_d^{Y,Z}$	6.13e-05 ± 2.29e-04	7.98e-05 ± 2.29e-04	-2.65e-04 ± 1.99e-04	-4.88e-05 ± 1.22e-04	-7.87e-05 ± 1.22e-04
$c_d^{XY,XY}$	-2.12e-04 ± 1.23e-04	8.33e-05 ± 1.17e-04	1.37e-04 ± 9.91e-05	5.84e-05 ± 6.28e-05	2.22e-05 ± 6.28e-05
$c_d^{X,Y}$	-6.67e-05 ± 6.50e-05	-2.09e-05 ± 5.72e-05	-6.40e-05 ± 5.11e-05	-4.27e-05 ± 3.25e-05	-5.00e-05 ± 3.25e-05

Table 5: Partially unblinded results in ee channel. The table shows the average and one standard deviation of 1000 fit results.

	2015+2016	2017	2018	full Run2	Run2-Comb. Fit
$d_u^{X,Z}$	6.06e-06 ± 7.42e-06	-2.01e-07 ± 6.46e-06	-5.75e-06 ± 5.71e-06	-1.07e-06 ± 3.77e-06	-7.33e-07 ± 3.79e-06
$d_u^{Y,Z}$	-2.57e-06 ± 7.12e-06	1.22e-05 ± 6.50e-06	-1.37e-05 ± 5.57e-06	-1.92e-06 ± 3.48e-06	-2.68e-06 ± 3.50e-06
$d_u^{XY,XY}$	4.92e-06 ± 3.56e-06	-6.05e-06 ± 3.48e-06	-8.44e-07 ± 2.84e-06	-2.02e-07 ± 1.84e-06	-8.82e-07 ± 1.84e-06
$d_u^{X,Y}$	-7.66e-07 ± 1.86e-06	1.71e-06 ± 1.67e-06	3.33e-06 ± 1.46e-06	1.93e-06 ± 9.46e-07	1.67e-06 ± 9.48e-07
$c_u^{X,Z}$	4.71e-06 ± 5.76e-06	-1.56e-07 ± 5.02e-06	-4.47e-06 ± 4.43e-06	-8.31e-07 ± 2.93e-06	-5.69e-07 ± 2.94e-06
$c_u^{Y,Z}$	-1.99e-06 ± 5.53e-06	9.46e-06 ± 5.05e-06	-1.06e-05 ± 4.33e-06	-1.49e-06 ± 2.70e-06	-2.09e-06 ± 2.72e-06
$c_u^{XY,XY}$	3.82e-06 ± 2.77e-06	-4.70e-06 ± 2.70e-06	-6.55e-07 ± 2.21e-06	-1.57e-07 ± 1.43e-06	-6.85e-07 ± 1.43e-06
$c_u^{X,Y}$	-5.95e-07 ± 1.44e-06	1.33e-06 ± 1.30e-06	2.59e-06 ± 1.13e-06	1.50e-06 ± 7.35e-07	1.30e-06 ± 7.37e-07
$c_d^{X,Z}$	2.10e-04 ± 2.57e-04	-6.94e-06 ± 2.24e-04	-1.99e-04 ± 1.97e-04	-3.70e-05 ± 1.31e-04	-2.54e-05 ± 1.31e-04
$c_d^{Y,Z}$	-8.88e-05 ± 2.46e-04	4.21e-04 ± 2.25e-04	-4.72e-04 ± 1.93e-04	-6.63e-05 ± 1.20e-04	-9.29e-05 ± 1.21e-04
$c_d^{XY,XY}$	1.70e-04 ± 1.23e-04	-2.09e-04 ± 1.20e-04	-2.92e-05 ± 9.83e-05	-6.97e-06 ± 6.38e-05	-3.05e-05 ± 6.38e-05
$c_d^{X,Y}$	-2.65e-05 ± 6.42e-05	5.92e-05 ± 5.78e-05	1.15e-04 ± 5.05e-05	6.69e-05 ± 3.27e-05	5.78e-05 ± 3.28e-05

Table 6: Partially unblinded results in $\mu\mu$ channel. The table shows the average and one standard deviation of 1000 fit results.

	2015+2016	2017	2018	full Run2	Run2-Comb. Fit
$d_u^{X,Z}$	7.48e-06 ± 7.22e-06	-7.07e-06 ± 6.44e-06	-6.56e-06 ± 5.72e-06	-3.23e-06 ± 3.68e-06	-2.98e-06 ± 3.68e-06
$d_u^{Y,Z}$	-5.06e-07 ± 7.16e-06	7.37e-06 ± 6.54e-06	-1.07e-05 ± 5.63e-06	-1.69e-06 ± 3.46e-06	-2.49e-06 ± 3.46e-06
$d_u^{XY,XY}$	-5.76e-07 ± 3.72e-06	-1.82e-06 ± 3.31e-06	1.69e-06 ± 2.81e-06	8.03e-07 ± 1.86e-06	-1.62e-07 ± 1.86e-06
$d_u^{X,Y}$	-1.35e-06 ± 1.93e-06	4.94e-07 ± 1.63e-06	7.10e-07 ± 1.43e-06	3.21e-07 ± 9.25e-07	1.05e-07 ± 9.27e-07
$c_u^{X,Z}$	5.81e-06 ± 5.61e-06	-5.49e-06 ± 5.01e-06	-5.09e-06 ± 4.44e-06	-2.51e-06 ± 2.86e-06	-2.31e-06 ± 2.86e-06
$c_u^{Y,Z}$	-3.93e-07 ± 5.57e-06	5.73e-06 ± 5.08e-06	-8.35e-06 ± 4.37e-06	-1.32e-06 ± 2.69e-06	-1.94e-06 ± 2.69e-06
$c_u^{XY,XY}$	-4.48e-07 ± 2.89e-06	-1.42e-06 ± 2.57e-06	1.31e-06 ± 2.18e-06	6.24e-07 ± 1.44e-06	-1.26e-07 ± 1.45e-06
$c_u^{X,Y}$	-1.05e-06 ± 1.50e-06	3.84e-07 ± 1.27e-06	5.51e-07 ± 1.11e-06	2.50e-07 ± 7.19e-07	8.17e-08 ± 7.20e-07
$c_d^{X,Z}$	2.59e-04 ± 2.50e-04	-2.44e-04 ± 2.23e-04	-2.27e-04 ± 1.98e-04	-1.12e-04 ± 1.27e-04	-1.03e-04 ± 1.27e-04
$c_d^{Y,Z}$	-1.75e-05 ± 2.48e-04	2.55e-04 ± 2.26e-04	-3.72e-04 ± 1.95e-04	-5.86e-05 ± 1.20e-04	-8.63e-05 ± 1.20e-04
$c_d^{XY,XY}$	-1.99e-05 ± 1.29e-04	-6.30e-05 ± 1.14e-04	5.85e-05 ± 9.73e-05	2.78e-05 ± 6.43e-05	-5.59e-06 ± 6.45e-05
$c_d^{X,Y}$	-4.66e-05 ± 6.66e-05	1.71e-05 ± 5.65e-05	2.46e-05 ± 4.93e-05	1.11e-05 ± 3.20e-05	3.64e-06 ± 3.21e-05

Table 7: Partially unblinded results in $ee + \mu\mu$ channel. The table shows the average and one standard deviation of 1000 fit results.

5.2.2 Spread VS Stats

In this section we compare Spread and average stats uncertainty (we call is Stats for simplisity) in each scrambled dataset. A spread of a dataset is the average of absolute residual ($\sum_{i=1}^{100}|1 - data_i|/100$ in a signal free dataset). Each dataset have 100 data points one point per bin. Therefore an average stats uncertainty is $\sum_{i=1}^{100}\sigma_i^{stat}/100$, where i the bin index. In a dataset, if data points are distributed around 1 due to it's statistical fluctuation, then the ratio, Spread/Stats, is around one.

The figure 70 shows the distribution of Spread/Stats from the 1000 scrambled datasets for $ee + \mu\mu$ channel. As we can see from the plots, the Spread is usually 10-20% larger compared to the Stats, and it is true for the dataset of different years and full Run 2.

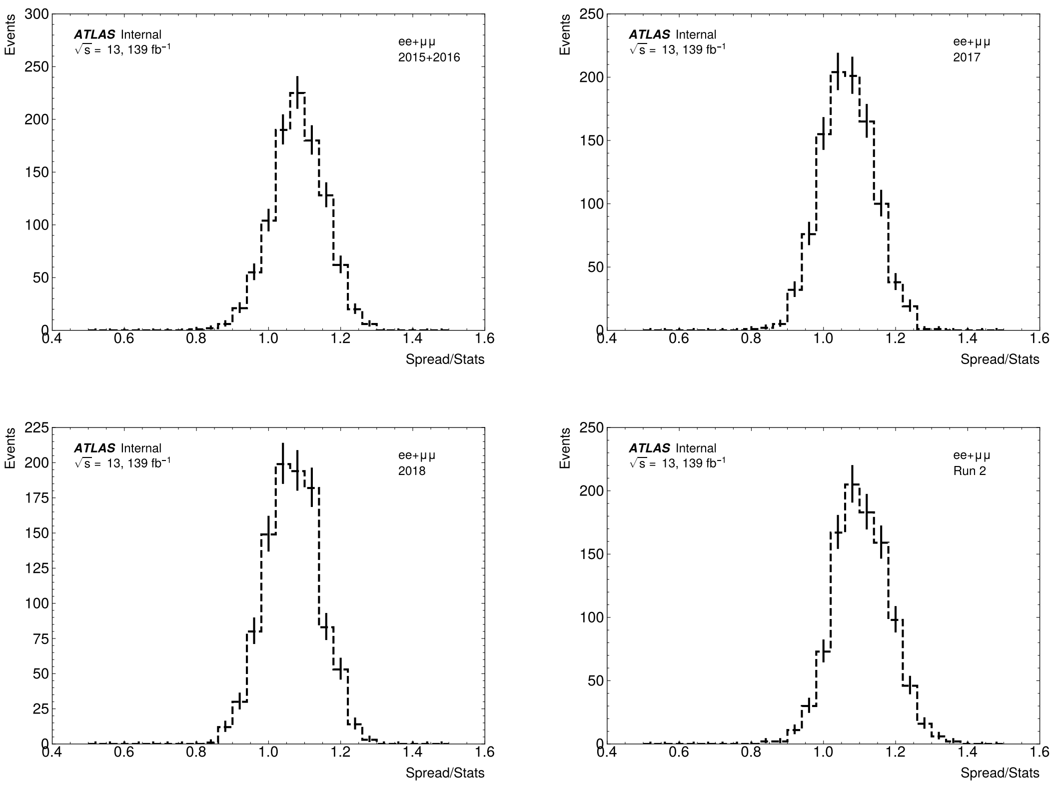

Figure 34: Spreads VS average stats. Plots show average results of 1000 scrambled datasets. An average statistic is the average statistical uncertainty in each scrambled dataset, and the spread of a dataset is an average of absolute residual. The 2015+2016 dataset(top left), the 2017 dataset (top right), the 2018 dataset(bottom left), and the Run 2 dataset (bottom right). The Run 2 dataset is the sum of all year's data. Spread is about 10-20% larger in most of the scrambled datasets. This indicates there is a small systematic effect or the way we are calculating the ZC error in the double ratio. This could be the reason why the average values shifted from zero

5.3 Increase number of scrambled datasets

In this Section we comapred results across small and large sample sizes to study the impact of varying the number of scrambled datasets, as denoted by the sample size. We used two sample sizes, 1000 and 10000.

Table 8 shows fit results of two different sample size. As we can see from the table that two set of results are consistent. Thus increasing number of scrambled datasets does not change results.

parameters	1K	10K
$d_u^{X,Z}$	-2.98e-06 ± 3.68e-06	-2.99e-06 ± 3.62e-06
$d_u^{Y,Z}$	-2.49e-06 ± 3.46e-06	-2.50e-06 ± 3.59e-06
$d_u^{XY,XY}$	-1.62e-07 ± 1.86e-06	-2.39e-07 ± 1.89e-06
$d_u^{X,Y}$	1.05e-07 ± 9.27e-07	1.41e-07 ± 9.40e-07
$c_u^{X,Z}$	-2.31e-06 ± 2.86e-06	-2.32e-06 ± 2.81e-06
$c_u^{Y,Z}$	-1.94e-06 ± 2.69e-06	-1.94e-06 ± 2.79e-06
$c_u^{XY,XY}$	-1.26e-07 ± 1.45e-06	-1.86e-07 ± 1.47e-06
$c_u^{X,Y}$	8.17e-08 ± 7.20e-07	1.09e-07 ± 7.31e-07
$c_d^{X,Z}$	-1.03e-04 ± 1.27e-04	-1.03e-04 ± 1.25e-04
$c_d^{Y,Z}$	-8.63e-05 ± 1.20e-04	-8.66e-05 ± 1.24e-04
$c_d^{XY,XY}$	-5.59e-06 ± 6.45e-05	-8.27e-06 ± 6.54e-05
$c_d^{X,Y}$	3.64e-06 ± 3.21e-05	4.86e-06 ± 3.25e-05

Table 8: Results from small and large number of scrambled datasets. Results are obtained using combined Run 2.

To compete our studies, we split 10000 samples into 10 batches. Each batche contain 1000 scrabled datasets. In this study we comared results of 10 batches. Table of results of 10 batches is shown in the appendix F.3. The results in each batch is the average results obtained from 1000 fit results in corresponding batch. Figure 35 show the results in each batch which is normalized to the result of first batch. The gray band in the plot is the uncertainty band of the first batch. From the plot we can see that results of 10 batches are consistent.

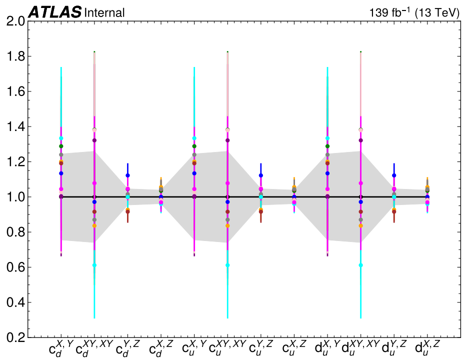

Figure 35: The results in each batchs

5.4 Investigate different probing periods

We checked scrambled datasets in various probing periods. As the results shown in Figure 36, we didn't see any strange behavior in various periods. The 7 hours period showed slitly larger (compared to other periods) deviations from zeros. Therefore, we checked 6.87 and 7.13 hours period to check if it is a coinsidence. The results of 6.87, 7, 7.13 hours periods show that the slitly larger deviations is just a coinsidence (not related to a specific period).

Figure 37 shows Spreads VS average stats in 1, 6.87, 7, 7.13, 24 hours and sidereal periods. Detail of Spreads and Average Stats are described in Section 5.2.2.

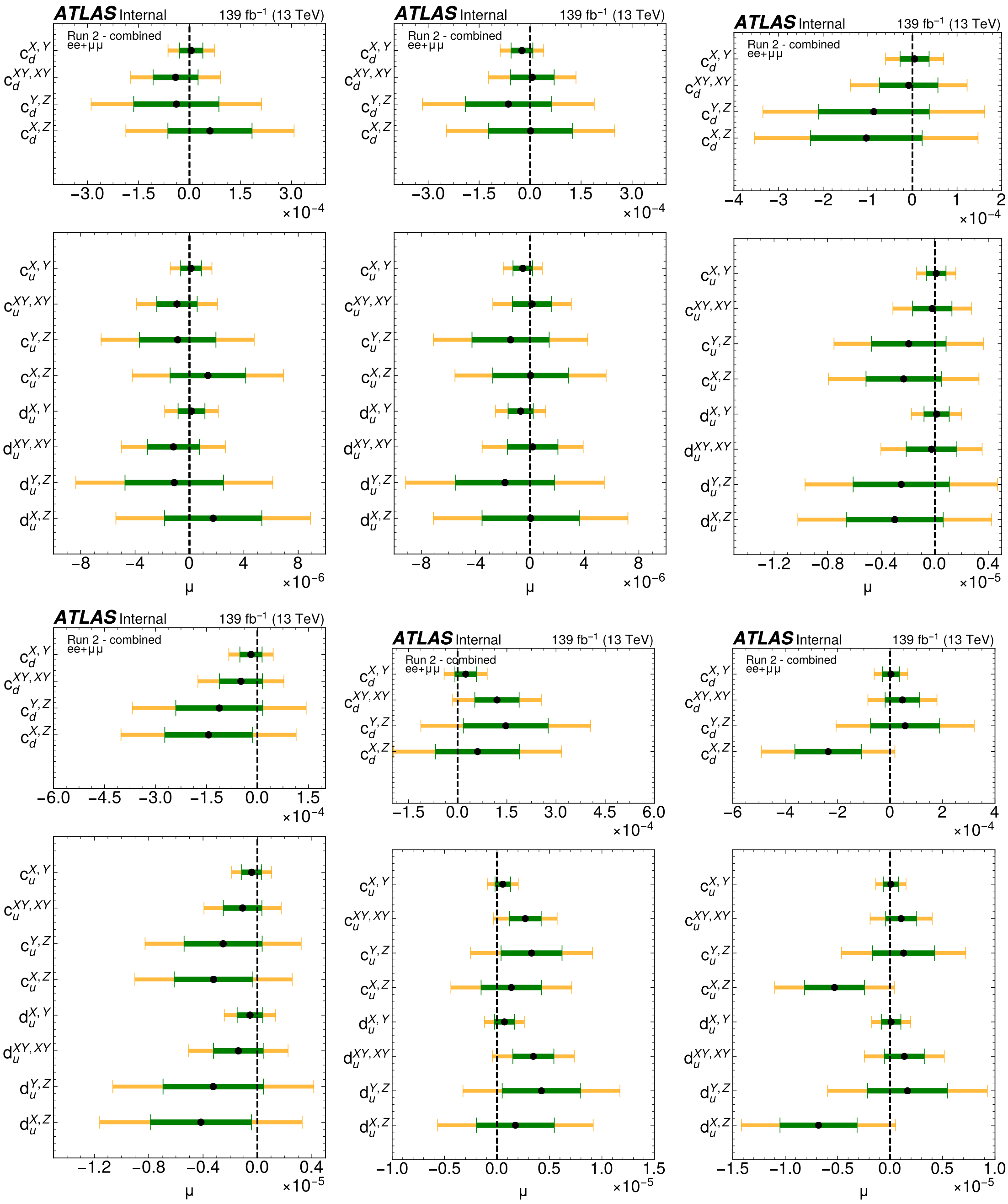

Figure 36: Results of scrambled datasets in 1 hour (top left), 24 hours (top middle), sidereal (top right), 6.87 hours (bottom left), 7 hours (bottom middle), and 7.13 hours (bottom right) periods. Results are the average of 1000 fit results (the sidereal period used 10000 fit results).

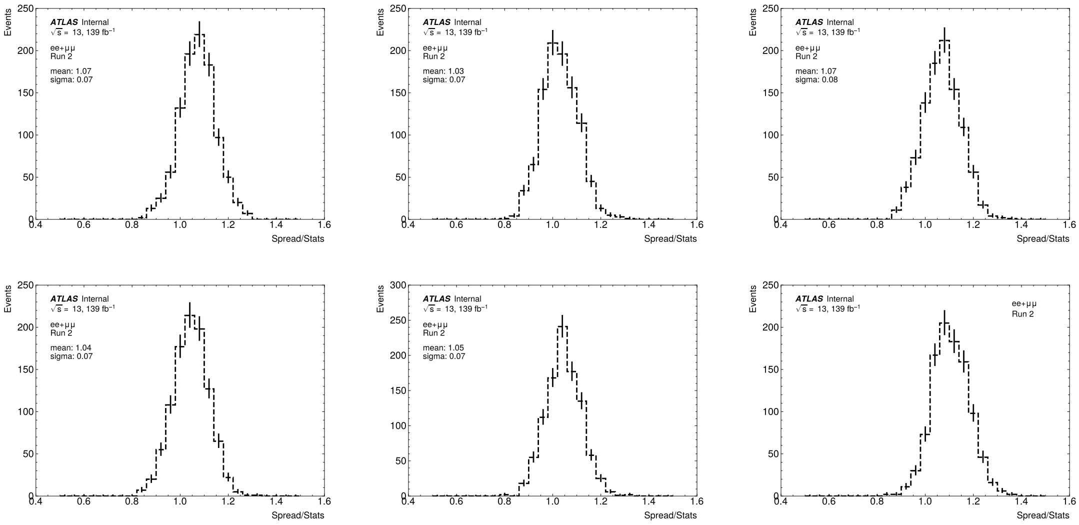

Figure 37: Left to right: 1, 6.87 and 7 hours perios (top plots); 7.13, 24 hours and sidereal perios (bottom plots). Spreads VS average stats. Plots show average results of 1000 scrambled datasets. An average statistic is the average statistical uncertainty in each scrambled dataset, and the spread of a dataset is an average of absolute residual. Spread is about 10-20% larger in most of the scrambled datasets. This indicates there is a small systematic effect or the way we are calculating the ZC error in the double ratio. This could be the reason why the average values shifted from zero

6 Time-dependent systematic uncertainties

6.1 Overview of systematic

Sources of systematic effects can be considered as nominal analysis in ATLAS, but following items are canceled out by taking the double ratio.

a Theory related items, i.e. cross section of all SM processes, QCD scale uncertainty, any kinematic mismodling including higher order calculation in the Matrix element or parton shawer, variation of parton distribution functions.

b Most of experimental items, i.e., object reconstruction and identification efficiencies, total luminosity, normalization effects.

c year by year normnalization effect, i.e., LUCID luminosity measurement variation.

d mis-measurement on Z counting or LUCID luminosity.

e Time dependent effects, i.e. $\langle\mu\rangle$ effect.

Only time dependent effects remained as systematic uncertainty, therefore items "a" and "b" are not the source of systematic on the LIV search.

Concerning to item "c", there are variation in the LUCID luminosity measurement, which effect are corrected by van-deer-mir scan. If we sum up all years (2015 to 2018) into one double ratio, this would introduce additional systematic effect, but as we discussed in Section 5, we build the double ratio each year separately and combine in the final fit. Therefore, this effect is not necessary to take into account.

Nevertheless, item "d", mis-measurement on the input of the double ratio could introduce the systematic effect. In order to exclude mis-measurement, LBs have large deviation, i.e. integrated luminosity would be 0 in one of input (either LUCID, electron channel or muon channel of Z counting luminosity) are excluded from the input LB as additional data quality requirement.

The item "e" is the systematic source we need to take into account. Actully, the Z counting luminosity measurement does takes into account variation of the trigger and the identification efficiencies for each LB. Therefore, even efficiency variation (in item "c") is NOT the dominat source of systematic effect for this analysis. The statistics uncertainty of the efficiency correction in each LB is taken into account and propagaed to the error bar of each phase bin of the double ratio.

There is a correction factor as function of $\langle\mu\rangle$ in the Z counting measurement, which is FMC correction factor. The $\langle\mu\rangle$ dependence of $F^{\mathrm{MC}}_{Z\to\ell^+\ell^-}$ is fitted with a second-order polynomial per decay channel and per data-taking period, and the fit results are shown in Figure 38.

The $Z \to e^+e^-$ correction factor is ¿10% below unity for all $\langle\mu\rangle$, implying that there are additional efficiency effects beyond those captured in the tag-and-probe procedure. The variation between $\langle\mu\rangle = 20$ and $\langle\mu\rangle = 50$ is around 2%, suggesting that pileup-dependent effects are mild.

The fits of $F^{\mathrm{MC}}_{Z\to\ell^+\ell^-}$, as shown in Figure 38, are used in the determination of the Z-counting based luminosity per data-taking period and as a function of the pileup parameter $\langle\mu\rangle$. The statistical uncertainties of the fits are small and are neglected.

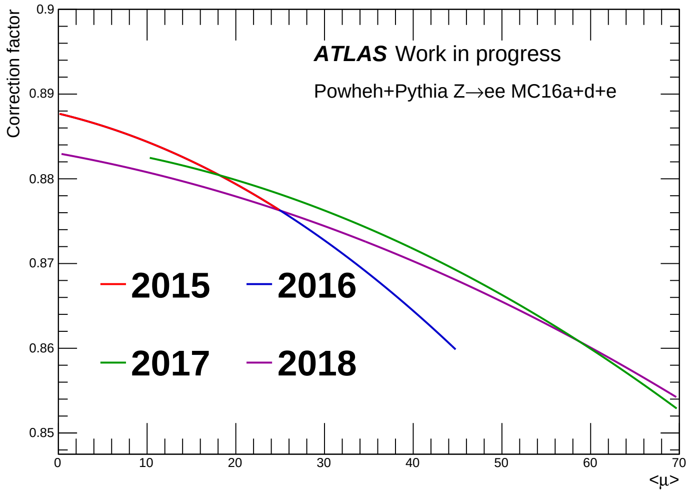

Figure 38: $Z \to e^+e^-$ correction factors used for each year of data produced using the dedicated Monte Carlo campaigns. The lines show second-order polynomial fits to the correction factor for the corresponding $\langle\mu\rangle$ range per year.

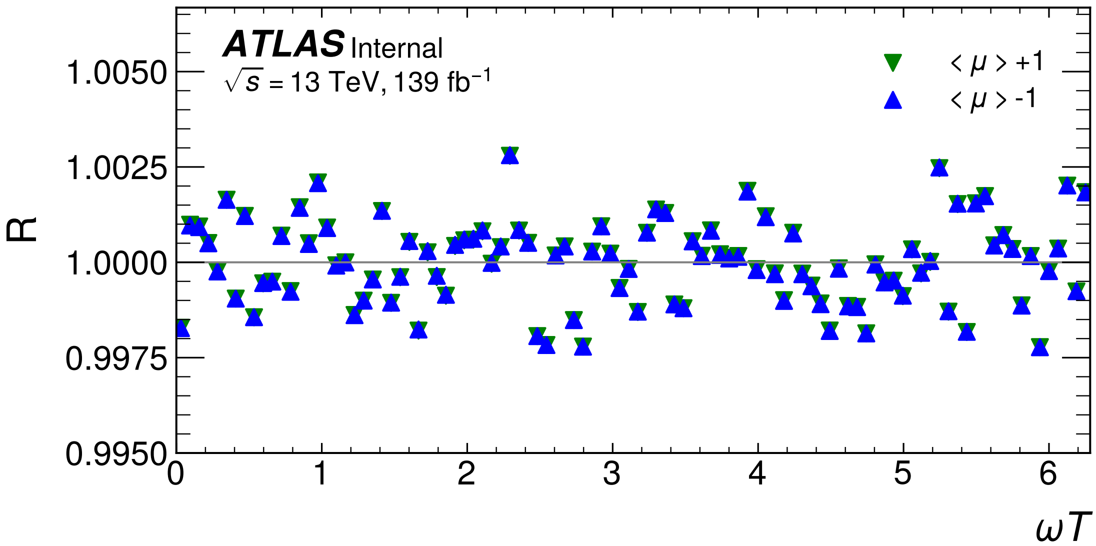

Figure 39: Comparison of the toy dataset with average pileup $< \mu >$ increased and decreased by 1. The toy dataset corresponds to four years of Run 2 luminosity is used. The X-axis is the sidereal phase, ωT.

The size of systematic on the FMC corrction is estimated by shifting $\langle\mu\rangle$ by ± 1 to apply correction. This is done by the TOY study described in Section. . Figure 39 shows the effect in the double ratio in a Toy experiment. Result with 1000 Toy experiments are summarized in Figure 40. We observed this effect is small with respect to statistical uncertainty.

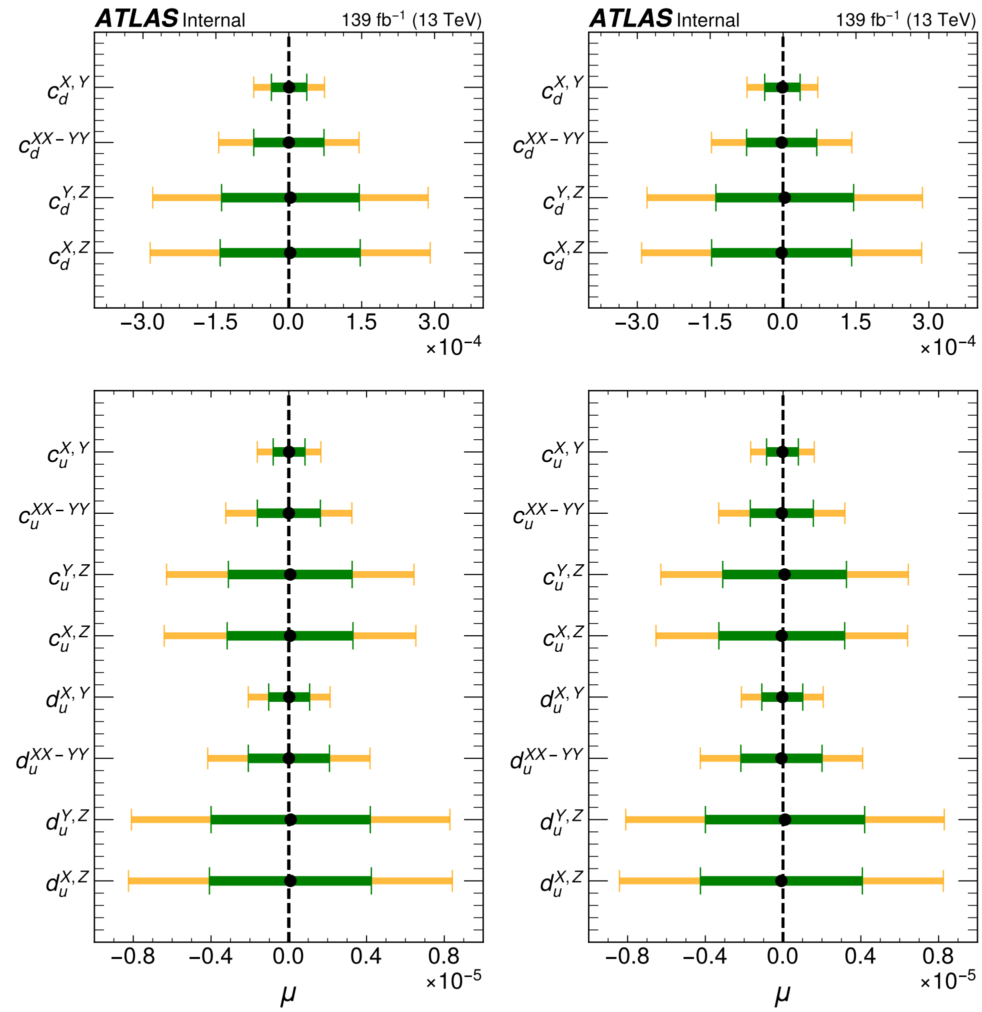

Figure 40: Comparison of the fit results with average pileup $<\mu>$ increased and decreased by 1; left (+1) and right (-1). Results are obtained from 1000 toy datasets corresponding to four years of Run 2 luminosity. The X-axis is the sidereal phase, ωT.

6.2 Residual systematic effects

In Section B we tested our scrambled method on our toy samples. The toys consist of SM-like toys (No signals), and signal-injected toys. In sidereal phase, the MS-like toys show data points are distributed according to only statistical fluctuation. The signal injected toys show some oscillations correspond to injected signal shape.

The figure 51 shows a comparison of a SM-like toy before (left) and and after scrambled (right). We can see that the scrambled toy is just like an original one, it's also SM-like toy. In the same way we tested the scrambling on 1000 SM-like toys. The figure 52 shows average of 1000 fitted results and their one/two sigma standard deviation. The left plot shows the results from SM-like toys before scrambling, and the right plot shows the results after apply scrambling. As we expected, in both case, average fit result on each coefficeint is in a good agreement with zero. Because, in SM-like toy datasets we have only simulated statistical uncertainties. Therefore, we expect no bias (average fit result consistent with zero).

In signal injected case, scrambling can completely washed the injected singal. The figure 55 show the compareson of before and after scrambling of an siganl injected toy sample. In the left plot we clearly see the oscillation pateren, but in the right plot after scrambling we didn't observe presence of any signals. After scramibing the fitted signal strength down to $-4.4E^{-7} \pm 3.8E^{-6}$; it was $5.3E^{-5} \pm 3.8E^{-6}$ before scrambling. This study concluded that event if a singal is present the scrambling can wash it. Also, if a dataset has only statistical uncertaity (e.g other systematics effects are negligible), we can still expect that the fitted signal strength after scrambling will be consistent with zere (e.g no bias will be seen).

However, when we check the data using scrambling (partially unblind), we observe that average values of fit results in some singal strengths (coefficents) deviate from zero (bias). This can be clearly seen in figure 69 where Run 2 - combined fit results are presented. In the plot, each black dot is an average fit results of 1000 scrabled datasets. In a few coefficents we observed the average fit results deviate from zero but still within one sigma range. The exact numbers can be read from the table 7. This kind of deviation indicate that there is a systematic effact other than statisticle fluctuation. Because, as we expected from our toy studies we discussed before, if the statistical fluctuation is the only source of uncertainty then we should observe results from scrambled dataset also consistent with zero.

The deviation from zero in each coefficeint shown in figure 69 can be seen as size of spurious signal uncertainty for corresponding coefficent. The spurious signal is models as Gaussian distribution centered at zero and it standard deviation is absolute values of average fit results (size of the deviation from zero) shown in figure 69. The size of spurious signals for Run 2 fit and Run 2 - combined fit are given in table 9.

	Run2	Run2-Comb. Fit
$d_u^{X,Z}$	-3.23e-06	-2.99e-06
$d_u^{Y,Z}$	-1.69e-06	-2.5e-06
$d_u^{XY,XY}$	8.03e-07	-2.39e-07
$d_u^{X,Y}$	3.21e-07	1.41e-07
$c_u^{X,Z}$	-2.51e-06	-2.32e-06
$c_u^{Y,Z}$	-1.32e-06	-1.94e-06
$c_u^{XY,XY}$	6.24e-07	-1.86e-07
$c_u^{X,Y}$	2.50e-07	1.09e-07
$c_d^{X,Z}$	-1.12e-04	-1.03e-04
$c_d^{Y,Z}$	-5.86e-05	-8.66e-05
$c_d^{XY,XY}$	2.78e-05	-8.27e-06
$c_d^{X,Y}$	1.11e-05	4.86e-06

Table 9: Size of spurious signal uncertainties in $ee + \mu\mu$ channel, for Run 2 fit (first column) and Run 2 - combined fit (second column, obtained from 10K samples).

6.3 Statistical analyses including systematics

6.3.1 Profiled χ^2 with the spurious signal uncertainty

We can include a spurious sisgnal uncertainty, s to the $\chi^2 = \sum \left(\frac{y_i - f_i(\mu)}{\sigma_i}\right)^2$ fit by replacing $f(\mu)$ with $F(\mu, \beta * s)$. Where β is a nuissance parameter coinstrained by a normal Gaussian $N(0,1)$, and s is the size of spurious signal uncertainty. We consider spurious singal uncertainty is symmetric. For the siganl shpape $d_u^{X,Z}$ we can write $F(\mu, \beta * s)$ as $F_{d,u}^{X,Z}(\mu, \beta * s_u^{X,Z}) = 1 + (\mu + \beta * s) * (6.28069 * cos(x) - 41.0569 * sin(x))$. Then the total χ^2 is $\chi^2 = \sum \left(\frac{y_i - F_{u,i}^{X,Z}(\mu, \beta * s)}{\sigma_i} \right)^2 + \beta^2$. Where i is bin index, the β^2 is due to the normal Gaussian constaint of β.

The figure 72 shows fit results from profiled χ^2 with a spurious signal uncertainty $s_u^{X,Z} = 3.01e - 06$. In the left plot, we fit a scrabked dataset. In the right plot, we fit a singal injected toy dataset. In both case, including spurious signal uncertainty enlargened the final fit uncertainty by 29% (28% for the singal injected toy dataset). The figure 42 show the signal injected toy dataset. The injected singal amplitude is $\sigma_{LIV}/\sigma_{SM} = 0.0009$.

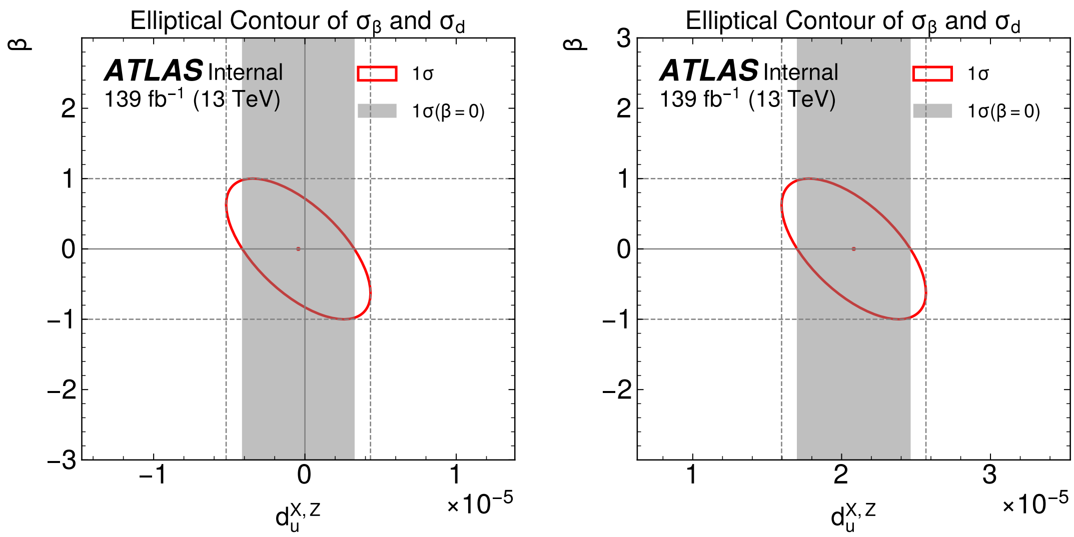

Figure 41: Combined fit results of profiled χ^2 fit, which includes the spurious signal uncertainty as a nuisance parameter β. The left plot is the result from one of the scrambled datasets. The right plot is a signal-injected toy dataset with strength $\sigma_{LIV}/\sigma_{SM} = 0.0009$; the signal model is $d_u^{X,Z}$. The grey band is the 68% uncertainty band without spurious signal. The fit uncertainty is 29%(28% for the toy) larger when we add the spurious signal uncertainty to the fit.

6.4 Summary of results

6.5 Scan periods with spurious signal uncertainties included

The significance of injected signals is studied as a function of probing periods using signal-injected toys. We performed this study using four types of SME signal shapes, and the injected maximum signal amplitude for each type of signal is 5%. The injected SME signal shapes are $d_u^{X,Z}$, $d_u^{Y,Z}$, d_u^{XX-YY}, $d_u^{X,Y}$.

The top left plot in Figure 43 shows the canned significance without systematic included in the fit. As we expected, a huge spike appeared when the probing period was equal to the sidereal period (23:56:04) because the signal is injected in the sidereal period. The top right plot shows the significance with systematic included in the fit. The significance is reduced when we add systematics. Profiled χ^2 fit with systematics does not change the value of parameters obtained from the fit, as shown in the bottom left plot.

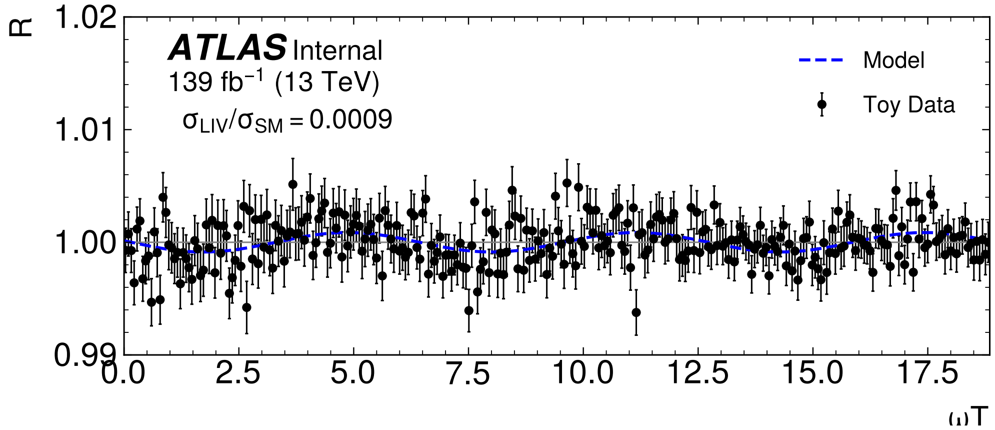

Figure 42: The signal-injected combined toy dataset with signal amplitude σ_{LIV}/σ_{SM} = 0.0009 and $d_u^{X,Z}$ model (post-fit). The blue line is the signal shape after the combined fit. The error bars on the data points are statistical errors.

The main contribution of the systematics is in the fit uncertainties. The bottom right plot shows the ratio of the fit uncertainties with/without systematic inclusion in the fit. The fit uncertainties are increased; for example, 12.5% increased in $d_u^{X,Z}$ signal shape. The impact of a systematic mainly depends on the input size. The corresponding spurious signal systematic are listed in the table 10

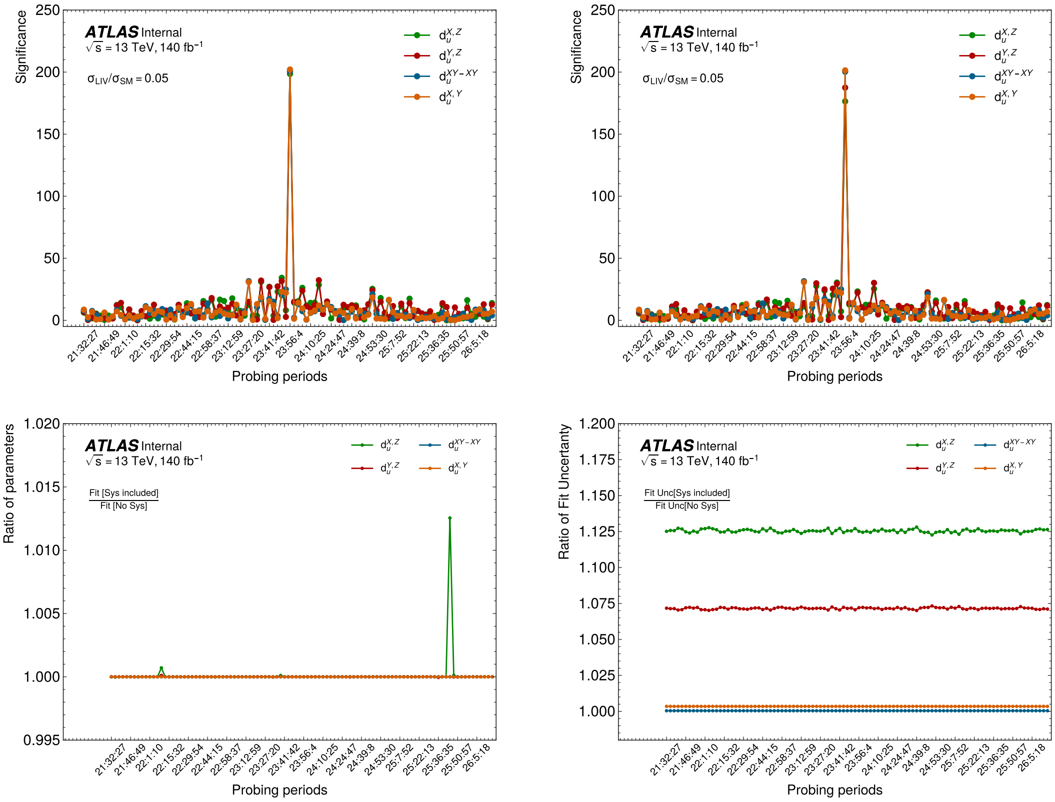

Figure 43: Scan probing periods on the signal-injected toys. Four types of SME signal shapes are injected at maximum amplitude 5% level, as shown in the plots. The top left plot shows significance without systematic is included in the fit. The Top right plot shows significance with systematic included in the fit. The bottom left plot shows the ratio of fitted parameters with/without systematic included in the fit. The bottom right plot shows the ratio of the fit uncertainties with/without systematic inclusion in the fit.

7 Model independent limits

7.1 Model-independent signal shapes

The SME predicts nth-order sinusoidal temporal oscillations of the forms $\cos(n\omega_\oplus T_\oplus)$ and $\sin(n\omega_\oplus T_\oplus)$. Recall from Sec. 2.2 that a dimension d SME operator is multiplied by a coefficient of dimension $4-d$. A given coefficient induces a tower of n-order sidereal harmonics $n\omega_\oplus T_\oplus$ the number of Lorentz indices of an SME coefficient. For example, revisiting again $\mathcal{L}_b$ from Eq. (2), the SME coefficient b_μ has one Lorentz index. We showed in Eq. (23) that only $n = 1$ and $\cos(\omega_\oplus T_\oplus), \sin(\omega_\oplus T_\oplus)$ appear. For the c- and d-type coefficients, with two Lorentz indices, we expect $n = 1$ and $n = 2$ harmonics. This is observed in the decomposition of c_f^{33} in Eq. (25). Note that unique linear combinations of SME coefficients (prefactors) are associated to a single harmonic of order n [**Kostelecky:2009zp**]. There are no "cross terms" involving, for example, a prefactor with time-dependent behavior controlled by both $\cos(\omega_\oplus T_\oplus)$ and $\cos(2\omega_\oplus T_\oplus)$. This is a generic result applying for all n.

For the $a^{(5)}$- and $b^{(5)}$-type coefficients one finds $n = 1, 2$ and 3 harmonics in general. The $a^{(5)}$ predictions for the Drell-Yan cross section evaluated in the ATLAS detector frame are found to be absent $3\omega_\oplus$ signatures. This contrasts with the cross section for deep inelastic scattering, where all harmonics appear. An intuitive explanation for this absence is that Drell-Yan is a more colliear or "symmetric" process than deep inelastic scattering. As a result, fewer coefficient components are present in the detector-frame cross section. The ones that do appear have laboratory indices which, upon replacing with the SCF coefficients, exhibit only first and second harmonics. This is likely to be a special case rather than a generic expectation for Drell-Yan. A significant number of nonminimal SME coefficients of low mass dimension (e.g. $d = 5, 6$) have distinct symmetry properties relative to those exhibited by the $a^{(5)}$-type coefficients. Anticipating that phenomenological predictions with higher than $2\omega_\oplus$ harmonics will appear in future provides motivation for developing the generic framework outlined here.

In this section, we outline a new method for capturing higher-order harmonics that encompass any new physics associated with sidereal oscillations. While the SME is a model-independent effective field theory, it makes specific predictions about the prefactors multiplying the sidereal functions. We generalize these types of signal shapes by leaving the prefactors as free fit parameters.

In particular, we consider three distinct generic model signal shapes

$$f_1(\omega_\oplus T_\oplus | p_1, p_2) = 1 + p_1 \cos(\omega_\oplus T_\oplus) + p_2 \sin(\omega_\oplus T_\oplus), \tag{52a}$$

$$f_2(2\omega_\oplus T_\oplus | p_1, p_2) = 1 + p_1 \cos(2\omega_\oplus T_\oplus) + p_2 \sin(2\omega_\oplus T_\oplus), \tag{52b}$$

$$f_3(3\omega_\oplus T_\oplus | p_1, p_2) = 1 + p_1 \cos(3\omega_\oplus T_\oplus) + p_2 \sin(3\omega_\oplus T_\oplus), \tag{52c}$$

and treat each function f_1, f_2 and f_3 with prefactor parameters p_1 and p_2 as mutually exclusive. As we limit attention to a finite number of free parameters, and make no assumption about the physics generating them, we refer to them as "generic models". Parameters p_1 and p_2 are to be determined from χ^2 fits to data. The results of these fits are the parameter values and a covariance matrix. The corresponding p_1 and p_2 confidence intervals are obtained from the covariance matrix.

7.2 Spurious signal uncertainties on p_1 and p_2

Figure 44 shows the estimated set of parameters, p_1 and p_2, corresponding to generic models with different frequencies, $\omega_\oplus T_\oplus$, $2\omega_\oplus T_\oplus$, $3\omega_\oplus T_\oplus$. These results are obtained from 10000 scrambled datasets by fitting

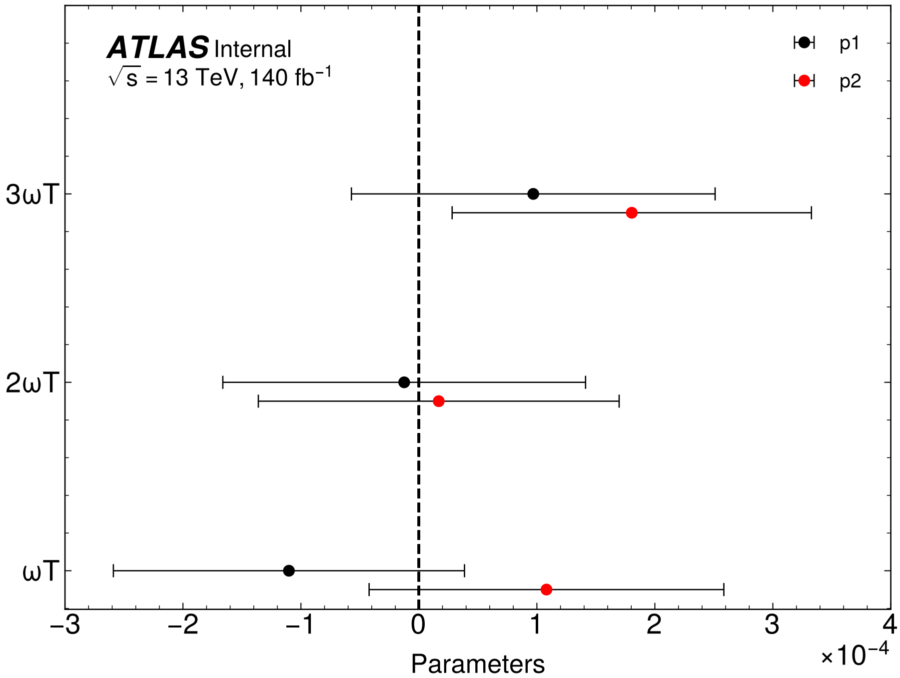

Figure 44: Estimated parameters and their uncertainties of generic signal shapes; obtained by fitting 10000 scrambled datasets. The plot shows the mean and one standard deviation of 10000 fit results. Note $\omega T = \omega_\oplus T_\oplus$ on the y-axis.

each dataset with the generic signal shapes. The three signal shapes are mutually exclusive, so a fit includes one signal shape at a time. The corresponding numerical values of the Fig. 44 are also given in the Tab. 10.

In the ideal case, we could expect the central value of the parameters to be zero because we took an average of the 10000 fit results from the scrambled datasets. In reality, the average values are not zero, as we have seen in Fig. 44. Therefore, we can take these non-zero values as spurious signal uncertainties and include them in the unblinded results.

	Generic models		
Pars	$\omega_\oplus T_\oplus$	$2\omega_\oplus T_\oplus$	$2\omega_\oplus T_\oplus$
p_1	-1.10e-04 ± 1.49e-04	-1.25e-05 ± 1.54e-04	9.69e-05 ± 1.54e-04
p_2	1.08e-04 ± 1.50e-04	1.68e-05 ± 1.53e-04	1.80e-04 ± 1.52e-04

	Generic models		
Pars	$\omega_\oplus T_\oplus$	$2\omega_\oplus T_\oplus$	$2\omega_\oplus T_\oplus$
p_1	$(-1.23 \pm 1.49) \times 10^{-4}$	$(-0.25 \pm 1.54) \times 10^{-4}$	$(0.84 \pm 1.54) \times 10^{-4}$
p_2	$(1.08 \pm 1.50) \times 10^{-4}$	$(0.16 \pm 1.53) \times 10^{-4}$	$(1.79 \pm 1.52) \times 10^{-4}$
$\mathrm{corr}(p_1, p_2)$	0.0014	−0.00086	0.0071

Table 10: Estimated parameters and their uncertainties of generic signal shapes; obtained by fitting 10000 scrambled datasets. The means and one standard deviation of the 10000 fit results.

7.3 Profiled χ^2 fit with generic models

Here we introduce, s_1 and s_2 as size of spurious signal uncertainties corresponds to the parameters p_1 and p_2. The values of s_1 and s_2 are the absolute values of p_1 and p_2 in the Table 10, as these values are obtained from scrambled datasets (we take these samples as signal-free datasets).

To add spurious signal uncertainties to the χ^2 fit, we rewrite the Eqs. (52) as

$$F(\omega_\oplus T_\oplus,\ p_1,\ p_2|\beta_1,\ \beta_2) = f(\omega_\oplus T_\oplus,\ p_1 + \beta_1 * s_1,\ p_2 + \beta_1 * s_2), \tag{53}$$

where β_1 and β_2 nuisance parameters constrained as they follow normal Gaussian centered at zero, and the standard deviation is one. s_1 and s_2 are constant in the equations. The profiled χ^2 fit is constructed using the function $F(\omega_\oplus T_\oplus,\ p_1,\ p_2|\beta_1,\ \beta_2)$.

7.4 Application of generic results

In this subsection we show that the model-independent constraints in Eqs. (52a–52c), together with the theory predictions given in Sec. 2.11 (see Eqs. (33-44)) allow to reproduce the results of the one-parameter fits presented in Table 8.

The starting point are the model-independent fits for the parameters (p_1, p_2) summarized in Table 10. Adopting a simple frequentist χ^2 fit based on the theoretical expressions in Sec. 2.11 we obtain the results in Table 11, where, for ease of comparison, we show direct and indirect determinations side by side (the former are taken directly from Table 8).

We see that the two procedures yield perfectly compatible results. Figure 45 shows the direct and indirect results in a plot for visualization.

Parameters	Direct	Indirect
$d_u^{X,Z}$	$-(2.99 \pm 3.62) \times 10^{-6}$	$-(3.02 \pm 3.62) \times 10^{-6}$
$d_u^{Y,Z}$	$-(2.50 \pm 3.59) \times 10^{-6}$	$-(2.55 \pm 3.59) \times 10^{-6}$
$d_u^{XY,XY}$	$-(0.24 \pm 1.89) \times 10^{-6}$	$-(0.24 \pm 1.89) \times 10^{-6}$
$d_u^{X,Y}$	$(1.41 \pm 9.40) \times 10^{-7}$	$(1.40 \pm 9.41) \times 10^{-7}$
$c_u^{X,Z}$	$-(2.32 \pm 2.81) \times 10^{-6}$	$-(2.35 \pm 2.81) \times 10^{-6}$
$c_u^{Y,Z}$	$-(1.94 \pm 2.79) \times 10^{-6}$	$-(1.98 \pm 2.79) \times 10^{-6}$
$c_u^{XY,XY}$	$-(0.19 \pm 1.47) \times 10^{-6}$	$-(0.19 \pm 1.47) \times 10^{-6}$
$c_u^{X,Y}$	$(1.09 \pm 7.31) \times 10^{-7}$	$(1.09 \pm 7.31) \times 10^{-7}$
$c_d^{X,Z}$	$-(1.03 \pm 1.25) \times 10^{-4}$	$-(1.05 \pm 1.25) \times 10^{-4}$
$c_d^{Y,Z}$	$-(0.87 \pm 1.24) \times 10^{-4}$	$-(0.82 \pm 1.24) \times 10^{-4}$
$c_d^{XY,XY}$	$-(0.83 \pm 6.54) \times 10^{-5}$	$-(0.82 \pm 6.54) \times 10^{-5}$
$c_d^{X,Y}$	$(0.49 \pm 3.25) \times 10^{-5}$	$(0.49 \pm 3.26) \times 10^{-5}$

Table 11: Comparison between the direct and indirect (using the results of the two-parameter fits) extraction of bounds on the c- and d-coefficients.

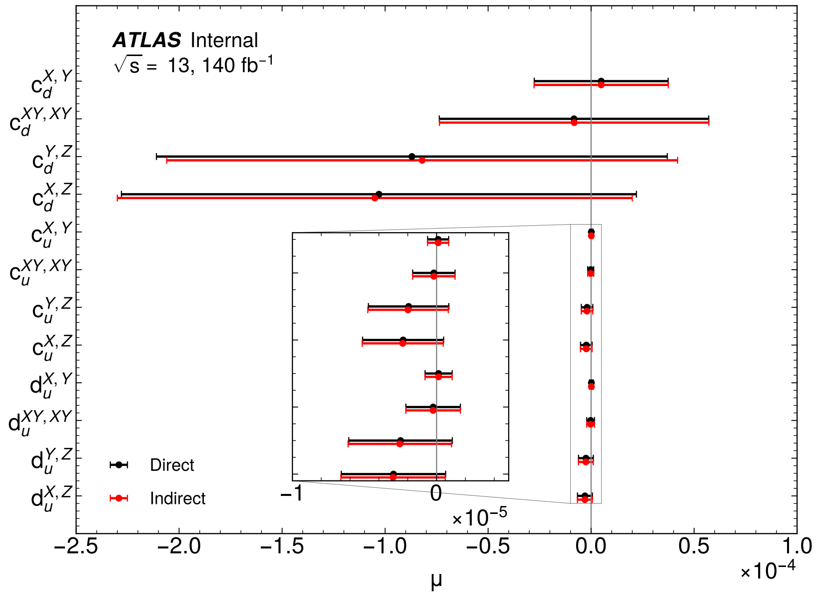

Figure 45: Comparison of the direct (fit results of SME predictions) and indirect (obtained using results of generic models). The indirect results are consistent with the direct results. Uncertainties correspond to 68% (one-sigma) confidence intervals.

7.5 Disentangling coefficient degeneracies

We briefly discuss the possibility of detecting a degenerate signal, meaning one where at least two SME coefficients could contribute to the signal. Recall Eqs. (33) and, for definiteness, consider a scenario where the signal matches "opposite phase" single-harmonic oscillations in the u-quark sector. For the minimal c- and d-type coefficients there are two possibilities:

$$\begin{aligned} & d_u^{XZ}\,[6.28069\cos(\omega_\oplus T_\oplus) - 41.0569\sin(\omega_\oplus T_\oplus)]\,, \\ & c_u^{XZ}\,[8.084\cos(\omega_\oplus T_\oplus) - 52.8451\sin(\omega_\oplus T_\oplus)] \end{aligned} \tag{54}$$

The small difference in magnitudes is a factor of 1.287. Both coefficients can contribute to the signal, but a single measurement at $Q = m_Z$ can not distinguish between the possibilities: a d_u dominated one, a c_u dominated one, or some combination of the two. A way to disentangle the signal is to perform another measurement at a different Q value. Referring to Fig. 1, one observes that the c_u-type coefficient curve in red and the d_u-type coefficient curve in green display distinct patterns of sensitivity as Q varies. The sensitivity to c_u increases from around 70 GeV and below, whereas d_u decreases from m_Z down to about 40 GeV. Focusing on the 40-45 GeV bin, the sensitivity to c_u decreases by about one order of magnitude, whereas d_u about three orders of magnitude (both relative to $Q = m_Z$). If the signal strengths measured at $Q = m_Z$ and $Q = 40 - 45$ GeV are found to differ by roughly an order of magnitude, this would disfavor the d_u-dominated scenario. This would suggest that Lorentz violation couples more strongly to the $L + R$ chiral combination, which is the combination that leaves the u-quark polarization properties unaffected.

8 Unblinded data studies

8.1 General method

8.1.1 95% C.L. using scrambled data

8.1.2 Discovery limits

8.2 Data results - to come -

9 Result

Place your results here.

10 Interpretation

You can find some text snippets that can be used in papers in `latex/atlassnippets.sty`. To use them, provide the `snippets` option to `atlasphysics`.

11 Conclusion

Place your conclusion here.

The supporting notes for the analysis should also contain a list of contributors. This information should usually be included in `mydocument-metadata.tex`. The list should be printed either here or before the Table of Contents.

List of contributions

Appendices

In an ATLAS note, use the appendices to include all the technical details of your work that are relevant for the ATLAS Collaboration only (e.g. dataset details, software release used). This information should be printed after the Bibliography.

A Theory details

A.1 Additional details regarding quark-sector modifications

The minimal SME is grounded in the generic principles of effective field theory. It retains all of the SM properties, including the $SU(3)\times SU(2)\times U(1)$ gauge structure $SU(2)\times U(1)$ breaking, renormalizability, microcausality, spin-statistics, canonical quantization, energy-momentum conservation, and observer Lorentz invariance. Its atypical properties are violations of particle Lorentz invariance and possibly CPT invariance [11]. The nonminimal SME relaxes the condition of renormalizability. As an effective field theory, the SME Lagrangian necessarily includes the SM Lagrangian, see Eq. (3). The additional Lorentz- and CPT-violating terms are treated as perturbations, which is experimentally justified. The SME is as a conservative extension of SM and can be thought of as a "test framework" for Lorentz and CPT violation.

Our current focus is capturing the dominant signatures in the quark sector of the minimal SME. To begin, consider a single-fermion field ψ. Gauge interactions of ψ include coupling constants. As we work in perturbation theory, any Lorentz-violating effects tied to coupling constants are expected to be suppressed relative to modified free-propagation effects, since the latter are absent of coupling constants. Loop contributions involve additional powers of coupling constants and are further suppressed. In this setting, the dominant Lorentz- and CPT-violating effects on ψ involve a generalized Dirac equation with several $d = 3$ and 4 operators [**Colladay:2001wk**]:

$$\mathcal{L} = \tfrac{1}{2}\bar{\psi} i \Gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi - \bar{\psi} M \psi, \tag{55}$$

where

$$\Gamma^{\mu} = \gamma^{\mu} + c^{\nu\mu}\gamma_{\nu} + d^{\nu\mu}\gamma_5\gamma_{\nu} + e^{\mu} + i f^{\mu}\gamma_5 + \tfrac{1}{2} g^{\alpha\beta\mu}\sigma_{\alpha\beta}, \tag{56}$$

$$M = a_{\mu}\gamma^{\mu} + b_{\mu}\gamma_5\gamma^{\mu} + \tfrac{1}{2} H_{\mu\nu}\sigma^{\mu\nu}. \tag{57}$$

The $-\bar{\psi}M\psi$ terms change sign under a CPT transformation (they are CPT odd). The operators are mass dimension $d =$ so the associated SME coefficients a_{μ}, $b_{\mu}\gamma_5\gamma^{\mu}$ and $H^{\mu\nu}$ have units of GeV^{-1}. As such, in a dimensionless observable these terms scale like (SME coefficient)/energy and are dominant in low-energy processes. Effects associated to these coefficients are expected to be small compared to those contained in Γ^{μ} and can be ignored here. Within Γ^{μ}, the coefficients e^{μ}, f^{μ}, $g^{\alpha\beta\mu}$ are inconsistent with the SM gauge structure and renormalizability [**ck98**], and can also be ignored here. This leaves the dimensionless and CPT-even c- and d-type discussed in the main text. As a brief aside, though these coefficients are CPT even, they introduce unconventional sources of other discrete-symmetry violations as shown in Tab. 12.

[11] Roughly half of the SME terms also violate CPT invariance.

Table 12: Coefficient properties under discrete symmetries where $(-1)^\mu = 1$ for $\mu = 0$ and $(-1)^\mu \equiv -1$ for $\mu = 1, 2, 3$.

Coefficient	C	P	T	CP	CT	PT	CPT
c_{00}, c_{jk}	+	+	+	+	+	+	+
c_{0j}, c_{j0}	+	−	−	−	−	+	+
d_{00}, d_{jk}	−	−	+	+	−	−	+
d_{0j}, d_{j0}	−	+	−	−	+	−	+

Considering now ψ as a quark field in terms of $SU(2)$ doublets and singlets, the $SU(2) \times U(1)$ invariant generalization can be written as

$$\mathcal{L} = \tfrac{1}{2} i c^{\mu\nu}_{Q_i} \overline{Q}_i \gamma_\mu \overset{\leftrightarrow}{D}_\nu Q_i + \tfrac{1}{2} i c^{\mu\nu}_{U_i} \overline{U}_i \gamma_\mu \overset{\leftrightarrow}{D}_\nu U_i + \tfrac{1}{2} i c^{\mu\nu}_{D_i} \overline{D}_i \gamma_\mu \overset{\leftrightarrow}{D}_\nu D_i \ , \tag{58}$$

where D_ν is the conventional gauge-covariant derivative. The fields have the usual definitions

$$Q_i = \begin{pmatrix} u_i \\ d_i \end{pmatrix}_L , \quad U_i = (u_i)_R \,, \quad D_i = (d_i)_R \,. \tag{59}$$

The index $i = 1, 2, 3$ denotes flavors $u_i = (u, c, t)$ and $d_i = (d, s, b)$ and the SME coefficients are given by $c^{\mu\nu}_{Q_i}$, $c^{\mu\nu}_{U_i}$, and $c^{\mu\nu}_{D_i}$.

Field redefinitions imply the antisymmetric parts of the coefficients are unobservable at first order in Lorentz violation [**Colladay:1998fq**, **Fittante:2012ua**, **Colladay:2002eh**]. The trace contributions are also unphysical because they can be absorbed into a rescaling of the fields. They are set to zero without loss of generality. This leaves nine observable components per coefficient type Q_i, U_i, D_i. It is customary and useful to define combinations of these coefficients:

$$\begin{aligned} c^{\mu\nu}_{u_i} &= (c^{\mu\nu}_{Q_i} + c^{\mu\nu}_{U_i})/2, \quad & c^{\mu\nu}_{d_i} &= (c^{\mu\nu}_{Q_i} + c^{\mu\nu}_{D_i})/2 \,, \\ d^{\mu\nu}_{u_i} &= (c^{\mu\nu}_{Q_i} - c^{\mu\nu}_{U_i})/2, \quad & d^{\mu\nu}_{d_i} &= (c^{\mu\nu}_{Q_i} - c^{\mu\nu}_{D_i})/2 \,. \end{aligned} \tag{60}$$

Since only three types of coefficients are initially present, the constraint $c^{\mu\nu}_{u_i} - c^{\mu\nu}_{d_i} = d^{\mu\nu}_{d_i} - d^{\mu\nu}_{u_i}$ eliminates a redundancy in Eq. (60). Expressing Eq. (58) in terms of the c- and d-type combinations introduced makes the comparison with the initial starting point Eq. (55) more direct.

A.2 Comments on expected coefficient sensitivities

Why are colliders suitable for probing the c- and d-type coefficients? As discussed in Sec. 1.2, directly probing the properties of quarks is often obscured by the strong interaction. However, the Drell-Yan process is one of a few where QCD factorization has been proven to all orders in the strong coupling constant. This enables analytical predictions for the scattering cross sections in terms of partonic matrix elements convoluted with parton distribution functions to be obtained.

An alternative path is to invoke effective theories like chiral perturbation theory (χPT) to relate parton-level coefficients to effective meson and baryon coefficients [**Altschul:2019beo**]. One could then use existing constraints on the effective proton and neutron coefficients $c^{\mu\nu}_{p,n}$ from, e.g. atomic-clock frequency comparisons to constrain the linear combinations of (valence) quark coefficients underlying the hadron coefficients. For example, in the absence of gluonic modifications, one obtains an effective c-type proton coefficient [**Kamand:2016xhv**]

$$c_p^{\mu\nu} = \left[\frac{1}{2}\alpha^{(1)} + \alpha^{(2)}\right](c_{uL}^{\mu\nu} + c_{uR}^{\mu\nu}) + \left[-\frac{1}{2}\alpha^{(1)} + \alpha^{(2)}\right](c_{dL}^{\mu\nu} + c_{dR}^{\mu\nu}), \tag{61}$$

where $\alpha^{(1)}, \alpha^{(2)}$ are low-energy constants. These coefficients are currently unknown, but could in principle be calculated using lattice QCD. Such route is not presently available and constraints on the quark-level parameters from χPT are technically unavailable. That said, by invoking considerations of naturalness, one could estimate that the quark constraints should be of a comparable order of magnitude to the hadron ones, unless some dramatic cancellation occurs.

On the extremely high-energy scale, and as mentioned in Sec. 2.3, astrophysical measurements have been suggested as potential probes of the c-type light-quark (u, d) coefficients. For the top quark, some constraints from threshold analyses imply the constraint $|c_t^{\mu\nu}| < 10^{-4}$, surpassing the those from $t\bar{t}$ production by ~2-3 orders of magnitude. Projections from upgrades at the LHC and the FCC claim to be able to surpass those from threshold analyses by 1-2 orders of magnitude [**Carle:2019ouy**]. Phenomenological constraints from radiative corrections have also been placed. Only the c-type coefficients have the correct tensor structure and symmetry properties for contributing to the one-loop photon vacuum polarization, leading to constraints on the purely timelike component $|c_t^{00}| \simeq 10^{-7}$ [**Satunin:2017wmk**].

In contrast, alternative probes of the d-type quark coefficients do not exist. The central complication is that these coefficients affect the quarks' polarization structure. This leads to significantly fewer observables and straightforward experimental probes. Virtual gluonic effects inside a bound hadron imply a given parton does not have a well-defined spin. For a spin-zero meson, the net spin from valence quarks must average to zero, rendering the d-type effects are unobservable in many processes [**Altschul:2020wkw**]. Reiterating some comments made in Sec. 2.5, the theory STA members have shown [**ls18**] that unpolarized hadronic processes mediated by a photon or gluon are completely insensitive to d-type coefficients. This apparent obstruction could be surmounted with polarized beamlines, but no existing collider has this capability. The key insight is to consider processes mediated instead by massive gauge bosons with distinct of left- and right-handed quark couplings [**lssv21**]. This is the only known route to probe the d-type quark coefficients. The Drell-Yan process is a perfect test process because of the wealth of existing data at invariant masses $Q \sim m_Z$, where the d-type coefficients display the largest cross-section enhancement. Deep inelastic scattering with t-channel Z exchange is another possibility, but is greatly suppressed.

In summary, with regard to the c-type coefficients, the expected constraints [**ls18**, **klsv19**] will not be competitive with existing first-generation (u, d) results, however, new linear coefficient combinations will be tested. Second-generation (s, c) constraints will exceed, by a few orders of magnitude [**lssv21**], the few existing constraints [**Karpikov:2016qvq**]. The d-type coefficients are much more motivated and suitable for our analysis.

B Singal injected and scrambled toys

B.1 Standard Model like toys

Standard Model like toys are signal free (no singal injection) in which we use Z boson cross section give SM to generate toys. In this type of models toy data points only contain statistical uncertainty. Thus, toy data points are randomly distributed around 1 according to statistical fluctuation. In figure 51, the error bar on each data point is statistical uncertainty. The left plot show one of the original toy dataset, and the right plot is it's corresponding scrambled version. The scrambled verions of SM like toys are also randomly distributed around 1 according to statistical fluctuation.

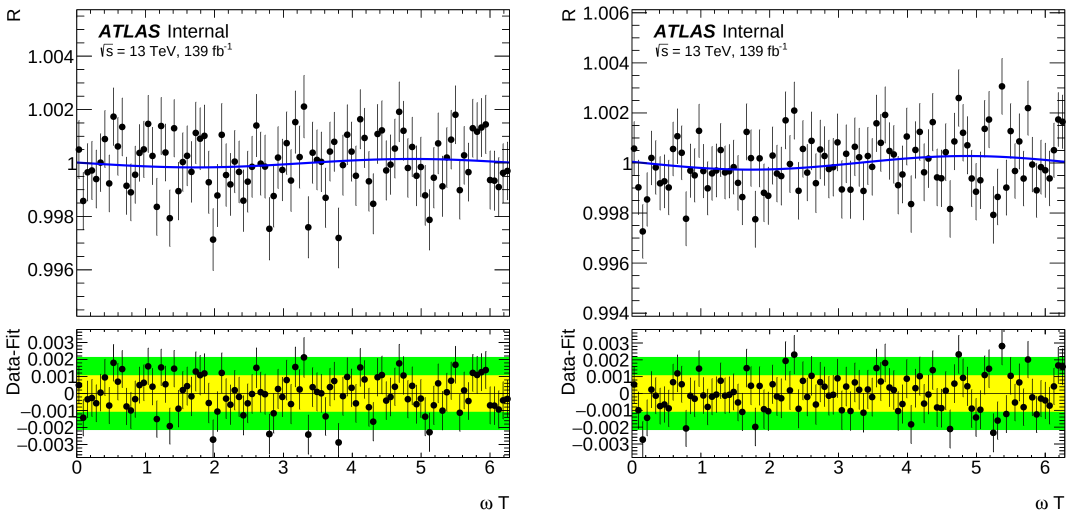

Figure 46: SM-like toy dataset (left), which corresponds to full Run 2 luminosity, and scrambled SM-like toy dataset (right). SM-like toys are signal-free datsets ($\sigma_{LIV}/\sigma_{SM} = 0$). The X-axis is the sidereal phase, ωT. The blue line corresponds to the $d_u^{X,Z}$ signal shape after fit to the toy dataset. Bottom panel in each plot is the residual (data-fit)

Figure 52 show fit results of SM like toys and their scrambled versions. The black points are average of 1000 fit results, and the green and yellow bands are corresponding one and two standard diviation. As we epected, the average fit results of MS like toys and their scrambled versions are in good agreement with each other and with zero since they are signal free toys.

B.2 Signal injected toys

In this section, we present toy datasets in which signal injected at different signal amplitude for a singal shape $d_u^{X,Z}$. The signal amplitude is σ_{LIV}/σ_{SM}. In figure 53, one of the 1000 toys with signal amplitude 0.0009 (left plot) and 0.0022 (right plot) is compared to a fitted model, $d_u^{X,Z}$.

In figure 54, we present average of 1000 fit results from signal injected toys. We can see that when $\sigma_{LIV}/\sigma_{SM} = 0.009$ we already seeing a few sigma deviation of the average value from zero (zero is SM expectation). And the deviation is even larger for larger signal. As we can see in both plots, there are three coefficients, $d_u^{X,Z}$, $c_u^{X,Z}$ and $c_d^{X,Z}$ show deviation. This is due to that these three signal shapes have very

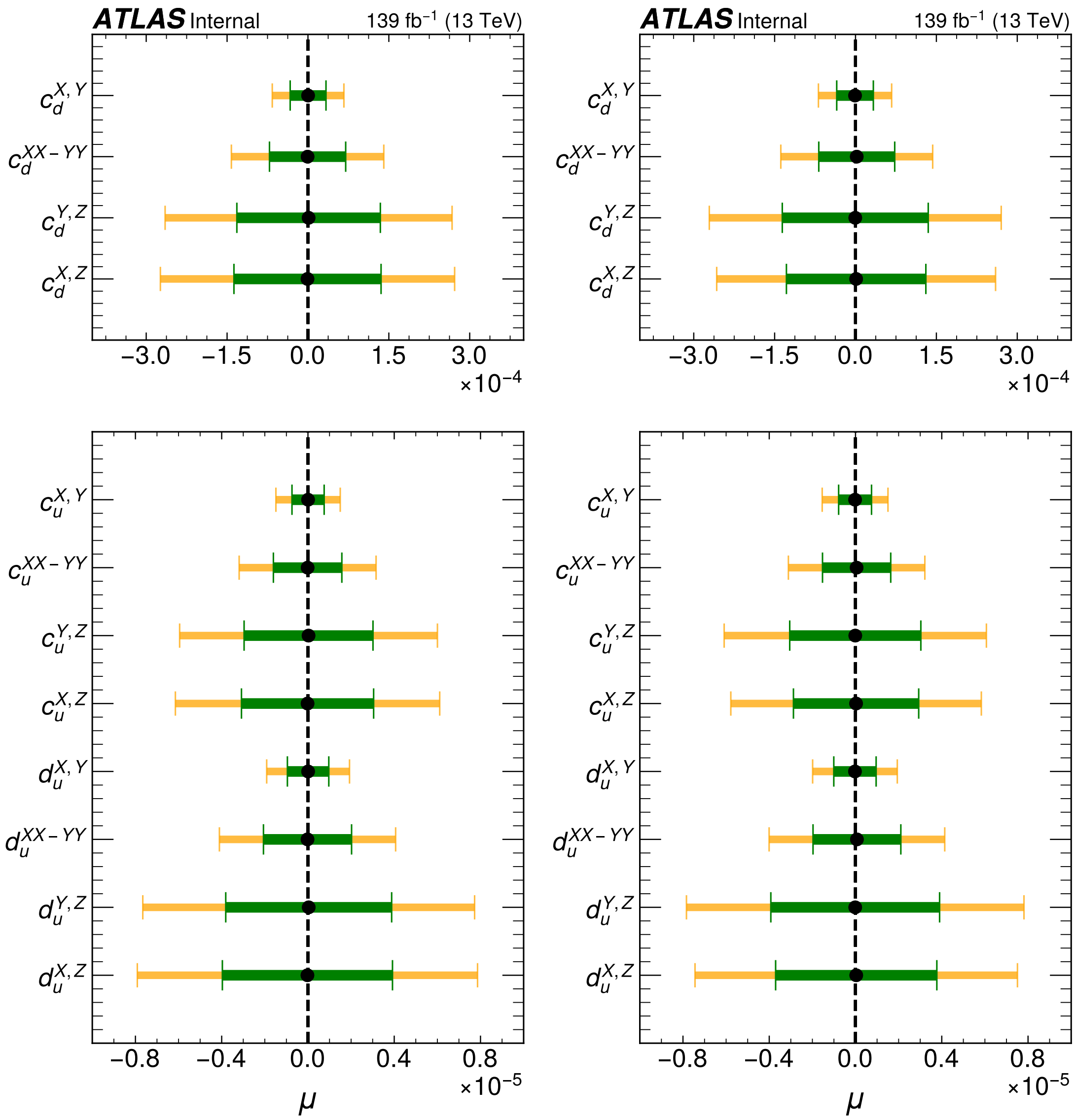

Figure 47: Fit results obtained from 1000 SM-like toy datasets. Results on SM-like toy datasets (left), which corresponds to full Run 2 luminosity, and scrambled SM-like toy datasets (right). SM-like toys are signal-free datsets ($\sigma_{LIV}/\sigma_{SM} = 0$). Y-axis corresponds to 12 exact signal shapes from SME. The X-axis is the signal strength. Yellow and green bends correspond to one and two signal uncertainty bands.

similar shape. Therefore, when the $d_u^{X,Z}$ is injected, other two coefficients also show some deviation from zero.

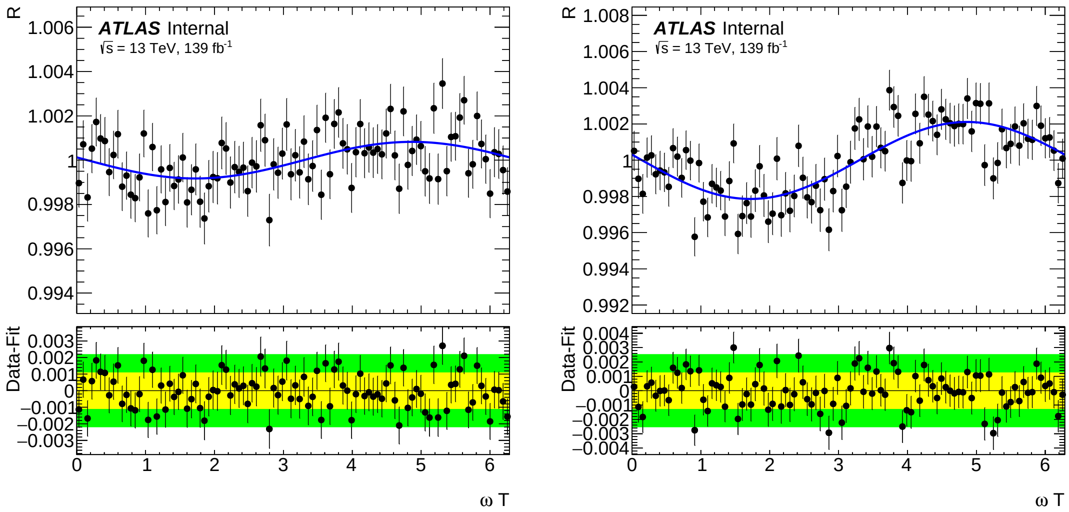

Figure 48: A signal injected toy dataset. The injected signal shape is $d_u^{X,Z}$ from SME. The njected signal fractions (σ_{LIV}/σ_{SM}) are 0.0009 (left) and 0.0022 (right). The blue line is the signal ($d_u^{X,Z}$) after fit to the toy dataset. Bottom panel in each plot is the residual (data-fit). The X-axis is the sidereal phase, ωT.

To proof the scrambling can perfectly wash away signals if they are in the real data, we applied scrambling to signal injected toys. In figure 55 shows a signal injected toy before (left) and after the scrambling (right). We can see from the left plot that signal is strong, and the fitted singal strenth is $5.3E^{-5} \pm 3.8E^{-6}$. Still, the scrambling method washed away such a strong signal. The fitted signal strength is resuced to $-4.4E^{-7} \pm 3.8E^{-6}$ which is very good agreement with zero. Since toy datasets are distributed accourding to only statistical fluctuation we expect that the toy dataset after scrambling give similar results as SM like toys.

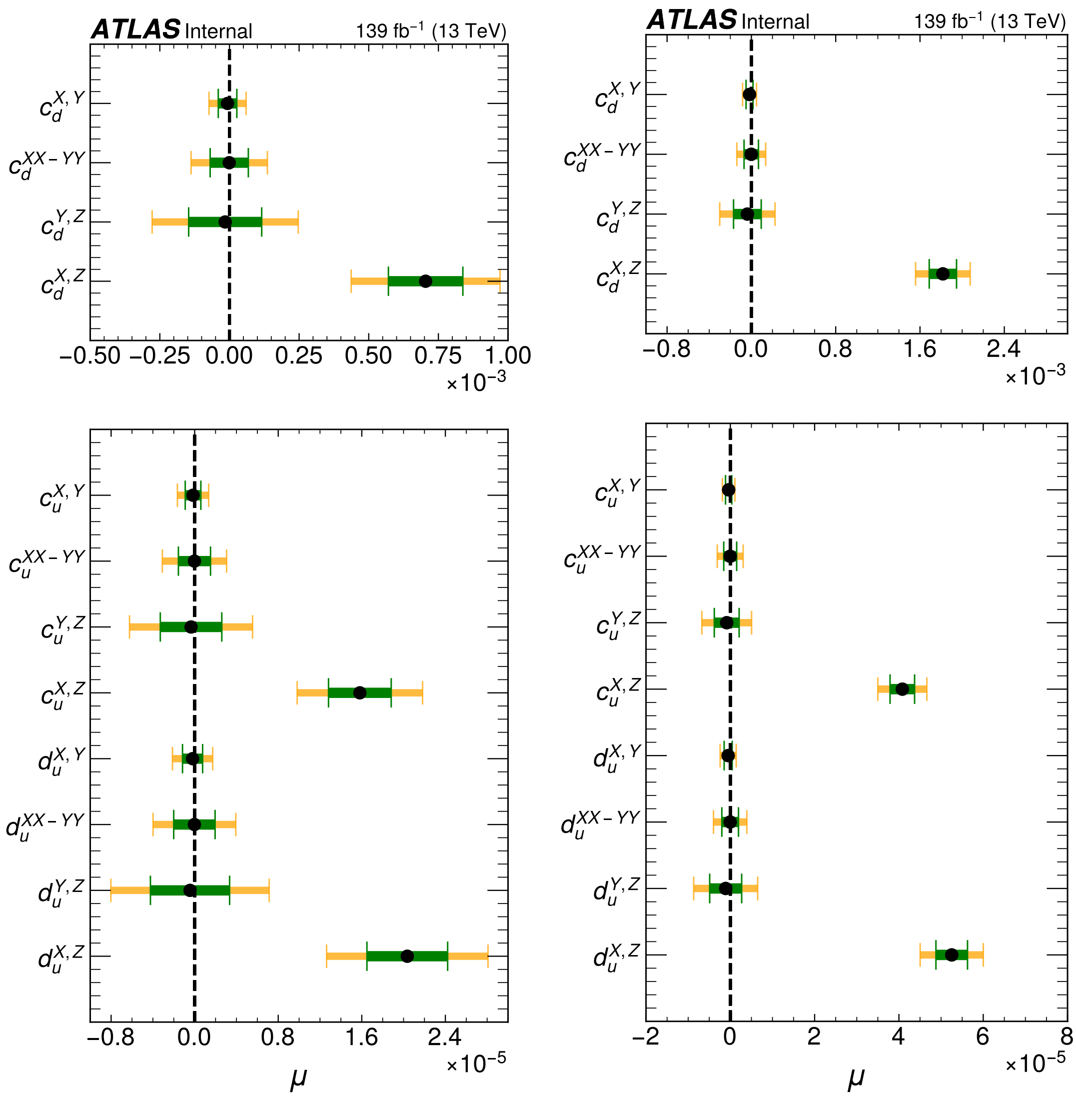

Figure 49: Fit results from 1000 signal injected toys. The injected signal shape is $d_u^{X,Z}$ from SME. The njected signal fractions (σ_{LIV}/σ_{SM}) are 0.0009 (left) and 0.0022 (right). Y-axis corresponds to 12 exact signal shapes from SME. The X-axis is the signal strength.

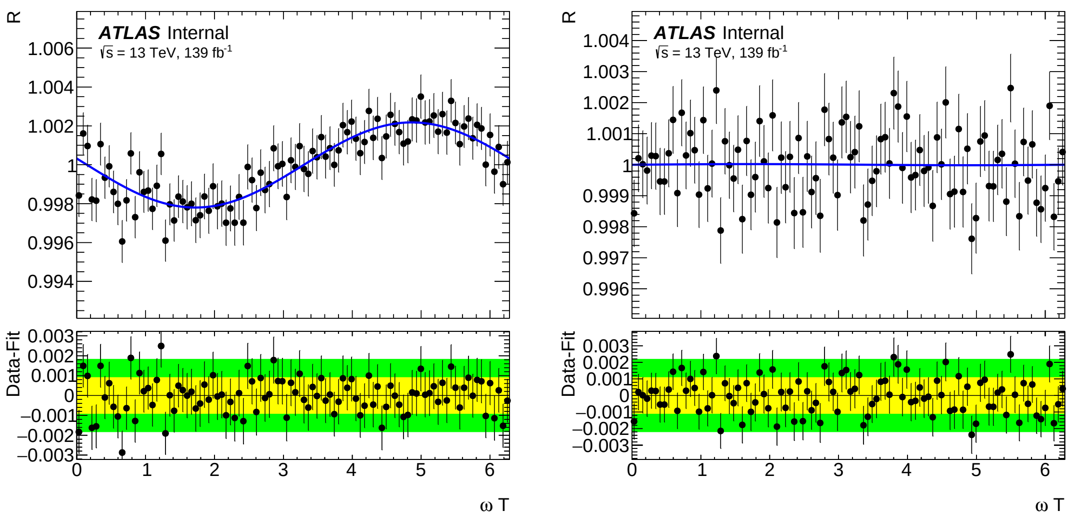

Figure 50: Comparison of a signal injected toy dataset before and after a scramble. The injected signal shape is $d_u^{X,Z}$ from SME. The injected signal fractions (σ_{LIV}/σ_{SM}) is 0.0022. The original toy dataset (left) and scrambled data (right). The blue line is the signal ($d_u^{X,Z}$) after fit to the toy dataset. The bottom panel in each plot is the residual (data - fit). The X-axis is the sidereal phase, ωT. The Fitted signal strength is $5.3E^{-5} \pm 3.8E^{-6}$ (left), and $-4.4E^{-7} \pm 3.8E^{-6}$ (right) after scrambling.

C Scramble

C.1 Standard Model like toys

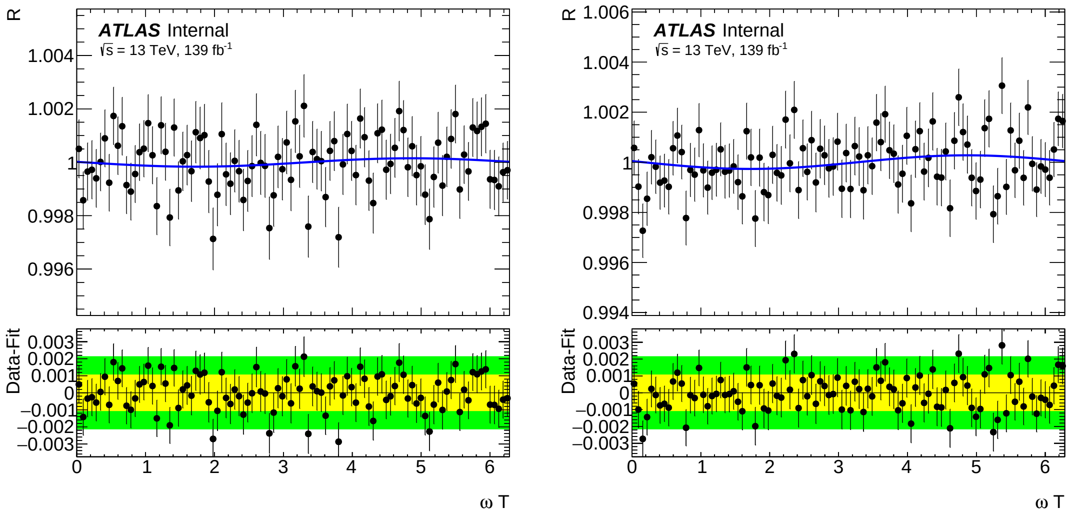

Figure 51: SM-like toy dataset (left), which corresponds to full Run 2 luminosity, and scrambled SM-like toy dataset (right). SM-like toys are signal-free datsets ($\sigma_{LIV}/\sigma_{SM} = 0$). The X-axis is the sidereal phase, ωT. The blue line corresponds to the $d_u^{X,Z}$ signal shape after fit to the toy dataset. Bottom panel in each plot is the residual (data-fit)

C.2 Signal injected toys

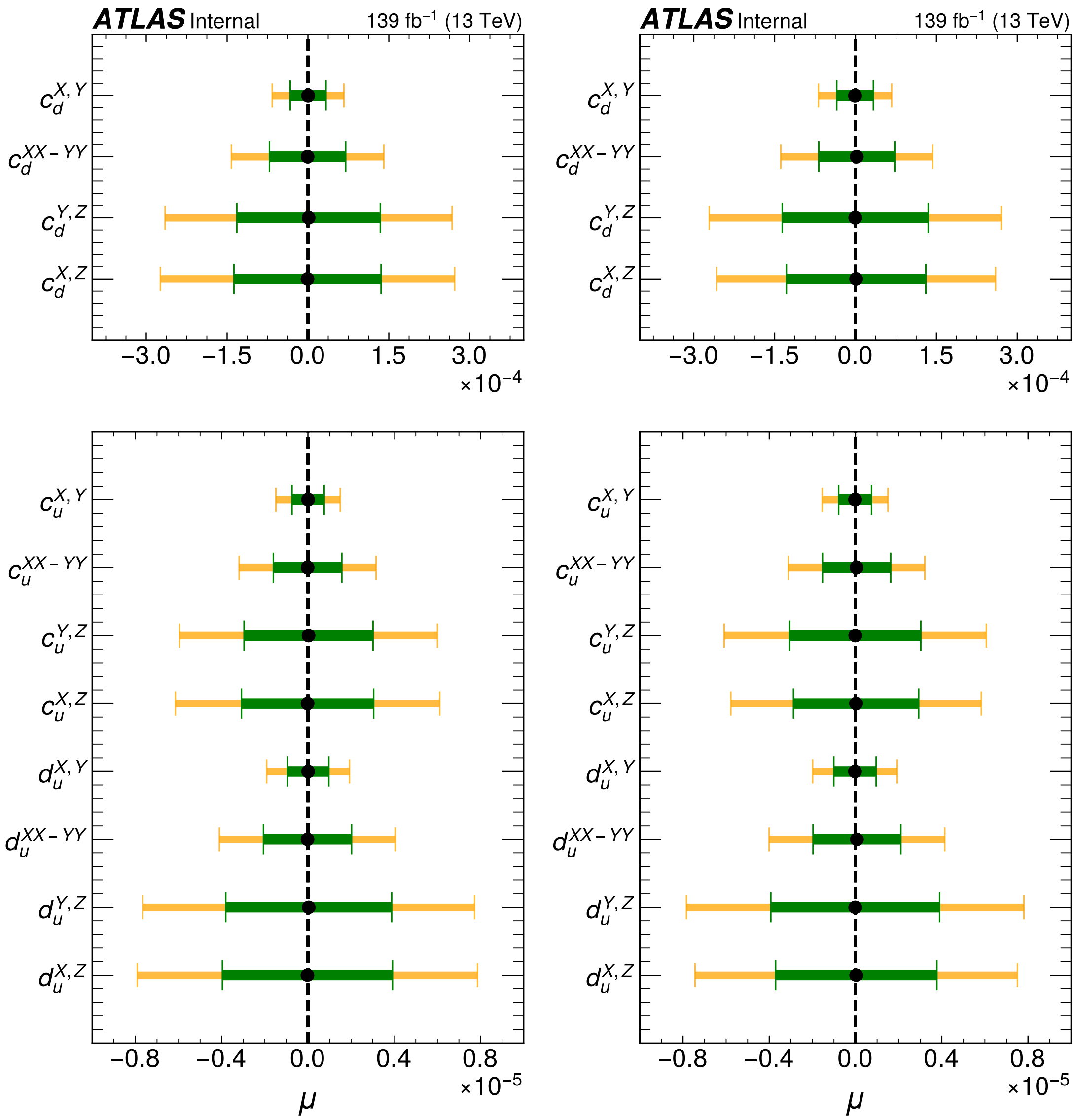

Figure 52: Fit results obtained from 1000 SM-like toy datasets. Results on SM-like toy datasets (left), which corresponds to full Run 2 luminosity, and scrambled SM-like toy datasets (right). SM-like toys are signal-free datsets ($\sigma_{LIV}/\sigma_{SM} = 0$). Y-axis corresponds to 12 exact signal shapes from SME. The X-axis is the signal strength. Yellow and green bends correspond to one and two signal uncertainty bands.

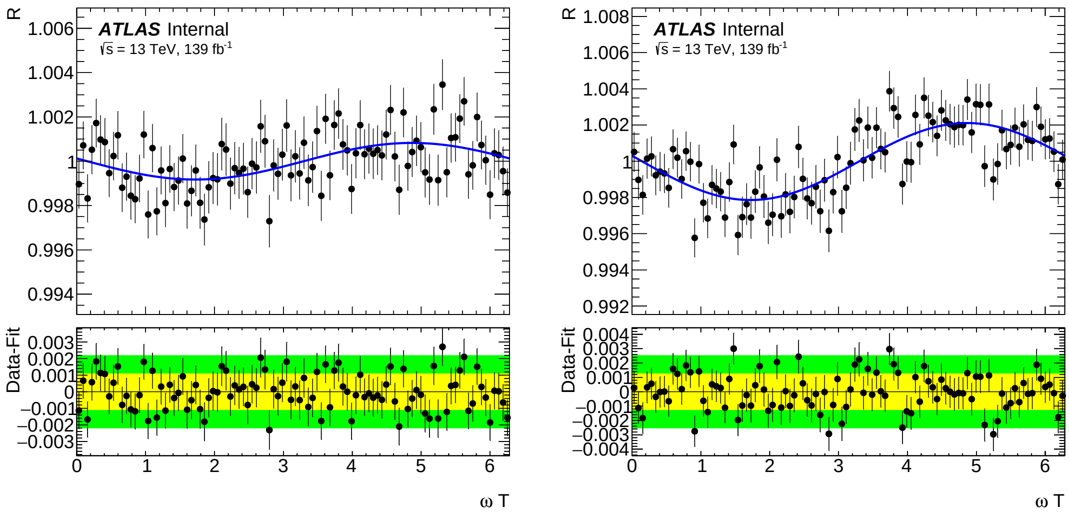

Figure 53: A signal injected toy dataset. The injected signal shape is $d_u^{X,Z}$ from SME. The njected signal fractions (σ_{LIV}/σ_{SM}) are 0.0015 (left) and 0.003 (right). The blue line is the signal ($d_u^{X,Z}$) after fit to the toy dataset. Bottom panel in each plot is the residual (data-fit). The X-axis is the sidereal phase, ωT.

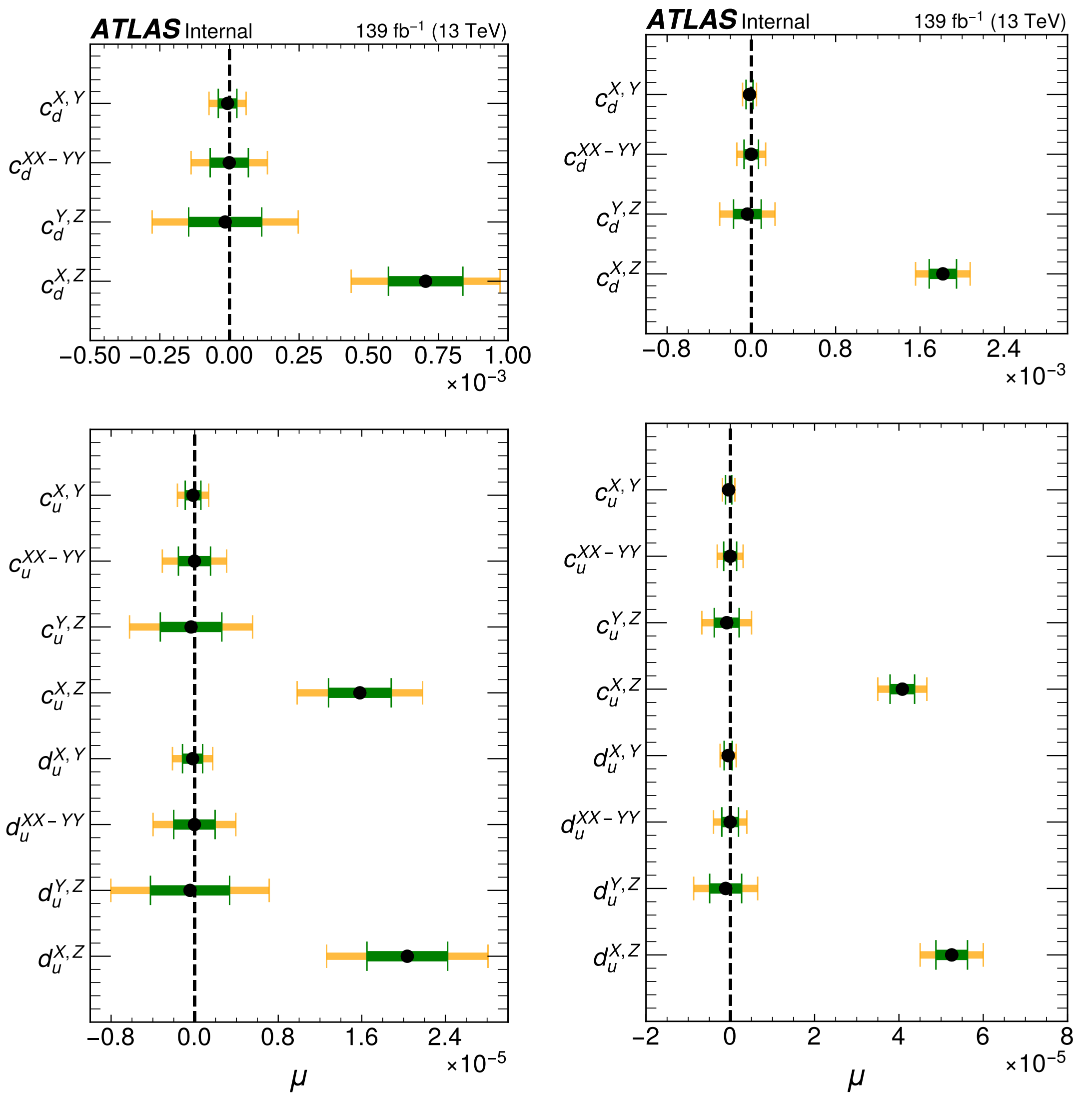

Figure 54: Fit results from 1000 signal injected toys. The injected signal shape is $d_u^{X,Z}$ from SME. The njected signal fractions (σ_{LIV}/σ_{SM}) are 0.0015 (left) and 0.003 (right). Y-axis corresponds to 12 exact signal shapes from SME. The X-axis is the signal strength.

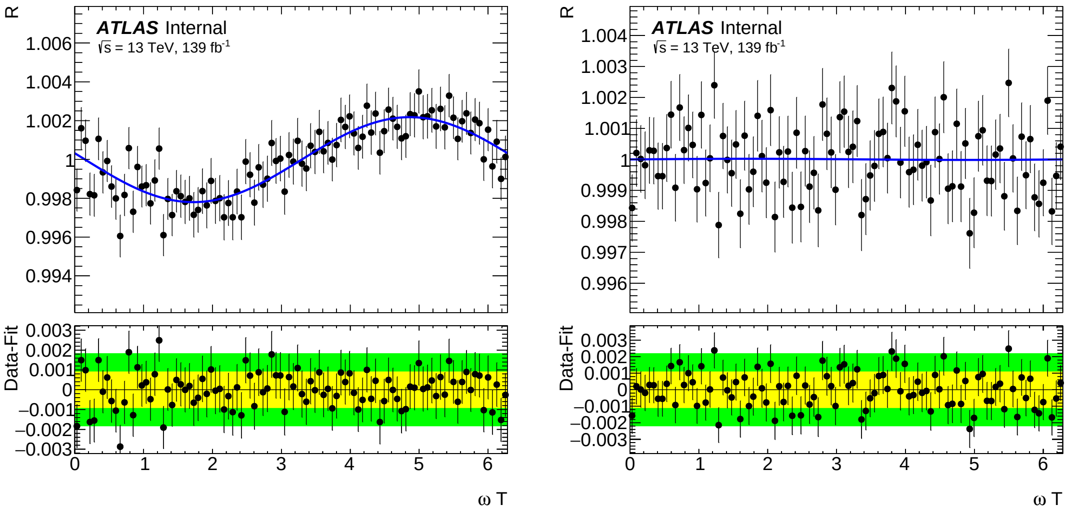

Figure 55: Comparison of a signal injected toy dataset before and after a scramble. The injected signal shape is $d_u^{X,Z}$ from SME. The injected signal fractions (σ_{LIV}/σ_{SM}) is 0.003. The original toy dataset (left) and scrambled data (right). The blue line is the signal ($d_u^{X,Z}$) after fit to the toy dataset. The bottom panel in each plot is the residual (data - fit). The X-axis is the sidereal phase, ωT. The Fitted signal strength is $5.3E^{-5} \pm 3.8E^{-6}$ (left), and $-4.4E^{-7} \pm 3.8E^{-6}$ (right) after scrambling.

D Year by year toys

D.1 2015 + 2016

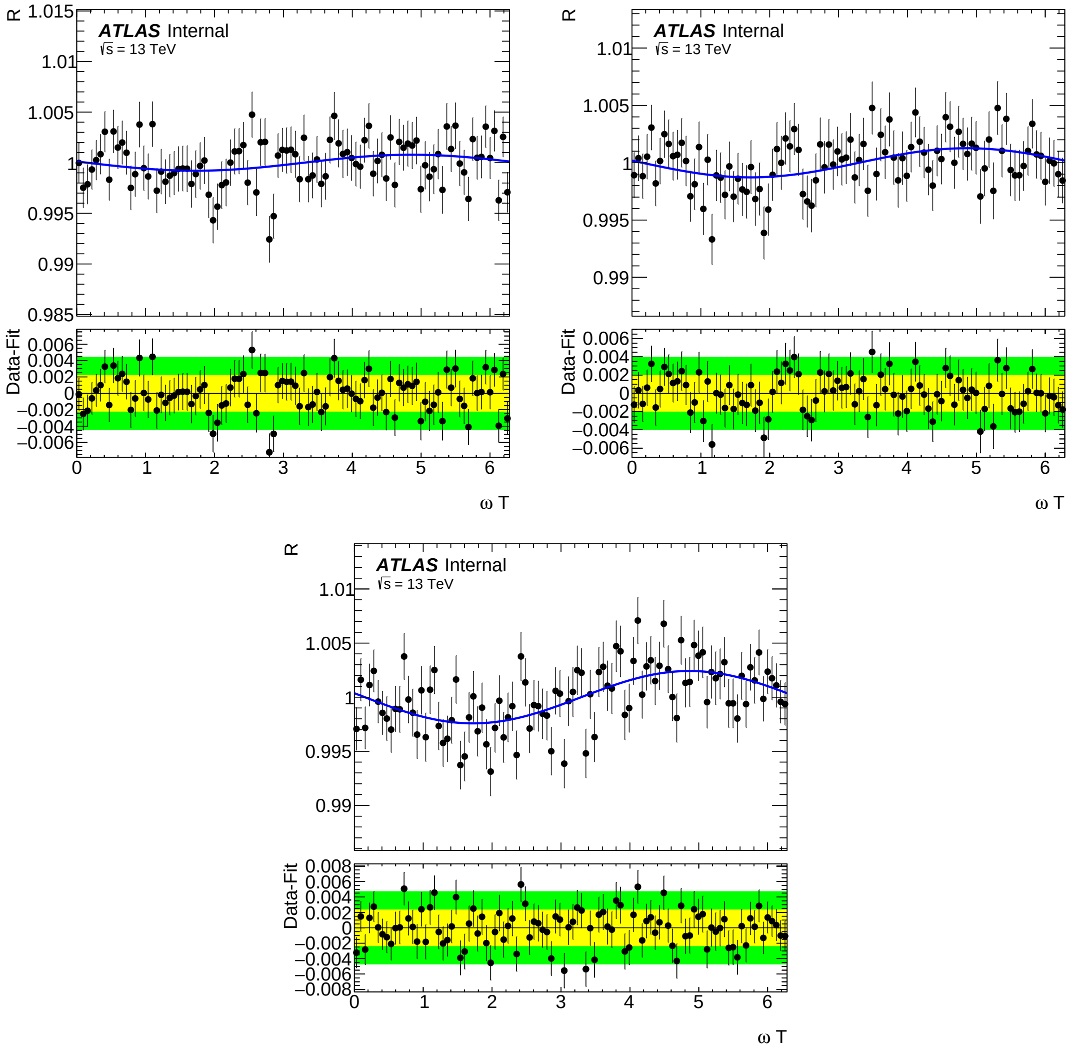

Figure 56: Comparison of 2015+2016 toy datasets with the fitted signal shapes. The SM-like toy (top left), and signal injected toys. The injected signal fraction (σ_{LIV}/σ_{SM}) is 0.0015 (top right) and 0.003 (bottom). The injected signal shape is $d_u^{X,Z}$ from SME. The blue line is the signal ($d_u^{X,Z}$) after fit to the toy dataset. The bottom panel in each plot is the residual (data - fit). The X-axis is the sidereal phase, ωT.

D.2 2017

D.3 2018

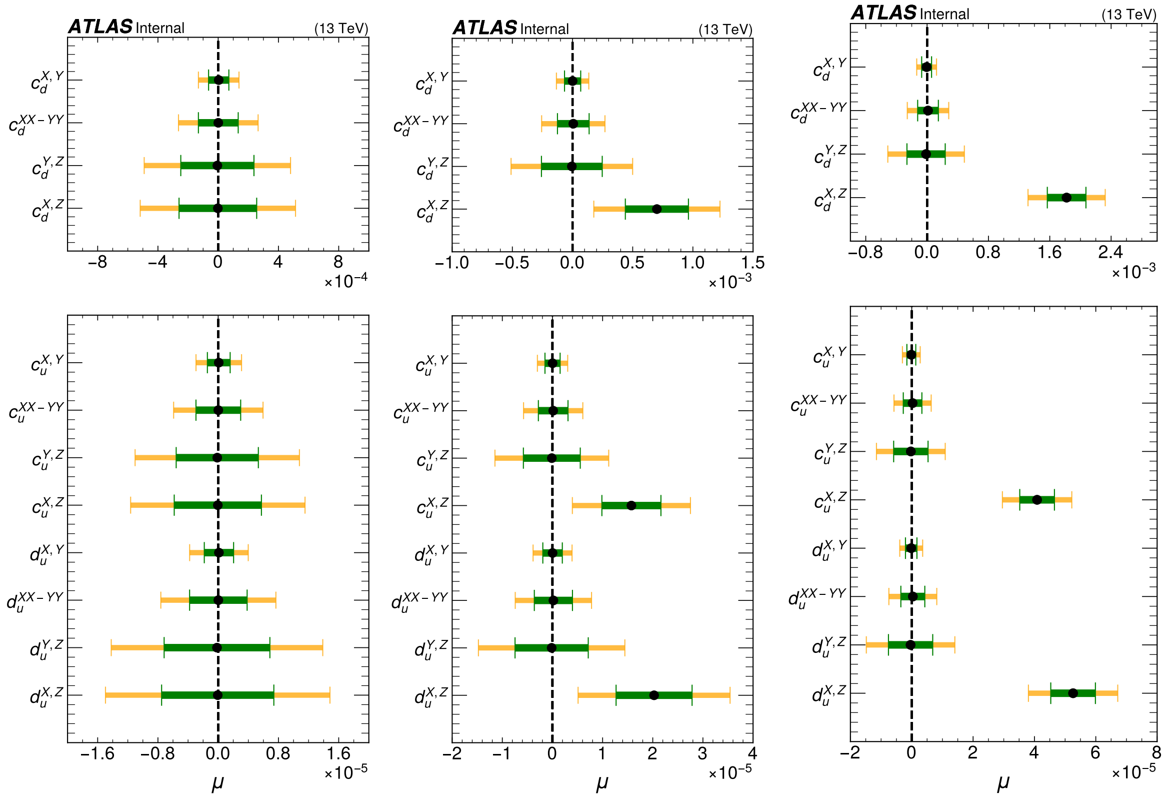

Figure 57: Fit results on 2015+2016 toy datasets.Fit results obtained from 1000 toys. The left one is SM-like, the signal injected toys with the signal fraction (σ_{LIV}/σ_{SM}) is 0.0015 (middle) and 0.003 (right). The injected signal shape is $d_u^{X,Z}$ from SME. Y-axis corresponds to 12 exact signal shapes from SME. The X-axis is the signal strength. Yellow and green bends correspond to one and two signal uncertainty bands.

year	2015+2016	2017	2018	full	full scrambled
σ_{LIV}/σ_{SM}	$d_u^{X,Z}$	$d_u^{X,Z}$	$d_u^{X,Z}$	$d_u^{X,Z}$	$d_u^{X,Z}$
0	-6.23-08 ±7.5e-06	-2.8e-06 ± 6.7e-06	1.9e-07 ± 5.8e-06	-2.5e-08 ± 3.9e-06	3.4e-08 ± 3.7e-06
0.0015	2.0e-05 ± 7.6e-06	2.0e-05 ±6.7e-06	2.1e-05 ± 5.7e-06	2.0e-05 ±3.9e-06	-
0.003	5.3e-05 ±7.3e-06	5.2e-05 ± 6.7e-06	5.3e-05 ± 5.5e-06	5.3e-05 ±3.7e-06	-

Table 13: Fit results from full toy datasets correspond to full Run 2 luminosity and from year-by-year datasets.

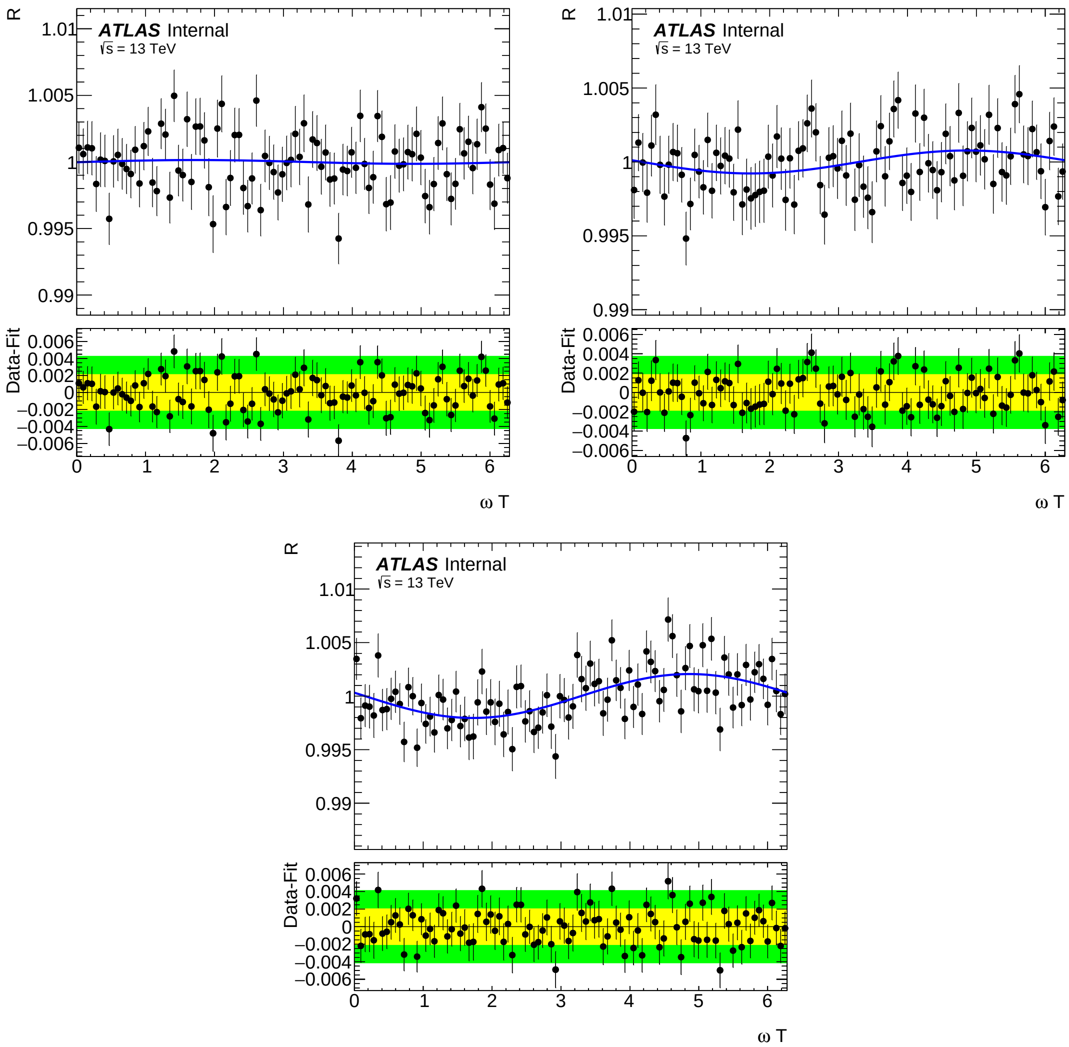

Figure 58: Comparison of 2017 toy datasets with the fitted signal shapes. The SM-like toy (top left), and signal injected toys. The injected signal fraction (σ_{LIV}/σ_{SM}) is 0.0015 (top right) and 0.003 (bottom). The injected signal shape is $d_u^{X,Z}$ from SME. The blue line is the signal ($d_u^{X,Z}$) after fit to the toy dataset. The bottom panel in each plot is the residual (data - fit). The X-axis is the sidereal phase, ωT.

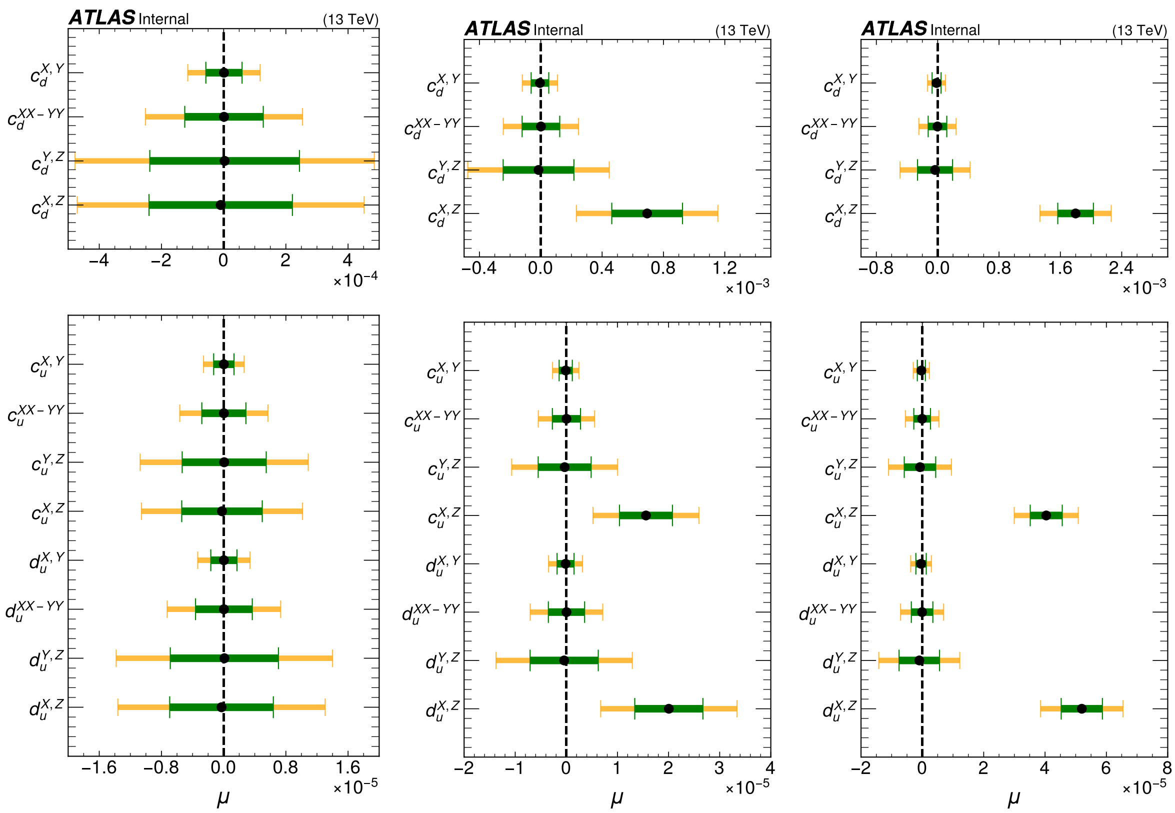

Figure 59: Fit results on 2017 toy datasets.Fit results obtained from 1000 toys. The left one is SM-like, the signal injected toys with the signal fraction (σ_{LIV}/σ_{SM}) is 0.0015 (middle) and 0.003 (right). The injected signal shape is $d_u^{X,Z}$ from SME. Y-axis corresponds to 12 exact signal shapes from SME. The X-axis is the signal strength. Yellow and green bends correspond to one and two signal uncertainty bands.

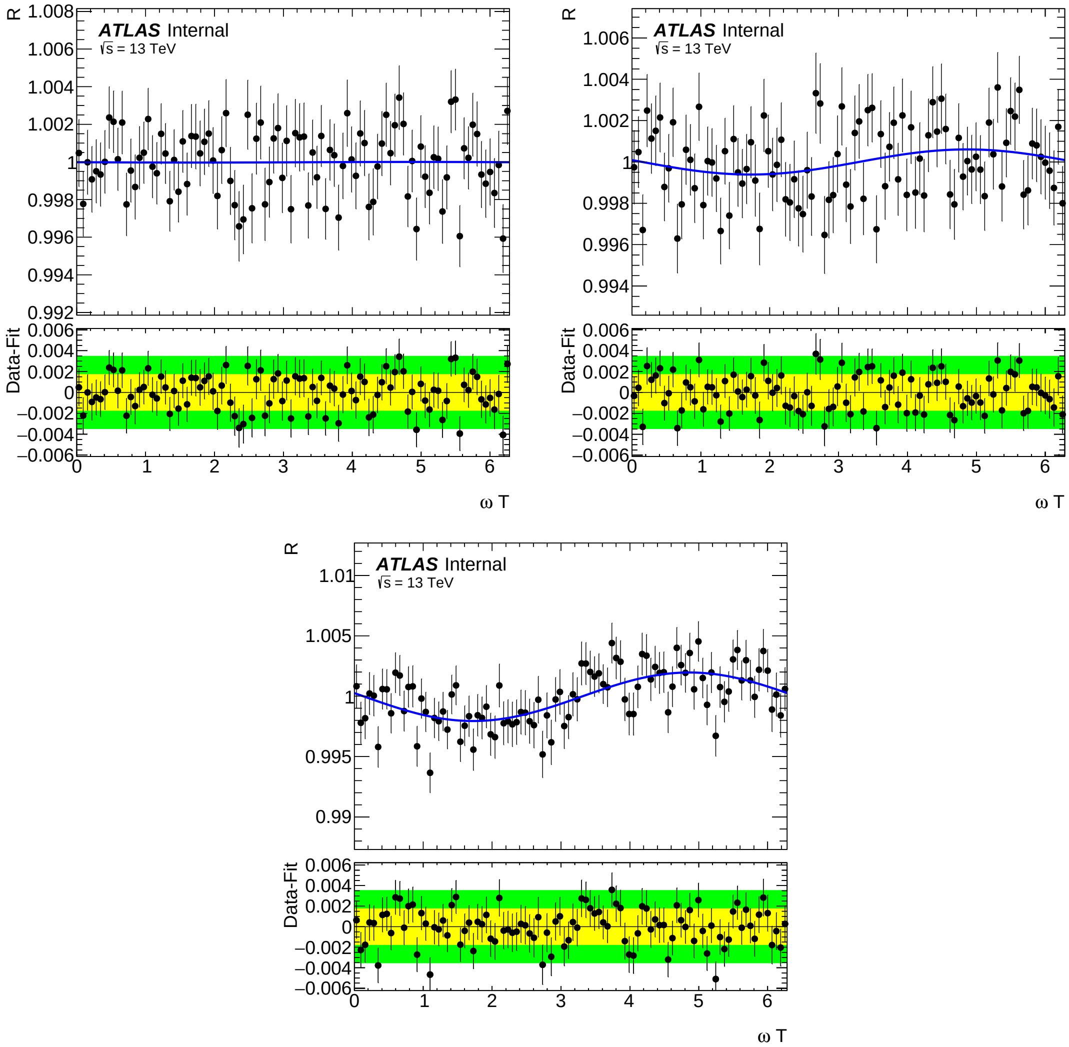

Figure 60: Comparison of 2018 toy datasets with the fitted signal shapes. The SM-like toy (top left), and signal injected toys. The injected signal fraction (σ_{LIV}/σ_{SM}) is 0.0015 (top right) and 0.003 (bottom). The injected signal shape is $d_u^{X,Z}$ from SME. The blue line is the signal ($d_u^{X,Z}$) after fit to the toy dataset. The bottom panel in each plot is the residual (data - fit). The X-axis is the sidereal phase, ωT.

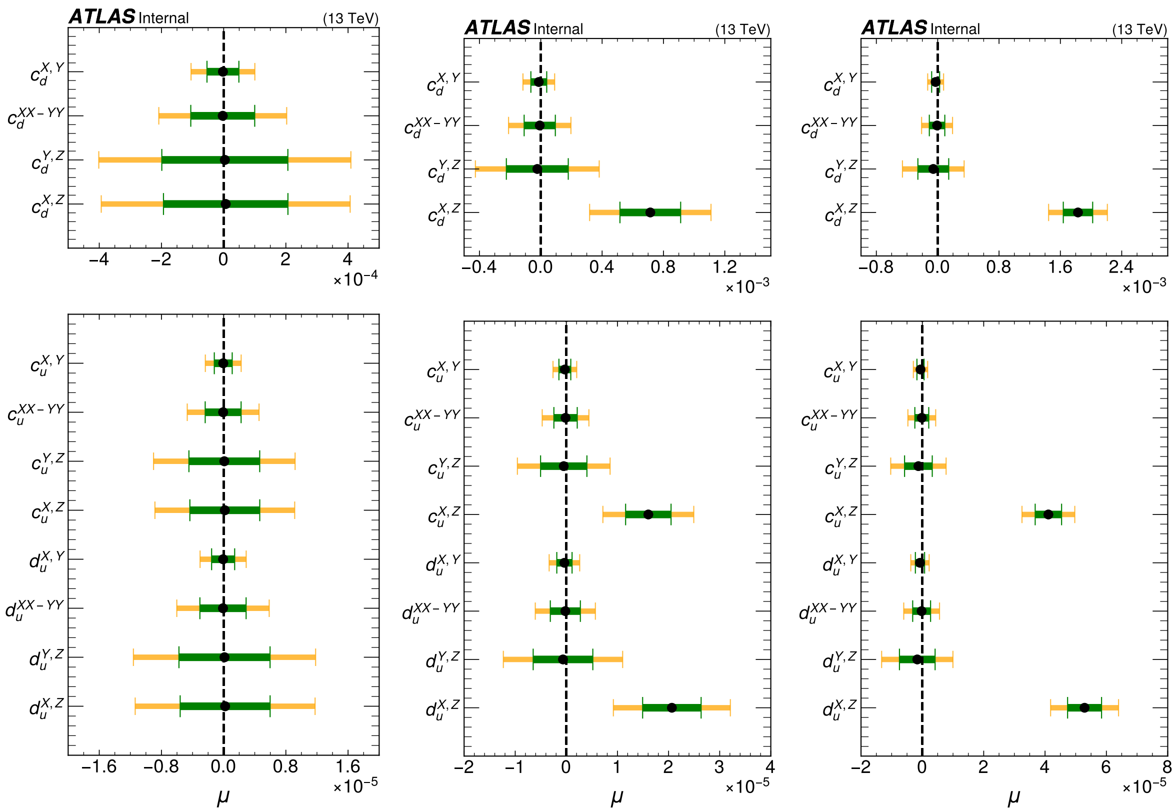

Figure 61: Fit results on 2018 toy datasets.Fit results obtained from 1000 toys. The left one is SM-like, the signal injected toys with the signal fraction (σ_{LIV}/σ_{SM}) is 0.0015 (middle) and 0.003 (right). The injected signal shape is $d_u^{X,Z}$ from SME. Y-axis corresponds to 12 exact signal shapes from SME. The X-axis is the signal strength. Yellow and green bends correspond to one and two signal uncertainty bands.

E Simultaneous fit

E.1 Simultaneous fit

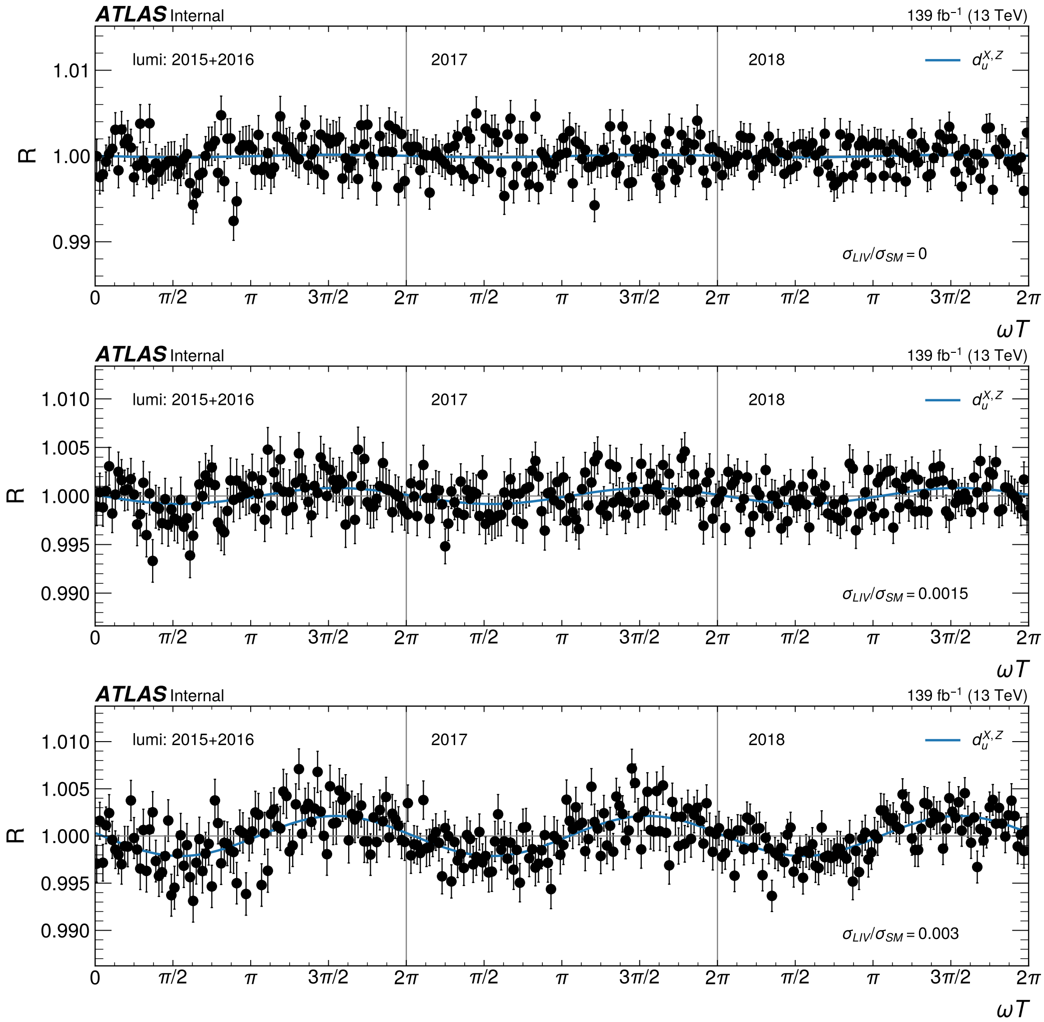

Figure 62: Simultaneous fit to three datasets; toy datasets correspond to 2012+2016, 2017 and 2018. The top plot is SM like toy dataset. The middl and bottom one are signal-injected ($d_u^{X,Z}$) toys with the signal fraction (σ_{LIV}/σ_{SM}) is 0.0015 (middle) and 0.003 (bottom). The X-axis is the sidereal phase, ωT. The blue line is fitted signal shape ($d_u^{X,Z}$).

	Fit to full dataset	Simultaneous fit
σ_{LIV}/σ_{SM}	$d_u^{X,Z}$	$d_u^{X,Z}$
0	2.5e-08 ± 3.9e-06	-2.1e-08 ± 3.9e-06
0.0015	2.0e-05 ± 3.9e-06	2.0e-05 ± 3.9e-06
0.003	5.3e-05 ± 3.7e-06	5.3e-05 ± 3.7e-06

Table 14: Comparison of the fit results from full toy datasets correspond to full Run 2 luminosity and simultaneous fit to year-by-year datasets.

F Partially unblinded results

F.1 Partially unblinded data

- year 2015+2016
- year 2017
- year 2018
- Run 2: sum of all years.
- Combined Run 2: stitch up three datasets in the range of $0 \to 6\pi$. The 2015+2016 dataset is in the $0 \to 2\pi$ range, the 2017 dataset is in the $2pi \to 4\pi$ range, and 2018 dataset is in the $4pi \to 6\pi$ range.

F.1.1 Fit results

	2015+2016	2017	2018	full Run2	Run2-Comb. Fit
$d_u^{X,Z}$	8.82e-06 ± 7.44e-06	-1.42e-05 ± 6.43e-06	-7.47e-06 ± 5.74e-06	-5.54e-06 ± 3.58e-06	-5.39e-06 ± 3.58e-06
$d_u^{Y,Z}$	1.77e-06 ± 6.61e-06	2.31e-06 ± 6.61e-06	-7.67e-06 ± 5.74e-06	-1.41e-06 ± 3.52e-06	-2.27e-06 ± 3.52e-06
$d_u^{XY,XY}$	-6.12e-06 ± 3.55e-06	2.41e-06 ± 3.39e-06	3.97e-06 ± 2.87e-06	1.69e-06 ± 1.81e-06	6.42e-07 ± 1.81e-06
$d_u^{X,Y}$	-1.93e-06 ± 1.88e-06	-6.03e-07 ± 1.65e-06	-1.85e-06 ± 1.48e-06	-1.24e-06 ± 9.41e-07	-1.45e-06 ± 9.41e-07
$c_u^{X,Z}$	6.85e-06 ± 5.78e-06	-1.10e-05 ± 5.00e-06	-5.80e-06 ± 4.46e-06	-4.31e-06 ± 2.78e-06	-4.19e-06 ± 2.78e-06
$c_u^{Y,Z}$	1.38e-06 ± 5.13e-06	1.79e-06 ± 5.13e-06	-5.96e-06 ± 4.46e-06	-1.10e-06 ± 2.74e-06	-1.77e-06 ± 2.74e-06
$c_u^{XY,XY}$	-4.76e-06 ± 2.76e-06	1.87e-06 ± 2.64e-06	3.08e-06 ± 2.23e-06	1.31e-06 ± 1.41e-06	4.99e-07 ± 1.41e-06
$c_u^{X,Y}$	-1.50e-06 ± 1.46e-06	-4.69e-07 ± 1.29e-06	-1.44e-06 ± 1.15e-06	-9.60e-07 ± 7.31e-07	-1.12e-06 ± 7.31e-07
$c_d^{X,Z}$	3.05e-04 ± 2.57e-04	-4.91e-04 ± 2.23e-04	-2.58e-04 ± 1.99e-04	-1.92e-04 ± 1.24e-04	-1.87e-04 ± 1.24e-04
$c_d^{Y,Z}$	6.13e-05 ± 2.29e-04	7.98e-05 ± 2.29e-04	-2.65e-04 ± 1.99e-04	-4.88e-05 ± 1.22e-04	-7.87e-05 ± 1.22e-04
$c_d^{XY,XY}$	-2.12e-04 ± 1.23e-04	8.33e-05 ± 1.17e-04	1.37e-04 ± 9.91e-05	5.84e-05 ± 6.28e-05	2.22e-05 ± 6.28e-05
$c_d^{X,Y}$	-6.67e-05 ± 6.50e-05	-2.09e-05 ± 5.72e-05	-6.40e-05 ± 5.11e-05	-4.27e-05 ± 3.25e-05	-5.00e-05 ± 3.25e-05

Table 15: Partially unblinded results in *ee* channel. The table shows the average and one standard deviation of 1000 fit results.

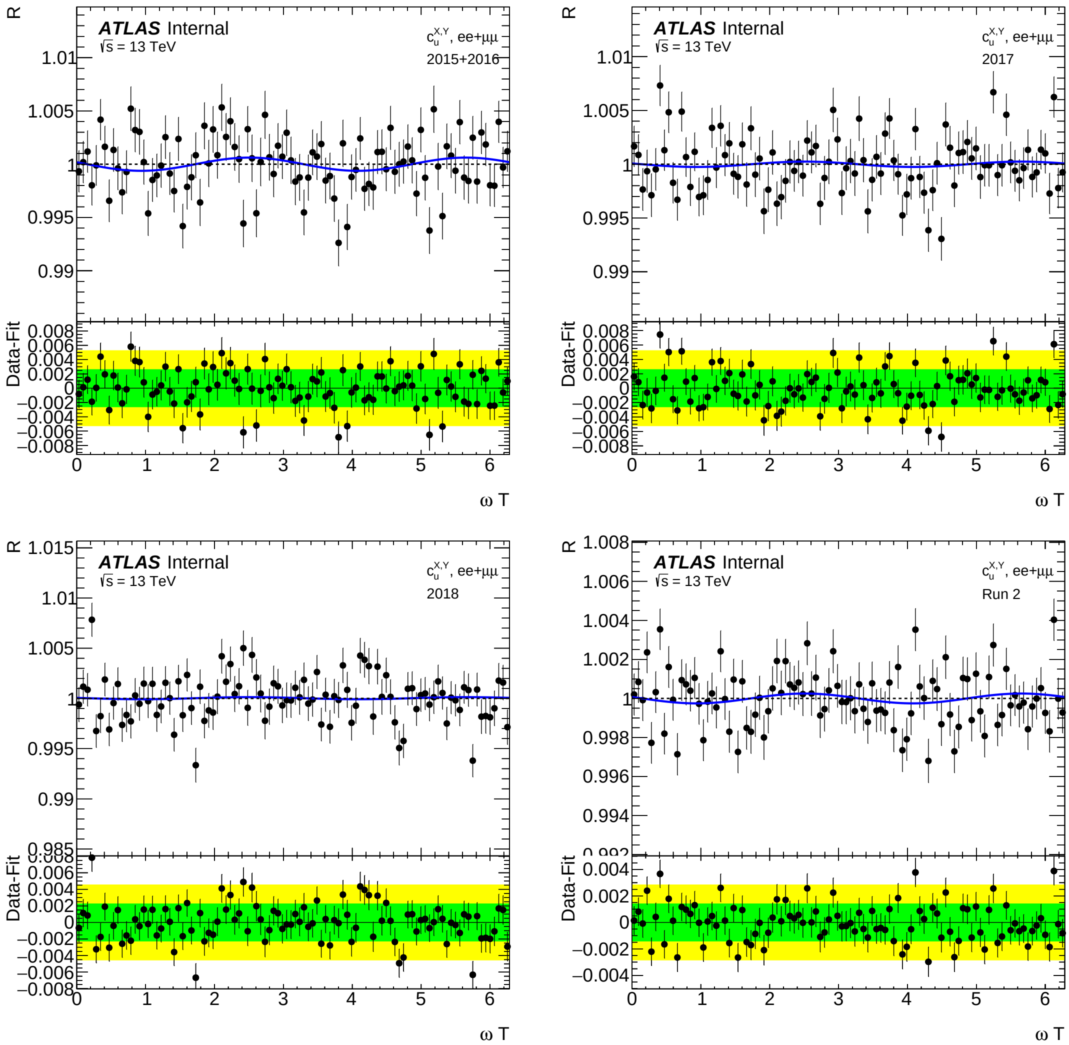

Figure 63: One of the scrambled datasets VS $c_u^{X,Y}$ model. The black dots are the scrambled data, and the blue lines are the fitted model. The 2015+2016 dataset(top left), the 2017 dataset (top right), the 2018 dataset(bottom left), and the Run 2 dataset (bottom right). The Run 2 dataset is the sum of all year's data. The bottom panel of the plots is the residual (data-model). The green and yellow bands correspond to one and two sigma uncertainty bands.

F.1.2 Spread VS Stats

F.1.3 Spurious signal uncertainty

	2015+2016	2017	2018	full Run2	Run2-Comb. Fit
$d_u^{X,Z}$	6.06e-06 ± 7.42e-06	-2.01e-07 ± 6.46e-06	-5.75e-06 ± 5.71e-06	-1.07e-06 ± 3.77e-06	-7.33e-07 ± 3.79e-06
$d_u^{Y,Z}$	-2.57e-06 ± 7.12e-06	1.22e-05 ± 6.50e-06	-1.37e-05 ± 5.57e-06	-1.92e-06 ± 3.48e-06	-2.68e-06 ± 3.50e-06
$d_u^{XY,XY}$	4.92e-06 ± 3.56e-06	-6.05e-06 ± 3.48e-06	-8.44e-07 ± 2.84e-06	-2.02e-07 ± 1.84e-06	-8.82e-07 ± 1.84e-06
$d_u^{X,Y}$	-7.66e-07 ± 1.86e-06	1.71e-06 ± 1.67e-06	3.33e-06 ± 1.46e-06	1.93e-06 ± 9.46e-07	1.67e-06 ± 9.48e-07
$c_u^{X,Z}$	4.71e-06 ± 5.76e-06	-1.56e-07 ± 5.02e-06	-4.47e-06 ± 4.43e-06	-8.31e-07 ± 2.93e-06	-5.69e-07 ± 2.94e-06
$c_u^{Y,Z}$	-1.99e-06 ± 5.53e-06	9.46e-06 ± 5.05e-06	-1.06e-05 ± 4.33e-06	-1.49e-06 ± 2.70e-06	-2.09e-06 ± 2.72e-06
$c_u^{XY,XY}$	3.82e-06 ± 2.77e-06	-4.70e-06 ± 2.70e-06	-6.55e-07 ± 2.21e-06	-1.57e-07 ± 1.43e-06	-6.85e-07 ± 1.43e-06
$c_u^{X,Y}$	-5.95e-07 ± 1.44e-06	1.33e-06 ± 1.30e-06	2.59e-06 ± 1.13e-06	1.50e-06 ± 7.35e-07	1.30e-06 ± 7.37e-07
$c_d^{X,Z}$	2.10e-04 ± 2.57e-04	-6.94e-06 ± 2.24e-04	-1.99e-04 ± 1.97e-04	-3.70e-05 ± 1.31e-04	-2.54e-05 ± 1.31e-04
$c_d^{Y,Z}$	-8.88e-05 ± 2.46e-04	4.21e-04 ± 2.25e-04	-4.72e-04 ± 1.93e-04	-6.63e-05 ± 1.20e-04	-9.29e-05 ± 1.21e-04
$c_d^{XY,XY}$	1.70e-04 ± 1.23e-04	-2.09e-04 ± 1.20e-04	-2.92e-05 ± 9.83e-05	-6.97e-06 ± 6.38e-05	-3.05e-05 ± 6.38e-05
$c_d^{X,Y}$	-2.65e-05 ± 6.42e-05	5.92e-05 ± 5.78e-05	1.15e-04 ± 5.05e-05	6.69e-05 ± 3.27e-05	5.78e-05 ± 3.28e-05

Table 16: Partially unblinded results in $\mu\mu$ channel. The table shows the average and one standard deviation of 1000 fit results.

	2015+2016	2017	2018	full Run2	Run2-Comb. Fit
$d_u^{X,Z}$	7.48e-06 ± 7.22e-06	-7.07e-06 ± 6.44e-06	-6.56e-06 ± 5.72e-06	-3.23e-06 ± 3.68e-06	-2.98e-06 ± 3.68e-06
$d_u^{Y,Z}$	-5.06e-07 ± 7.16e-06	7.37e-06 ± 6.54e-06	-1.07e-05 ± 5.63e-06	-1.69e-06 ± 3.46e-06	-2.49e-06 ± 3.46e-06
$d_u^{XY,XY}$	-5.76e-07 ± 3.72e-06	-1.82e-06 ± 3.31e-06	1.69e-06 ± 2.81e-06	8.03e-07 ± 1.86e-06	-1.62e-07 ± 1.86e-06
$d_u^{X,Y}$	-1.35e-06 ± 1.93e-06	4.94e-07 ± 1.63e-06	7.10e-07 ± 1.43e-06	3.21e-07 ± 9.25e-07	1.05e-07 ± 9.27e-07
$c_u^{X,Z}$	5.81e-06 ± 5.61e-06	-5.49e-06 ± 5.01e-06	-5.09e-06 ± 4.44e-06	-2.51e-06 ± 2.86e-06	-2.31e-06 ± 2.86e-06
$c_u^{Y,Z}$	-3.93e-07 ± 5.57e-06	5.73e-06 ± 5.08e-06	-8.35e-06 ± 4.37e-06	-1.32e-06 ± 2.69e-06	-1.94e-06 ± 2.69e-06
$c_u^{XY,XY}$	-4.48e-07 ± 2.89e-06	-1.42e-06 ± 2.57e-06	1.31e-06 ± 2.18e-06	6.24e-07 ± 1.44e-06	-1.26e-07 ± 1.45e-06
$c_u^{X,Y}$	-1.05e-06 ± 1.50e-06	3.84e-07 ± 1.27e-06	5.51e-07 ± 1.11e-06	2.50e-07 ± 7.19e-07	8.17e-08 ± 7.20e-07
$c_d^{X,Z}$	2.59e-04 ± 2.50e-04	-2.44e-04 ± 2.23e-04	-2.27e-04 ± 1.98e-04	-1.12e-04 ± 1.27e-04	-1.03e-04 ± 1.27e-04
$c_d^{Y,Z}$	-1.75e-05 ± 2.48e-04	2.55e-04 ± 2.26e-04	-3.72e-04 ± 1.95e-04	-5.86e-05 ± 1.20e-04	-8.63e-05 ± 1.20e-04
$c_d^{XY,XY}$	-1.99e-05 ± 1.29e-04	-6.30e-05 ± 1.14e-04	5.85e-05 ± 9.73e-05	2.78e-05 ± 6.43e-05	-5.59e-06 ± 6.45e-05
$c_d^{X,Y}$	-4.66e-05 ± 6.66e-05	1.71e-05 ± 5.65e-05	2.46e-05 ± 4.93e-05	1.11e-05 ± 3.20e-05	3.64e-06 ± 3.21e-05

Table 17: Partially unblinded results in $ee + \mu\mu$ channel. The table shows the average and one standard deviation of 1000 fit results.

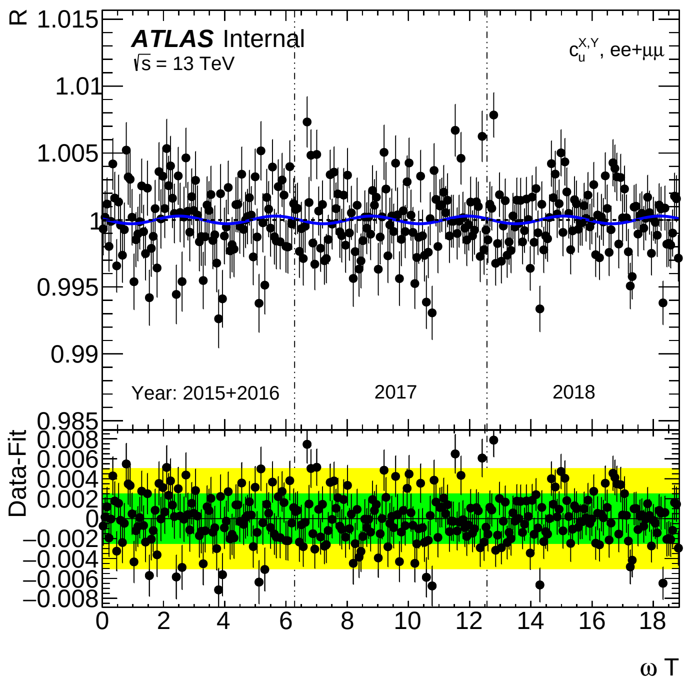

Figure 64: The Run 2 combined dataset (ne of the scrambled datasets) VS $c_u^{X,Y}$ model. The black dots are the scrambled data, and the blue lines are the fitted model. The bottom panel of the plots is the residual (data-model). The green and yellow bands correspond to one and two sigma uncertainty bands.

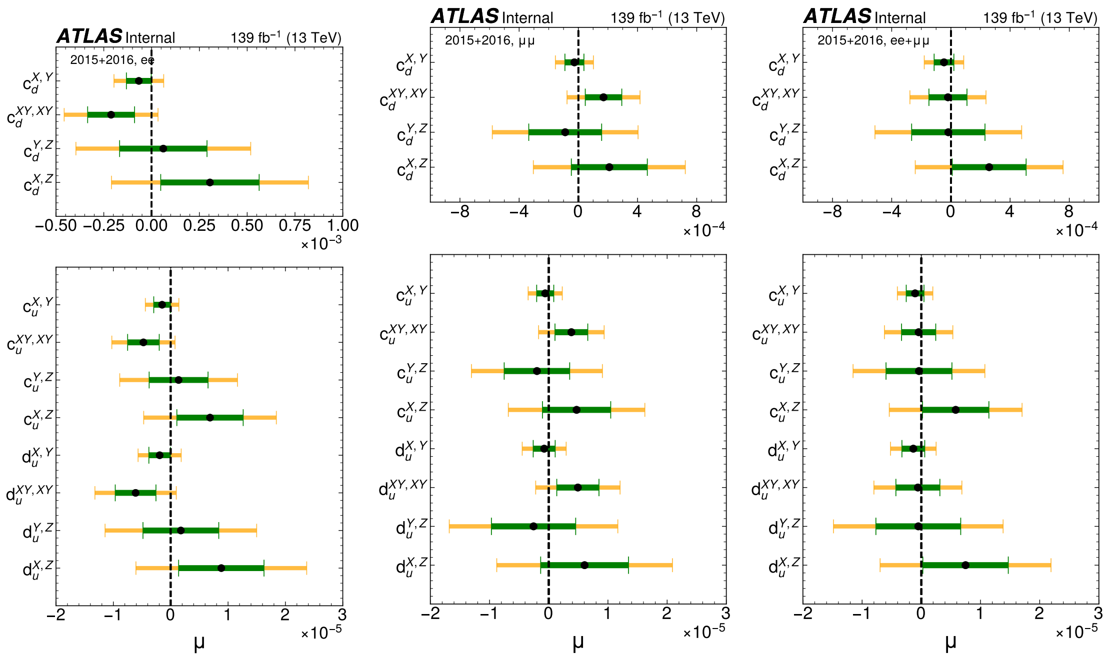

Figure 65: Scrambled datasets collected in 2015 and 2016. Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The ee channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

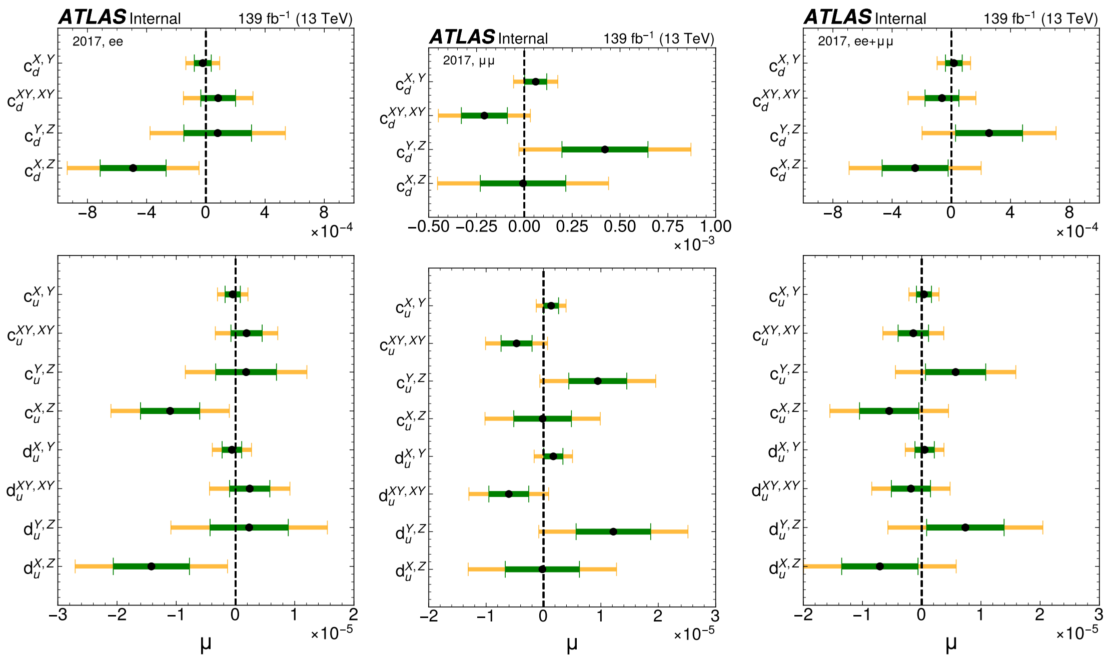

Figure 66: The scrambled datasets collected in 2017. Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The ee channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

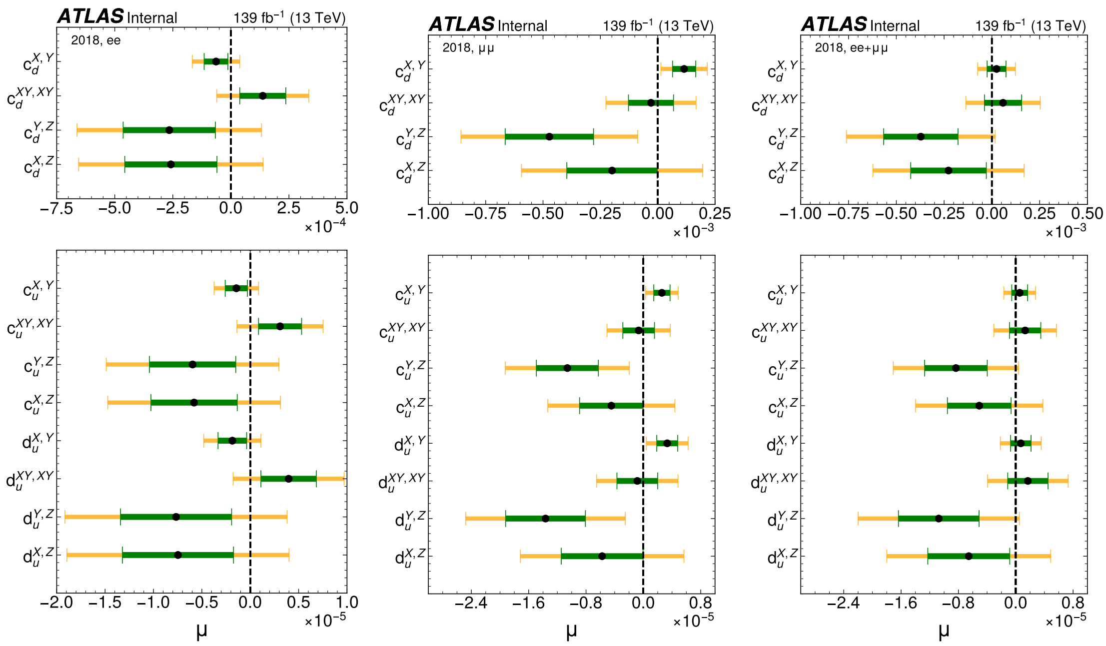

Figure 67: The scrambled datasets collected in 2018. Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The ee channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

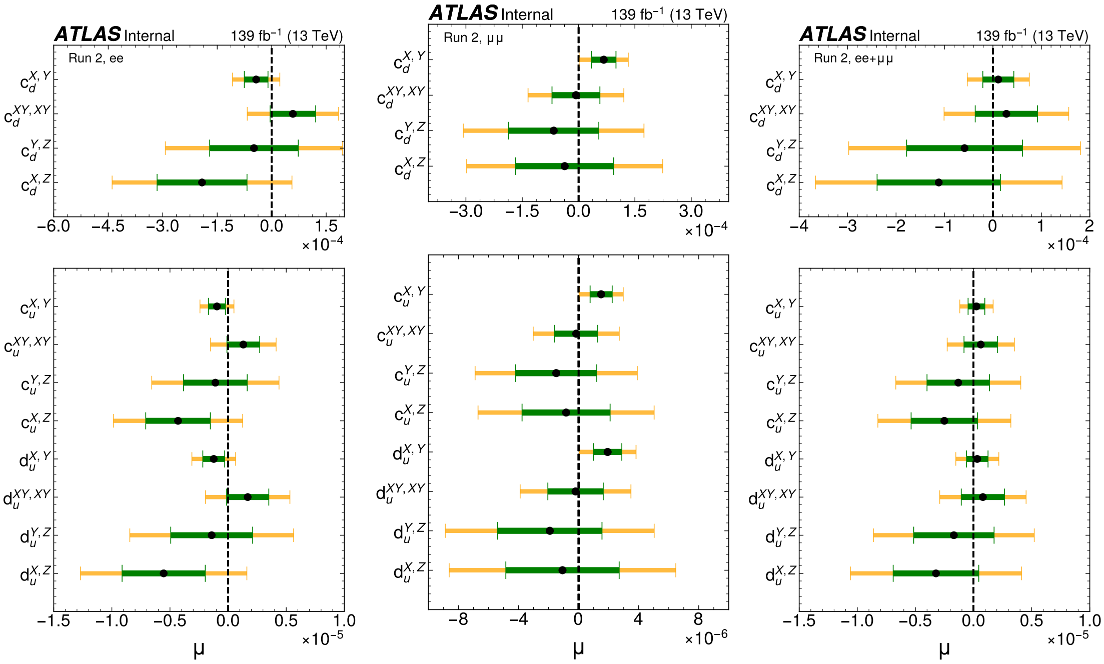

Figure 68: The scrambled datasets in full Run 2 (sum of all years). Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The *ee* channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

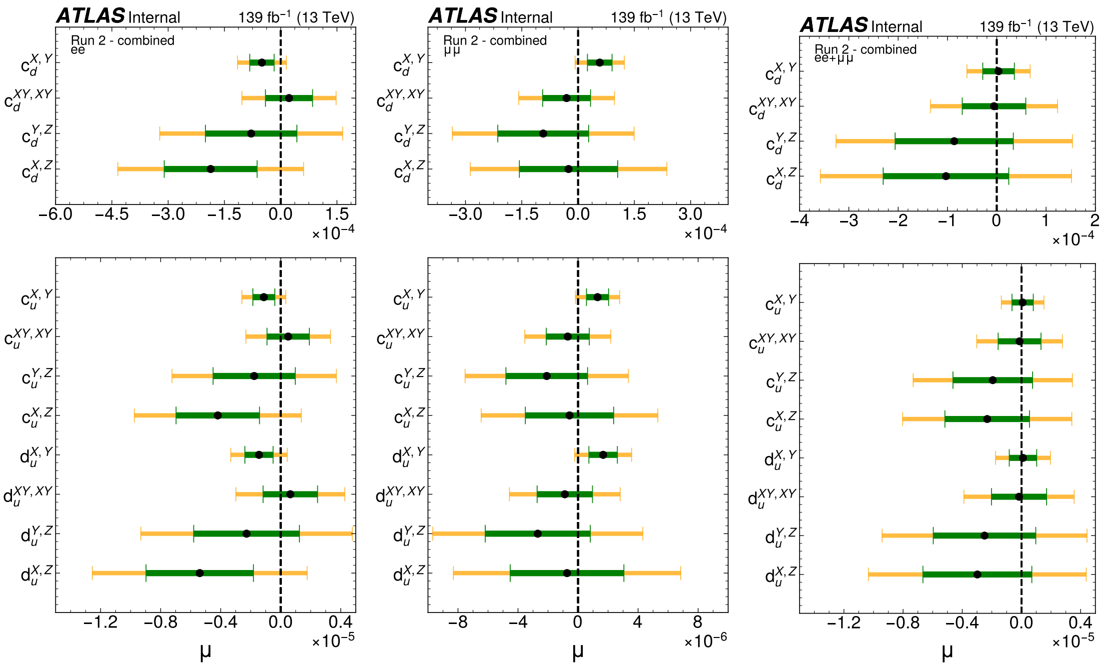

Figure 69: Run2-combined scrambled datasets. Plots show average and one (two) standard deviation of 1000 fit results. The black dots are the average values of 1000 fit results, and the green (yellow) band is the corresponding one (two) sigma band. The *ee* channel is in the left, the $\mu\mu$ channel is in the middle, and the $ee + \mu\mu$ is in the right.

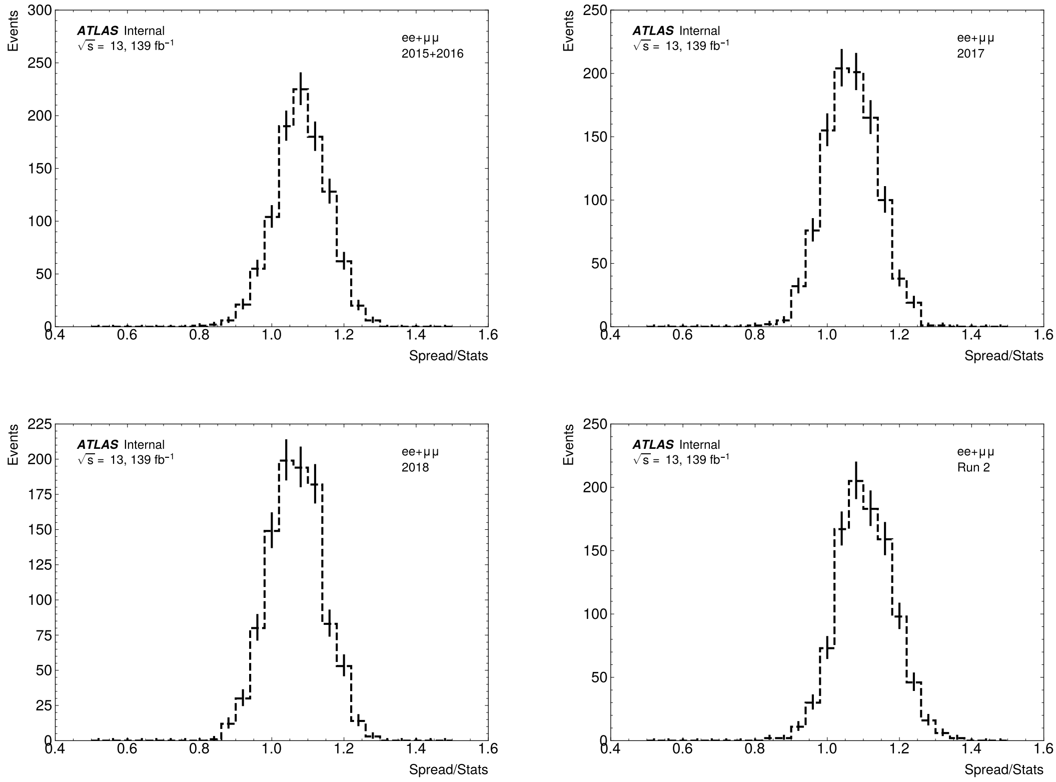

Figure 70: Spreads VS average stats. Plots show average results of 1000 scrambled datasets. An average statistic is the average statistical uncertainty in each scrambled dataset, and the spread of a dataset is an average of absolute residual, e.g., $|1 - data|4$. The 2015+2016 dataset(top left), the 2017 dataset (top right), the 2018 dataset(bottom left), and the Run 2 dataset (bottom right). The Run 2 dataset is the sum of all year's data. Spread is about 10-20% larger in most of the scrambled datasets. This indicates there is a small systematic effect or the way we are calculating the ZC error in the double ratio. This could be the reason why the average values shifted from zero

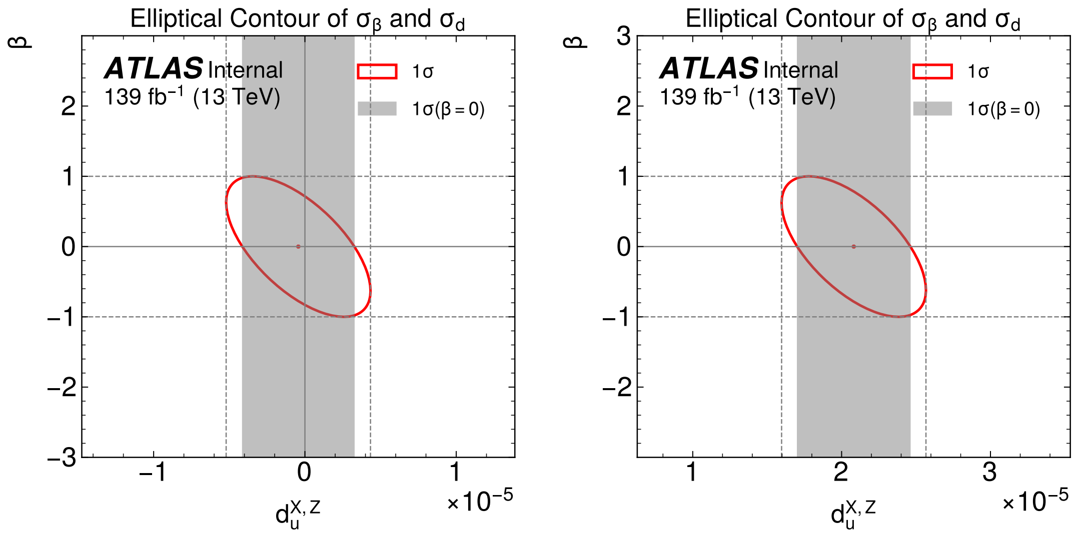

Figure 71: Combined fit results of profiled χ^2 fit, which includes the spurious signal uncertainty as a nuisance parameter β. The left plot is the result from one of the scrambled datasets. The right plot is a signal-injected toy dataset with strength $\sigma_{LIV}/\sigma_{SM} = 0.0009$; the signal model is $d_u^{X,Z}$. The grey band is the 68% uncertainty band without spurious signal. The fit uncertainty is 29(28 for the toy)% larger when we add the spurious signal uncertainty to the fit.

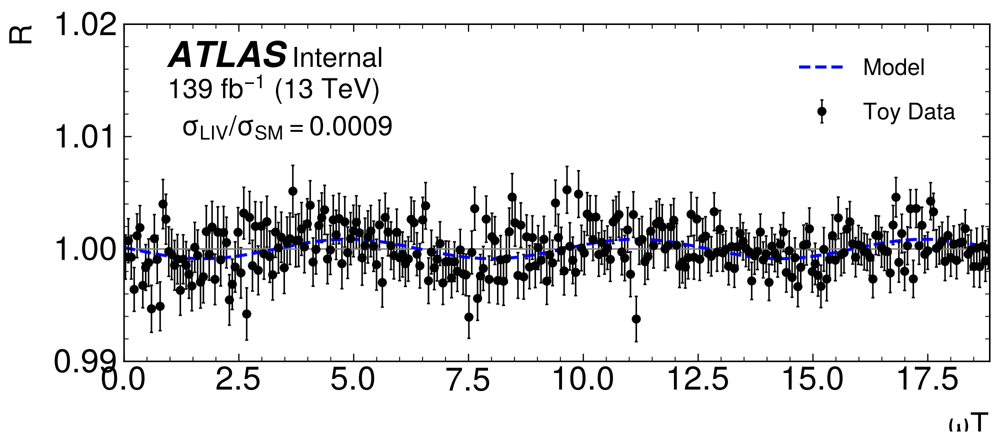

Figure 72: The signal-injected combined toy dataset with signal strength $\sigma_{LIV}/\sigma_{SM} = 0.0009$ and $d_u^{X,Z}$ model (post-fit). The blue line is the signal shape after the combined fit. The error bars on the data points are statistical errors.

	2015+2016	2017	2018	Combined
$d_u^{X,Z}$	-1.39e-05 ± 7.12e-06	6.46e-06 ± 6.58e-06	8.55e-06 ± 5.68e-06	1.74e-06 ± 3.58e-06
$d_u^{Y,Z}$	-6.42e-06 ± 6.79e-06	5.64e-06 ± 6.41e-06	-2.04e-06 ± 5.81e-06	-1.12e-06 ± 3.62e-06
$d_u^{XY,XY}$	-4.40e-08 ± 3.76e-06	7.80e-07 ± 3.36e-06	-3.06e-06 ± 3.03e-06	-1.18e-06 ± 1.91e-06
$d_u^{X,Y}$	6.88e-07 ± 1.84e-06	-1.75e-07 ± 1.70e-06	-3.07e-09 ± 1.48e-06	1.50e-07 ± 9.86e-07
$c_u^{X,Z}$	-1.08e-05 ± 5.53e-06	5.02e-06 ± 5.11e-06	6.64e-06 ± 4.41e-06	1.35e-06 ± 2.78e-06
$c_u^{Y,Z}$	-4.99e-06 ± 5.28e-06	4.38e-06 ± 4.98e-06	-1.58e-06 ± 4.51e-06	-8.69e-07 ± 2.81e-06
$c_u^{XY,XY}$	-3.42e-08 ± 2.92e-06	6.06e-07 ± 2.61e-06	-2.38e-06 ± 2.35e-06	-9.18e-07 ± 1.49e-06
$c_u^{X,Y}$	5.34e-07 ± 1.43e-06	-1.36e-07 ± 1.32e-06	-2.39e-09 ± 1.15e-06	1.16e-07 ± 7.66e-07
$c_d^{X,Z}$	-4.80e-04 ± 2.46e-04	2.24e-04 ± 2.28e-04	2.96e-04 ± 1.97e-04	6.03e-05 ± 1.24e-04
$c_d^{Y,Z}$	-2.22e-04 ± 2.35e-04	1.95e-04 ± 2.22e-04	-7.05e-05 ± 2.01e-04	-3.87e-05 ± 1.25e-04
$c_d^{XY,XY}$	-1.52e-06 ± 1.30e-04	2.70e-05 ± 1.16e-04	-1.06e-04 ± 1.05e-04	-4.09e-05 ± 6.62e-05
$c_d^{X,Y}$	2.38e-05 ± 6.36e-05	-6.05e-06 ± 5.90e-05	-1.07e-07 ± 5.14e-05	5.17e-06 ± 3.41e-05

Table 18: Results fo scrampled dataset in 1 hour period; results are obtained from combinded Run 2 fit.

F.2 Partially unblinded data in deffirent periods

F.3 Scrambled datasets: 10 batches

The 10000 scrambled datasets split into 10 batches. Each batch has 1000 scrambled datasets. Table 24 shows the results of 10 batches from combined Run 2 fit. The Each result in the table reppresent an average result of a 1000 fit reaults in corresponding batches; the corresponding uncertainty is the one standard

	2015+2016	2017	2018	Combined
$d_u^{X,Z}$	1.18e-06 ± 6.78e-06	-8.37e-06 ± 6.71e-06	-4.42e-06 ± 5.73e-06	-4.16e-06 ± 3.72e-06
$d_u^{Y,Z}$	-6.04e-06 ± 6.89e-06	4.05e-06 ± 6.50e-06	-6.26e-06 ± 5.90e-06	-3.25e-06 ± 3.70e-06
$d_u^{XY,XY}$	-3.65e-07 ± 3.74e-06	-3.97e-06 ± 3.30e-06	2.09e-07 ± 2.96e-06	-1.41e-06 ± 1.83e-06
$d_u^{X,Y}$	-6.08e-07 ± 1.87e-06	-2.89e-06 ± 1.66e-06	1.22e-06 ± 1.42e-06	-5.49e-07 ± 9.44e-07
$c_u^{X,Z}$	9.19e-07 ± 5.26e-06	-6.50e-06 ± 5.21e-06	-3.43e-06 ± 4.45e-06	-3.23e-06 ± 2.89e-06
$c_u^{Y,Z}$	-4.69e-06 ± 5.36e-06	3.15e-06 ± 5.05e-06	-4.87e-06 ± 4.58e-06	-2.52e-06 ± 2.87e-06
$c_u^{XY,XY}$	-2.84e-07 ± 2.90e-06	-3.08e-06 ± 2.57e-06	1.62e-07 ± 2.30e-06	-1.09e-06 ± 1.42e-06
$c_u^{X,Y}$	-4.72e-07 ± 1.45e-06	-2.24e-06 ± 1.29e-06	9.45e-07 ± 1.11e-06	-4.27e-07 ± 7.33e-07
$c_d^{X,Z}$	4.09e-05 ± 2.34e-04	-2.89e-04 ± 2.32e-04	-1.53e-04 ± 1.98e-04	-1.44e-04 ± 1.29e-04
$c_d^{Y,Z}$	-2.09e-04 ± 2.39e-04	1.40e-04 ± 2.25e-04	-2.17e-04 ± 2.04e-04	-1.12e-04 ± 1.28e-04
$c_d^{XY,XY}$	-1.26e-05 ± 1.29e-04	-1.37e-04 ± 1.14e-04	7.23e-06 ± 1.02e-04	-4.86e-05 ± 6.33e-05
$c_d^{X,Y}$	-2.10e-05 ± 6.47e-05	-9.99e-05 ± 5.74e-05	4.21e-05 ± 4.92e-05	-1.90e-05 ± 3.27e-05

Table 19: Results fo scrampled dataset in 6.87 hours period; results are obtained from combinded Run 2 fit.

	2015+2016	2017	2018	Combined
$d_u^{X,Z}$	8.10e-06 ± 7.12e-06	-1.58e-06 ± 6.43e-06	1.94e-07 ± 5.71e-06	1.75e-06 ± 3.71e-06
$d_u^{Y,Z}$	-1.93e-06 ± 7.00e-06	1.68e-06 ± 6.69e-06	1.10e-05 ± 6.07e-06	4.24e-06 ± 3.74e-06
$d_u^{XY,XY}$	-1.13e-06 ± 3.71e-06	1.80e-06 ± 3.31e-06	8.12e-06 ± 2.86e-06	3.46e-06 ± 1.95e-06
$d_u^{X,Y}$	6.74e-07 ± 1.84e-06	1.94e-06 ± 1.63e-06	-2.57e-07 ± 1.43e-06	7.05e-07 ± 9.49e-07
$c_u^{X,Z}$	6.30e-06 ± 5.53e-06	-1.23e-06 ± 5.00e-06	1.51e-07 ± 4.44e-06	1.36e-06 ± 2.88e-06
$c_u^{Y,Z}$	-1.50e-06 ± 5.44e-06	1.30e-06 ± 5.20e-06	8.53e-06 ± 4.72e-06	3.29e-06 ± 2.91e-06
$c_u^{XY,XY}$	-8.81e-07 ± 2.88e-06	1.40e-06 ± 2.58e-06	6.31e-06 ± 2.22e-06	2.69e-06 ± 1.52e-06
$c_u^{X,Y}$	5.24e-07 ± 1.43e-06	1.51e-06 ± 1.27e-06	-2.00e-07 ± 1.11e-06	5.48e-07 ± 7.38e-07
$c_d^{X,Z}$	2.80e-04 ± 2.46e-04	-5.46e-05 ± 2.22e-04	6.71e-06 ± 1.98e-04	6.06e-05 ± 1.28e-04
$c_d^{Y,Z}$	-6.69e-05 ± 2.42e-04	5.80e-05 ± 2.32e-04	3.80e-04 ± 2.10e-04	1.47e-04 ± 1.29e-04
$c_d^{XY,XY}$	-3.92e-05 ± 1.28e-04	6.23e-05 ± 1.15e-04	2.81e-04 ± 9.90e-05	1.20e-04 ± 6.76e-05
$c_d^{X,Y}$	2.33e-05 ± 6.38e-05	6.72e-05 ± 5.66e-05	-8.90e-06 ± 4.95e-05	2.44e-05 ± 3.28e-05

Table 20: Results fo scrampled dataset in 7 hours period; results are obtained from combinded Run 2 fit.

	2015+2016	2017	2018	Combined
$d_u^{X,Z}$	-8.00e-06 ± 7.06e-06	-6.57e-06 ± 6.28e-06	-6.15e-06 ± 5.90e-06	-6.82e-06 ± 3.67e-06
$d_u^{Y,Z}$	-7.00e-06 ± 7.35e-06	5.72e-06 ± 6.72e-06	5.02e-06 ± 5.87e-06	1.66e-06 ± 3.81e-06
$d_u^{XY,XY}$	-4.21e-06 ± 3.84e-06	5.75e-06 ± 3.27e-06	2.04e-06 ± 3.01e-06	1.36e-06 ± 1.90e-06
$d_u^{X,Y}$	-5.23e-07 ± 1.80e-06	-2.51e-07 ± 1.62e-06	7.32e-07 ± 1.41e-06	9.90e-08 ± 9.29e-07
$c_u^{X,Z}$	-6.21e-06 ± 5.48e-06	-5.11e-06 ± 4.88e-06	-4.78e-06 ± 4.58e-06	-5.30e-06 ± 2.85e-06
$c_u^{Y,Z}$	-5.44e-06 ± 5.71e-06	4.44e-06 ± 5.22e-06	3.90e-06 ± 4.56e-06	1.29e-06 ± 2.96e-06
$c_u^{XY,XY}$	-3.27e-06 ± 2.99e-06	4.47e-06 ± 2.54e-06	1.58e-06 ± 2.34e-06	1.06e-06 ± 1.48e-06
$c_u^{X,Y}$	-4.07e-07 ± 1.40e-06	-1.95e-07 ± 1.26e-06	5.69e-07 ± 1.10e-06	7.69e-08 ± 7.22e-07
$c_d^{X,Z}$	-2.77e-04 ± 2.44e-04	-2.27e-04 ± 2.17e-04	-2.13e-04 ± 2.04e-04	-2.36e-04 ± 1.27e-04
$c_d^{Y,Z}$	-2.42e-04 ± 2.54e-04	1.98e-04 ± 2.33e-04	1.74e-04 ± 2.03e-04	5.73e-05 ± 1.32e-04
$c_d^{XY,XY}$	-1.46e-04 ± 1.33e-04	1.99e-04 ± 1.13e-04	7.05e-05 ± 1.04e-04	4.70e-05 ± 6.59e-05
$c_d^{X,Y}$	-1.81e-05 ± 6.22e-05	-8.68e-06 ± 5.62e-05	2.53e-05 ± 4.89e-05	3.42e-06 ± 3.21e-05

Table 21: Results fo scrampled dataset in 7.13 hours period; results are obtained from combinded Run 2 fit.

	2015+2016	2017	2018	Combined
	2015+2016	2017	2018	Combined
$d_u^{X,Z}$	-8.76e-06 ± 7.24e-06	1.11e-05 ± 6.49e-06	-2.64e-06 ± 5.58e-06	4.22e-08 ± 3.58e-06
$d_u^{Y,Z}$	-4.61e-06 ± 6.96e-06	-1.05e-05 ± 6.51e-06	7.20e-06 ± 5.76e-06	-1.85e-06 ± 3.65e-06
$d_u^{XY,XY}$	-1.46e-06 ± 3.76e-06	4.86e-06 ± 3.27e-06	-1.96e-06 ± 2.85e-06	1.88e-07 ± 1.86e-06
$d_u^{X,Y}$	-2.54e-07 ± 1.86e-06	-1.27e-06 ± 1.61e-06	-5.87e-07 ± 1.49e-06	-6.93e-07 ± 9.24e-07
$c_u^{X,Z}$	-6.81e-06 ± 5.63e-06	8.62e-06 ± 5.05e-06	-2.05e-06 ± 4.33e-06	3.28e-08 ± 2.78e-06
$c_u^{Y,Z}$	-3.58e-06 ± 5.41e-06	-8.13e-06 ± 5.06e-06	5.60e-06 ± 4.47e-06	-1.44e-06 ± 2.84e-06
$c_u^{XY,XY}$	-1.14e-06 ± 2.92e-06	3.78e-06 ± 2.54e-06	-1.53e-06 ± 2.21e-06	1.46e-07 ± 1.44e-06
$c_u^{X,Y}$	-1.98e-07 ± 1.44e-06	-9.90e-07 ± 1.25e-06	-4.56e-07 ± 1.15e-06	-5.39e-07 ± 7.18e-07
$c_d^{X,Z}$	-3.03e-04 ± 2.51e-04	3.84e-04 ± 2.25e-04	-9.12e-05 ± 1.93e-04	1.46e-06 ± 1.24e-04
$c_d^{Y,Z}$	-1.60e-04 ± 2.41e-04	-3.62e-04 ± 2.25e-04	2.49e-04 ± 1.99e-04	-6.39e-05 ± 1.26e-04
$c_d^{XY,XY}$	-5.07e-05 ± 1.30e-04	1.68e-04 ± 1.13e-04	-6.79e-05 ± 9.84e-05	6.49e-06 ± 6.43e-05
$c_d^{X,Y}$	-8.80e-06 ± 6.42e-05	-4.41e-05 ± 5.56e-05	-2.03e-05 ± 5.14e-05	-2.40e-05 ± 3.20e-05

Table 22: Results fo scrampled dataset in 24 hours period; results are obtained from combinded Run 2 fit.

	2015+2016	2017	2018	Combined
$d_u^{X,Z}$	7.35e-06 ± 7.32e-06	-7.07e-06 ± 6.61e-06	-6.50e-06 ± 5.70e-06	-2.99e-06 ± 3.62e-06
$d_u^{Y,Z}$	-4.18e-07 ± 7.22e-06	7.34e-06 ± 6.53e-06	-1.08e-05 ± 5.82e-06	-2.50e-06 ± 3.59e-06
$d_u^{XY,XY}$	-6.28e-07 ± 3.70e-06	-1.91e-06 ± 3.36e-06	1.60e-06 ± 2.90e-06	-2.39e-07 ± 1.89e-06
$d_u^{X,Y}$	-1.28e-06 ± 1.86e-06	5.44e-07 ± 1.66e-06	7.14e-07 ± 1.46e-06	1.41e-07 ± 9.40e-07
$c_u^{X,Z}$	5.71e-06 ± 5.69e-06	-5.49e-06 ± 5.13e-06	-5.05e-06 ± 4.43e-06	-2.32e-06 ± 2.81e-06
$c_u^{Y,Z}$	-3.25e-07 ± 5.61e-06	5.70e-06 ± 5.07e-06	-8.39e-06 ± 4.52e-06	-1.94e-06 ± 2.79e-06
$c_u^{XY,XY}$	-4.88e-07 ± 2.88e-06	-1.48e-06 ± 2.61e-06	1.25e-06 ± 2.26e-06	-1.86e-07 ± 1.47e-06
$c_u^{X,Y}$	-9.96e-07 ± 1.45e-06	4.22e-07 ± 1.29e-06	5.55e-07 ± 1.14e-06	1.09e-07 ± 7.31e-07
$c_d^{X,Z}$	2.54e-04 ± 2.53e-04	-2.45e-04 ± 2.29e-04	-2.25e-04 ± 1.97e-04	-1.03e-04 ± 1.25e-04
$c_d^{Y,Z}$	-1.45e-05 ± 2.50e-04	2.54e-04 ± 2.26e-04	-3.73e-04 ± 2.01e-04	-8.66e-05 ± 1.24e-04
$c_d^{XY,XY}$	-2.17e-05 ± 1.28e-04	-6.60e-05 ± 1.16e-04	5.55e-05 ± 1.00e-04	-8.27e-06 ± 6.54e-05
$c_d^{X,Y}$	-4.43e-05 ± 6.45e-05	1.88e-05 ± 5.76e-05	2.47e-05 ± 5.06e-05	4.86e-06 ± 3.25e-05

Table 23: Results fo scrampled dataset in sidereal hours period; results are obtained from combinded Run 2 fit.

deviation obtained from 1000 fit results. Figure 73 show averate and one (two) standard deviation of 1000 fit results in each batch.

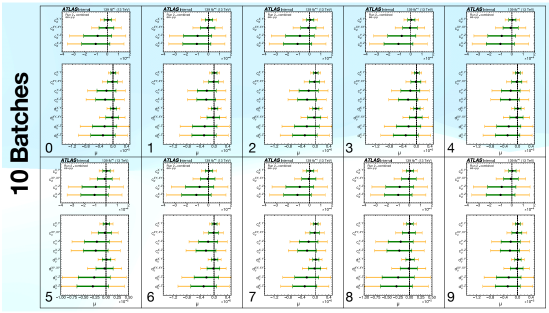

Figure 73: Plots of 10 batches. Black dots are the average of 1000 fit results in a corresponding batch. The green (yellow) band is the corresponding one (two) standard deviation of 1000 fit results.

	0	1	2	3	4	5	6	7	8	9
$c_d^{X,Y}$	4.24e-06 ± 3.28e-05	4.81e-06 ± 3.31e-05	5.46e-06 ± 3.22e-05	5.09e-06 ± 3.24e-05	4.26e-06 ± 3.19e-05	4.36e-06 ± 3.24e-05	5.05e-06 ± 3.32e-05	5.26e-06 ± 3.21e-05	5.66e-06 ± 3.22e-05	4.43e-06 ± 3.28e-05
$c_d^{XY,XY}$	-7.98e-06 ± 6.54e-05	-7.75e-06 ± 6.49e-05	-1.10e-05 ± 6.74e-05	-6.67e-06 ± 6.53e-05	-1.05e-05 ± 6.72e-05	-1.10e-05 ± 6.55e-05	-7.30e-06 ± 6.48e-05	-6.94e-06 ± 6.25e-05	-4.88e-06 ± 6.53e-05	-8.60e-06 ± 6.53e-05
$c_d^{Y,Z}$	-8.55e-05 ± 1.24e-04	-9.60e-05 ± 1.25e-04	-8.58e-05 ± 1.27e-04	-7.93e-05 ± 1.24e-04	-8.93e-05 ± 1.23e-04	-8.98e-05 ± 1.22e-04	-7.83e-05 ± 1.25e-04	-8.68e-05 ± 1.22e-04	-8.55e-05 ± 1.23e-04	-8.95e-05 ± 1.25e-04
$c_d^{X,Z}$	-1.03e-04 ± 1.25e-04	-1.03e-04 ± 1.24e-04	-1.06e-04 ± 1.26e-04	-1.08e-04 ± 1.28e-04	-1.01e-04 ± 1.25e-04	-1.01e-04 ± 1.21e-04	-1.07e-04 ± 1.31e-04	-1.07e-04 ± 1.23e-04	-9.85e-05 ± 1.22e-04	-9.95e-05 ± 1.27e-04
$c_u^{X,Y}$	9.52e-08 ± 7.36e-07	1.08e-07 ± 7.44e-07	1.23e-07 ± 7.24e-07	1.14e-07 ± 7.28e-07	9.56e-08 ± 7.15e-07	9.78e-08 ± 7.28e-07	1.13e-07 ± 7.46e-07	1.18e-07 ± 7.22e-07	1.27e-07 ± 7.24e-07	9.95e-08 ± 7.38e-07
$c_u^{XY,XY}$	-1.79e-07 ± 1.47e-06	-1.74e-07 ± 1.46e-06	-2.48e-07 ± 1.51e-06	-1.50e-07 ± 1.47e-06	-2.37e-07 ± 1.51e-06	-2.47e-07 ± 1.47e-06	-1.64e-07 ± 1.46e-06	-1.56e-07 ± 1.40e-06	-1.10e-07 ± 1.47e-06	-1.93e-07 ± 1.47e-06
$c_u^{Y,Z}$	-1.92e-06 ± 2.79e-06	-2.16e-06 ± 2.81e-06	-1.93e-06 ± 2.84e-06	-1.78e-06 ± 2.79e-06	-2.01e-06 ± 2.76e-06	-2.02e-06 ± 2.74e-06	-1.76e-06 ± 2.81e-06	-1.95e-06 ± 2.75e-06	-1.92e-06 ± 2.76e-06	-2.01e-06 ± 2.81e-06
$c_u^{X,Z}$	-2.31e-06 ± 2.81e-06	-2.30e-06 ± 2.78e-06	-2.38e-06 ± 2.82e-06	-2.43e-06 ± 2.87e-06	-2.27e-06 ± 2.81e-06	-2.26e-06 ± 2.72e-06	-2.40e-06 ± 2.94e-06	-2.41e-06 ± 2.76e-06	-2.21e-06 ± 2.74e-06	-2.23e-06 ± 2.86e-06
$d_u^{X,Y}$	1.23e-07 ± 9.48e-07	1.39e-07 ± 9.57e-07	1.58e-07 ± 9.31e-07	1.47e-07 ± 9.36e-07	1.23e-07 ± 9.21e-07	1.26e-07 ± 9.38e-07	1.46e-07 ± 9.60e-07	1.52e-07 ± 9.29e-07	1.63e-07 ± 9.32e-07	1.28e-07 ± 9.49e-07
$d_u^{XY,XY}$	-2.31e-07 ± 1.89e-06	-2.24e-07 ± 1.88e-06	-3.19e-07 ± 1.95e-06	-1.93e-07 ± 1.89e-06	-3.05e-07 ± 1.94e-06	-3.18e-07 ± 1.89e-06	-2.11e-07 ± 1.87e-06	-2.01e-07 ± 1.81e-06	-1.41e-07 ± 1.89e-06	-2.49e-07 ± 1.89e-06
$d_u^{Y,Z}$	-2.47e-06 ± 3.60e-06	-2.77e-06 ± 3.62e-06	-2.48e-06 ± 3.66e-06	-2.29e-06 ± 3.59e-06	-2.58e-06 ± 3.55e-06	-2.60e-06 ± 3.53e-06	-2.26e-06 ± 3.62e-06	-2.51e-06 ± 3.54e-06	-2.47e-06 ± 3.55e-06	-2.59e-06 ± 3.62e-06
$d_u^{X,Z}$	-2.97e-06 ± 3.61e-06	-2.97e-06 ± 3.58e-06	-3.07e-06 ± 3.63e-06	-3.13e-06 ± 3.69e-06	-2.92e-06 ± 3.61e-06	-2.91e-06 ± 3.50e-06	-3.09e-06 ± 3.78e-06	-3.10e-06 ± 3.55e-06	-2.85e-06 ± 3.53e-06	-2.88e-06 ± 3.68e-06

Table 24: Results of 10 batches. Results are the average of 1000 fit results in a corresponding batch.

G RooFit VS SciPy fits

To include spurious signal uncertainties in the χ^2, we need to add the Gaussian constraint term of nuisance parameters to the χ^2 minimization. This is easily achievable if we use the SciPy package to χ^2 fits.

In the chapter, we compared fit results obtained from RooFit and SciPy packages. We used 10000 scrambled datasets, and checked using both SME and Generic models. Figure 74 shows the comparison of RooFit and SciPy fits on an SME model ($d_{u,X,Z}$). The results show very good agreement between RooFit's and SciPy's results. The difference of the minimum χ^2 (bottom left plot) also indicates that in most cases, SciPy found slightly smaller minimum χ^2 values. Figure 75 compares RooFit and SciPy fits on the Generic signal models. The results show very good agreement between RooFit's and SciPy's results as well.

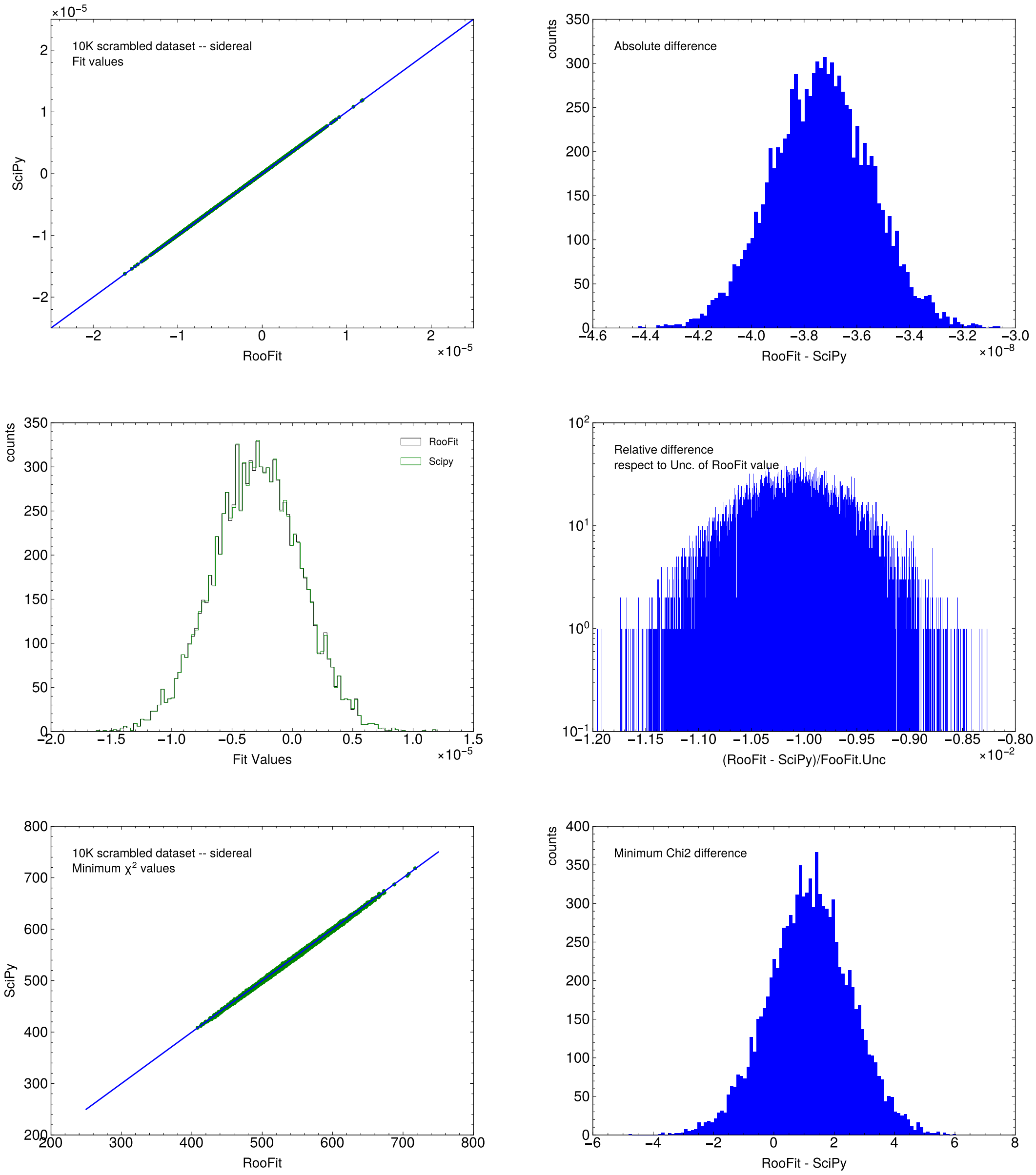

Figure 74: Comparison of the SME fit results from RooFit and SciPy on the 10K scrambled datasets. The top left plot shows the fit values; the x-axis is RooFit results, and the y-axis is SciPy results. The top right plot shows the difference between the RooFit and SciPy results. The middle left plot shows the distribution of the 10K fit results from RooFit and SciPy. The middle right plot shows the difference between RooFit and SciPy normalized to-fit errors from RooFit. The bottom left plot shows the minimum χ^2 values from RooFit and SciPy fits. The bottom right plot shows the difference in minimum χ^2.

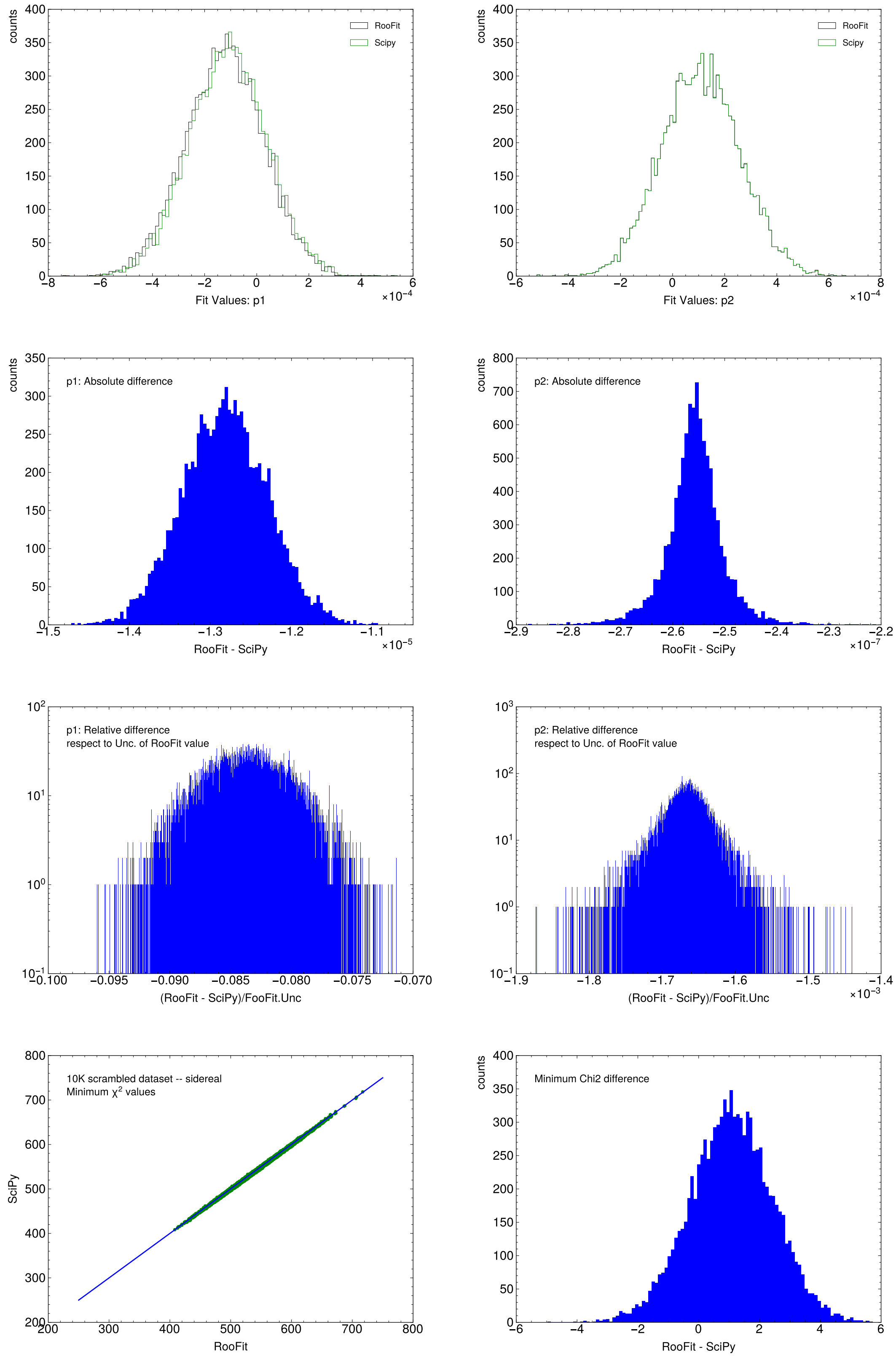

Figure 75: Comparison of the fit results of generic signals from RooFit and SciPy on the 10K scrambled datasets. The top plots show the fit values of p_1 and p_2. The plots in the second raw show the difference of the fit values from RooFit and SciPy fits. The plots in the third raw show the difference normalized to the fit errors from RooFit. The bottom left plot show the minimum χ^2 values from RooFit and SciPy fits; and the bottom right plot show the difference of the minimum χ^2 values.